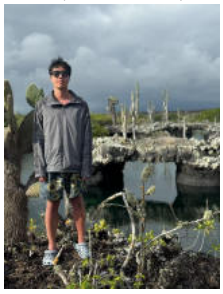
Game Design Document

DOS

TEAM - DOS

Team Members

Name	Roles	E-mail & GitHub ID
Anirudh Ramadurgam Venkat 	Team Captain, Systems	anirudhrv-usc
Wyatt(Yeefong) Wu 	Tools, User interface	yeefongwu
Aditya Prasanna Mogare 	Product Manager	AdityaMogare
Saheel Mayekar	Analytics, Stenographer,	

	Physics	Enfysyz
Vedant Jayesh Oza 	Game Level Ideation, Design and Development	Vedant-Jayesh-Oza

Note: Everyone is expected to code. It's a good idea for more than one member to take notes, especially in the event the stenographer is absent.
Reference: *Roles & Role Resources*

<Optional Group Photo>

Links

Unity Project GitHub Repository	https://github.com/CSCI-526/main-team-dos/
Playable Alpha Progress Check Build	
Playable Alpha Build	
Alpha Gameplay Video	https://www.youtube.com/watch?v=ITzqlka0WpI
Alpha Survey	https://docs.google.com/forms/d/e/1FAIpQLScNweaWz9dwilFscJGKp3AFGEQcqkkl_mWysNlvi3clCZWAtQ/viewform?usp=dialog
Playable Beta Progress Check Build	

Playable Beta Build	
Beta Gameplay Video	https://youtu.be/Wb2_Lu-b4kg
Beta Survey	https://docs.google.com/forms/d/e/1FAIpQLSejQC6ZR3kVCFgUxfFb_dy0krtLSVLsaJKcd27_K_wwjkyicg/viewform?usp=header
Playable Gold Progress Check Build	
Playable Gold Build	
Gold Gameplay Video	*https://youtu.be/N0JXVwXlidE*

*Note: It is required to have **different build links** for each of these sections. **Do NOT overwrite** a previous build with a new one as it will cause issues during grading.*

Game Introduction

Logline

2D-platformer + Portals
A 2D platformer with Portals where the player can move around and use Portals to clear the levels.

Player Goals / Win Conditions

Find the fastest way to get through the level.

Game Description

DOS, is a 2D platformer with the ability to shoot portals. The player can move around, jump and shoot portals. The blue portal is shot using the left click on the mouse and the orange portal is shot using the right click on the mouse. Repeated shooting of the portals will replace the previous portal.

The player will be able to teleport through the portals preserving their momentum. They can move obstacles and enemies around with portals in order to solve the puzzle, which encourages portal strategies over just jumping over them like in a normal platformer.

The objective of this game is to make it from the start to the end in the shortest amount of time.

Controls

Movement: A, W, D / ←,↑,→
Portals: Left Click / Right Click / C / V
Clear Portals: “R”
Reset Obstacles: ”O”
Restart Level: “L”
Exit Level: “ESC” (Double if in Full Screen Mode)

Detailed Design

Game Mechanics (How to Play)

Movement mechanic: Use the keyboard to move the player
Shooting mechanic: Use the mouse to aim and shoot portals from the Gun.
Score mechanic: Clear the level in the fastest time possible.

Game Mechanic Research

<Leaderboard / Level Timing> - Anirudh Ramadurgam Venkat

Game Title	TrackMania
Mechanic/System	Level Timing / Leaderboard
Interest	The game came out in 2003 and due to the way their level and timing for each track was designed, it created the perfect atmosphere for replayability and the game is still active to this day where players from all over the world try to beat world records for each track.
Interaction with other mechanics	The leaderboard and the level timing system makes the game more interesting and transforms it from any other racing game to a E-sports like game where players compete fiercely for the top spot. The global leaderboard makes it very evident that you are

	competing with the world and gaining even a fraction of a second feels incredibly rewarding.
Innovation	Our game not only uses a 2D platformer + Portal twist but also introduces a global leaderboard which makes the game replayable and creates a sense of competition when playing the game. Finding a new shortcut by solving the level in a different way and cutting your time by even a fraction of a second becomes rewarding.

<Portal / Puzzle> Wyatt Wu

Game Title	*Portals 2*
Mechanic/System	*Portals*
Interest	The portal mechanic is fascinating because it redefines how players perceive space and movement. Instead of moving through a level in a linear way, players must think about geometry, physics, and momentum in a creative manner. This transforms level design into a puzzle-solving exercise where players must carefully plan their actions to progress.
Interaction with other mechanics	Portals interact seamlessly with the environment, physics, and other puzzle elements. For example, momentum is conserved when entering and exiting portals, which creates challenges where players must build speed to launch themselves across gaps. Portals also work with buttons, cubes, lasers, and moving platforms, forcing the player to combine multiple systems together to solve a puzzle.
Innovation	Our game builds on the portal mechanic but twists it into a puzzle-solving action platformer. Instead of only solving static puzzles, players must also face enemies, hazards, and combat situations while using the portal gun strategically. This creates innovation by combining puzzle mechanics with combat and survival elements. Players might use portals not only to solve puzzles but also to outmaneuver enemies, redirect attacks, or escape dangerous scenarios. The global leaderboard then adds replayability, rewarding players who discover the most efficient and creative solutions.

<Enemy & Gates> Aditya Prasanna Mogare

Game Title	Prince of Persia
Mechanic/System	Enemy and Gates
Interest	In the original Prince of Persia, enemies served as direct combat challenges. These enemies required precise timing, reflexes, and mastery of the sword-fighting mechanic. Additionally, some gates were connected to pressure plates that opened only while weight was applied, forcing players to race against time before the gates closed again.
Interaction with other mechanics	In our version, this concept is reimagined. The **Enemy Mechanic** is currently lethal on contact; the player dies immediately, and the mission fails. However, like Prince of Persia, our environment still relies on **pressure plates** that open gates only when constant weight is applied. Instead of sword duels, our gameplay emphasizes **timing, positioning, and environmental awareness**.
Innovation	Unlike Prince of Persia, where enemies primarily test combat skill, our enemies contribute to the **puzzle and traversal** challenge.

<Movable Obstacles> Saheel Mayekar

Game Title	Diamond Rush
Mechanic/System	Movable Obstacles
Interest	In the 2D dungeon-crawler Diamond Rush, players navigate maze-like levels filled with enemies and traps. Movable obstacles add a significant strategic layer to this exploration by introducing environmental puzzles. These heavy objects can only be pushed horizontally, often blocking critical paths. This simple rule gives them a dual nature: they are both a tool and a hazard. If positioned correctly, an obstacle can help you solve a puzzle or crush an enemy. However, one wrong move can permanently trap you or block your path, forcing you to sacrifice a life and restart the level. This creates a high stakes puzzle element where players must think ahead before every push.

Interaction with other mechanics	Obstacles can be pushed around by the player, fall down due to gravity, block enemy paths, defeat enemies when dropped on them, defeat you if the player is under the obstacle for too long
Innovation	Our game builds on this by allowing the obstacles to be moved around using portable and giving them the ability to gain momentum similar to other entities in our game

<Level Design / Vertical Platforming> - Vedant Jayesh Oza

Game Title	Celeste
Mechanic/System	Vertical Platforming / Multi-Level Design
Interest	Celeste uses vertical level design to create challenging platforming sequences requiring precise movement and spatial thinking. The verticality creates a strong sense of progression as players ascend through increasingly difficult sections, making each climb feel rewarding.
Interaction with other mechanics	Vertical platforming interacts with dash mechanics, momentum conservation, and environmental hazards to create complex traversal challenges. The vertical space allows multiple solution paths where players can take high routes, low routes, or discover shortcuts that reward skilled play.
Innovation	Our game combines vertical multi-level platforming with portal mechanics to create unique traversal puzzles. Level_3_Final requires players to use portals strategically across four vertical tiers, thinking spatially about portal placement at different heights. The vertical design with portals creates non-linear solutions where skilled players can discover faster routes using portal physics and momentum, making levels replayable and rewarding experimentation.

Matrices

Twist & Mechanics Matrix

Mechanics	**Description**	**Interaction with Twist**	**Affected Genre Elements**	**Supports**
Portal Gun	The Gun can shoot Blue and Orange portals	The movable entities can move around the 2D map by creating portals and phase through them.	Movement, Level Design, Shooting	Momentum conservation, player, enemies, movable obstacles
Movable Obstacles	An obstacle that can be moved around	This obstacles can be can be teleported through portals and will carry a similar momentum mechanic like the player	Movement, shooting, level design	Player, enemies, portals, momentum conservation
Enemy and Gates	Gates Enemies act as hazards, and gates open via pressure plates needing constant weight.	Removes combat; enemies affect movement and timing.	Platformer & Puzzle focuses on timing and strategy.	Promotes problem-solving ,environmental awareness.
Momentum conservation	Movable entities gain momentum when falling from a height	Momentum is carried when travelling through the portals	Movement	Portal Gun, player, enemies, movable obstacles

Mechanic Interaction Matrix

Subject / Object	Portal Gun	Movable Obstacles	Enemy and Gates
Portal Gun			
Movable Obstacles	The portal Gun can be used on the Movable Obstacles to clear jumps, open gates and block enemies		
Enemy and Gates	The Portal gun can be used on the Enemies to get them to the gates for them to open or outsmart the enemy to get across them	Movable obstacles can be used to open gates and obstruct enemies.	

Single Mechanic Challenge Matrix (Beta)

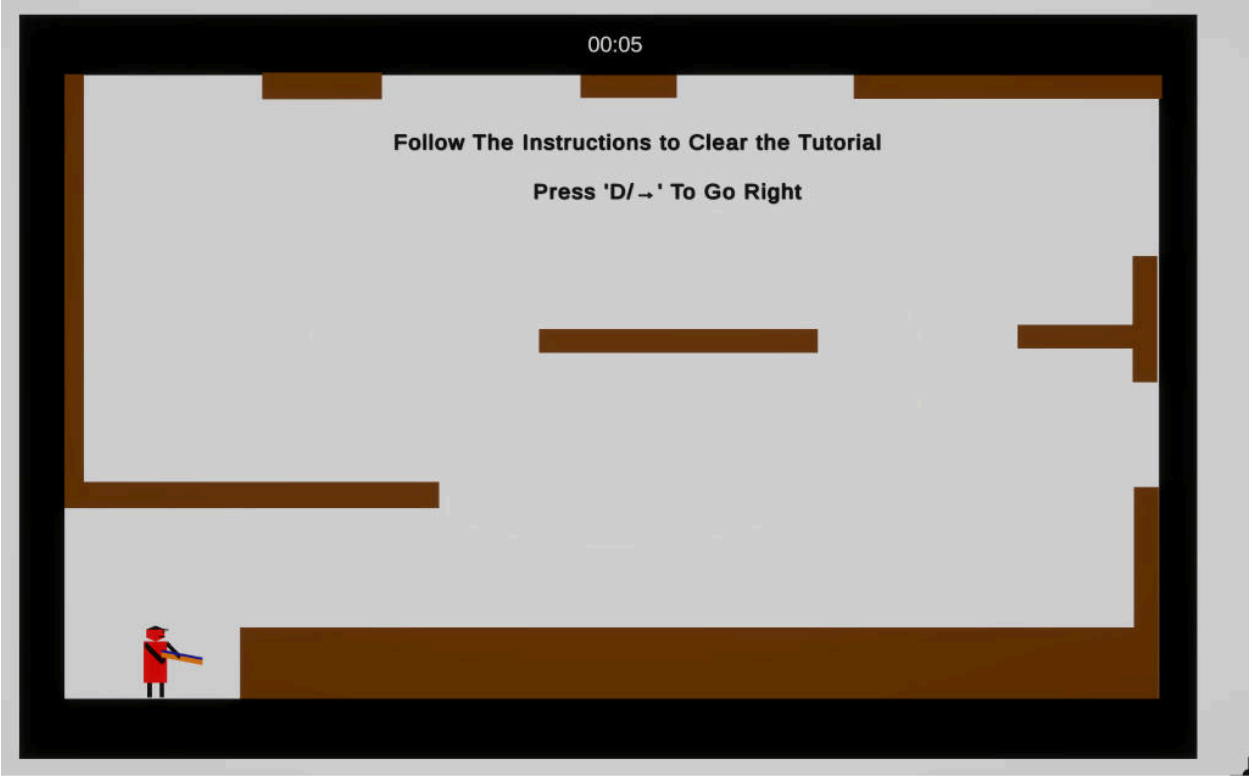
Challenge	Location	Difficulty	Player Must Learn	Instructions	Required Skills
Movement	Tutorial 1	Tutorial	Basic Controls	Tutorial 1 takes the player through how to do each movement one by one	Basic Controls
Enemy	Tutorial 2	Tutorial/Easy	How Portals Work	Tutorial 2 goes over how an enemy can be tackled by sending it through a portal	Movement, Portals
Gate Trigger	Tutorial 2	Tutorial/Easy	How Portals Work	Tutorial 2 goes into how the gate trigger works	Movement, Portals
Spikes	Level 3	Easy	Basic Controls	The color contrast of the player and the spikes with its shape make it evident that the player must not touch the spikes	Movement, Portals
Momentum	Level 4	Medium	That Jumping Into Portals conserves your momentum and lets you access parts that otherwise would be impossible	Although Level 4 explicitly revolves around Momentum, Momentum as a mechanic is usable from the beginning. We will show hints of how momentum can be used.	Movement, Portals, Timing
Moving Platforms	Level 5	Medium	Basic Controls & Timing	The platform moves in the level	Movement, Portals, Timing
Laser Gun Enemy	Level 6	Hard	Basic Controls & Timing	The laser has a Warning shot before it actually shoots to let the player know that they need to dodge	Movement, Portals, Timing

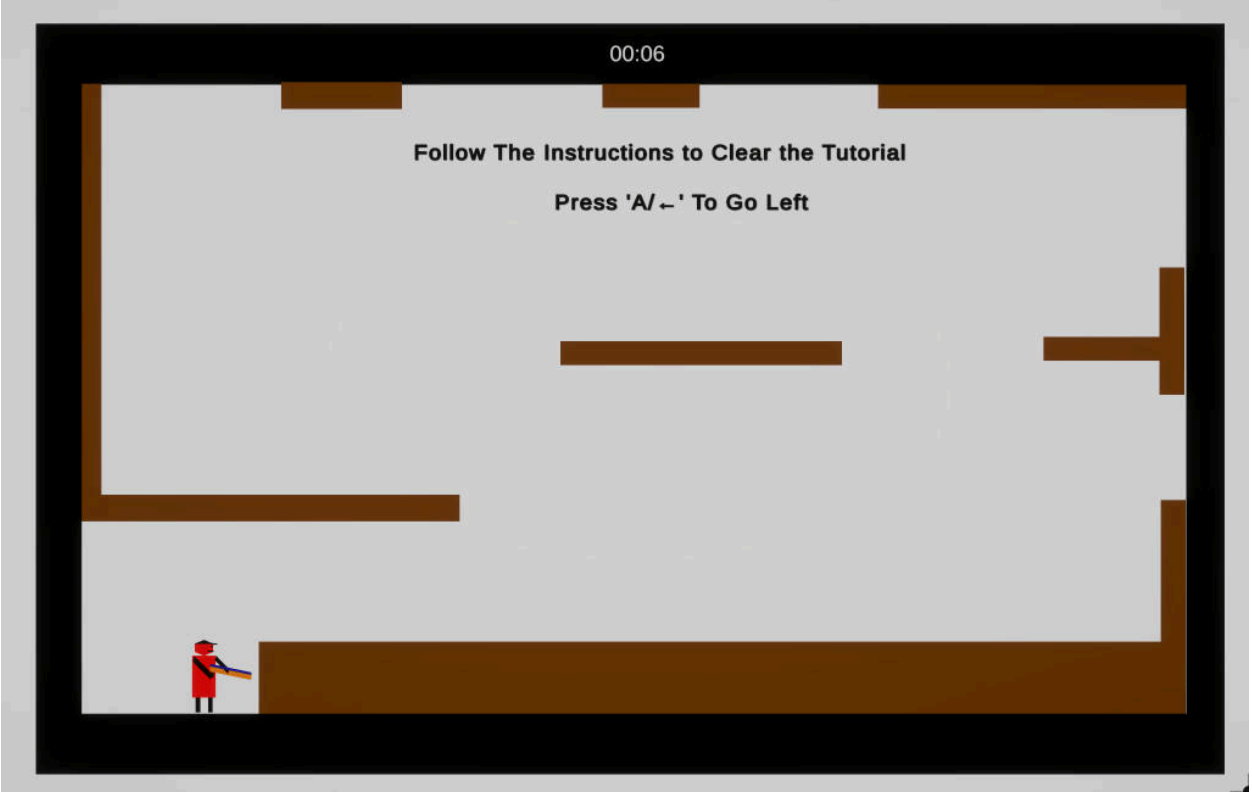
Reference: *Example Matrices*

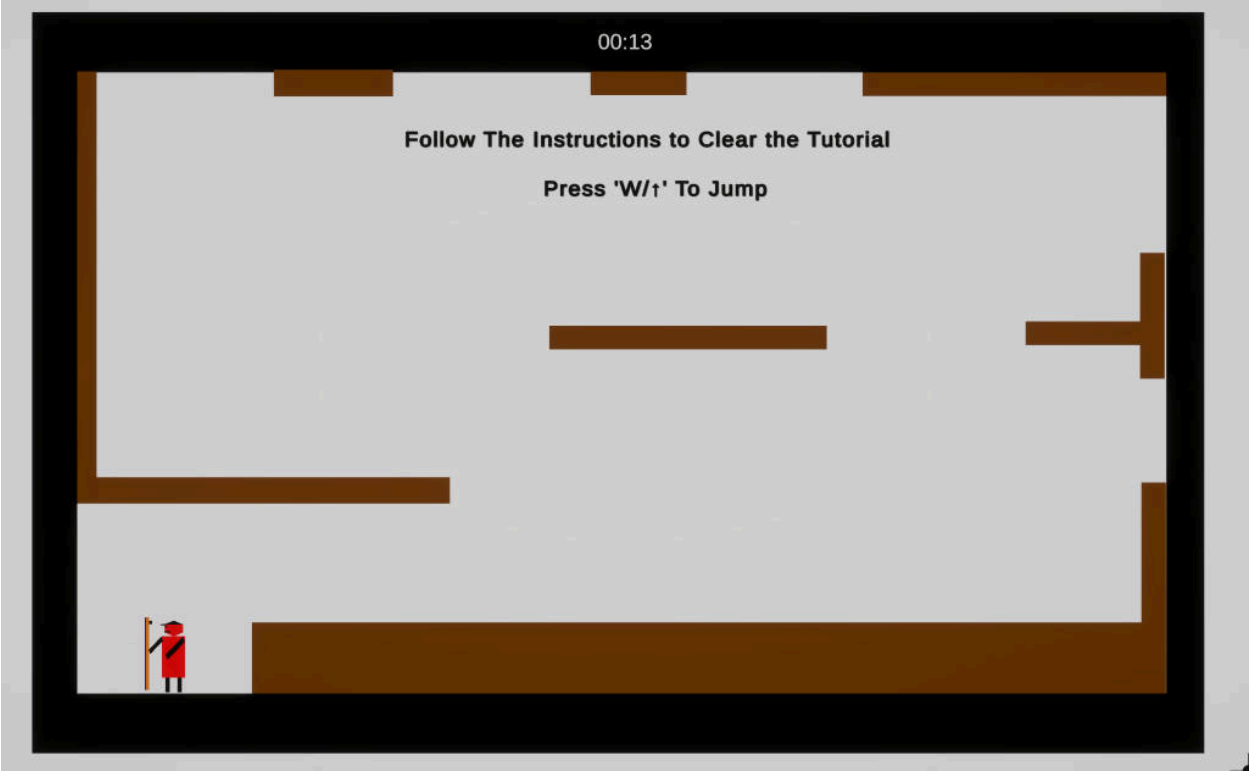
Tutorialization implementation for Mechanics (Beta)

Tutorial 1:
Helps the player get used to:

1. Moving around
2. Shooting portals
3. Jumping into the portals




00:06
Follow The Instructions to Clear the Tutorial
Press 'A/←' To Go Left


00:13
Follow The Instructions to Clear the Tutorial
Press 'W/↑' To Jump


00:23
Follow The Instructions to Clear the Tutorial
Shoot Blue Portal Using Left Click


00:33
Follow The Instructions to Clear the Tutorial
Shoot Orange Portal Using Right Click

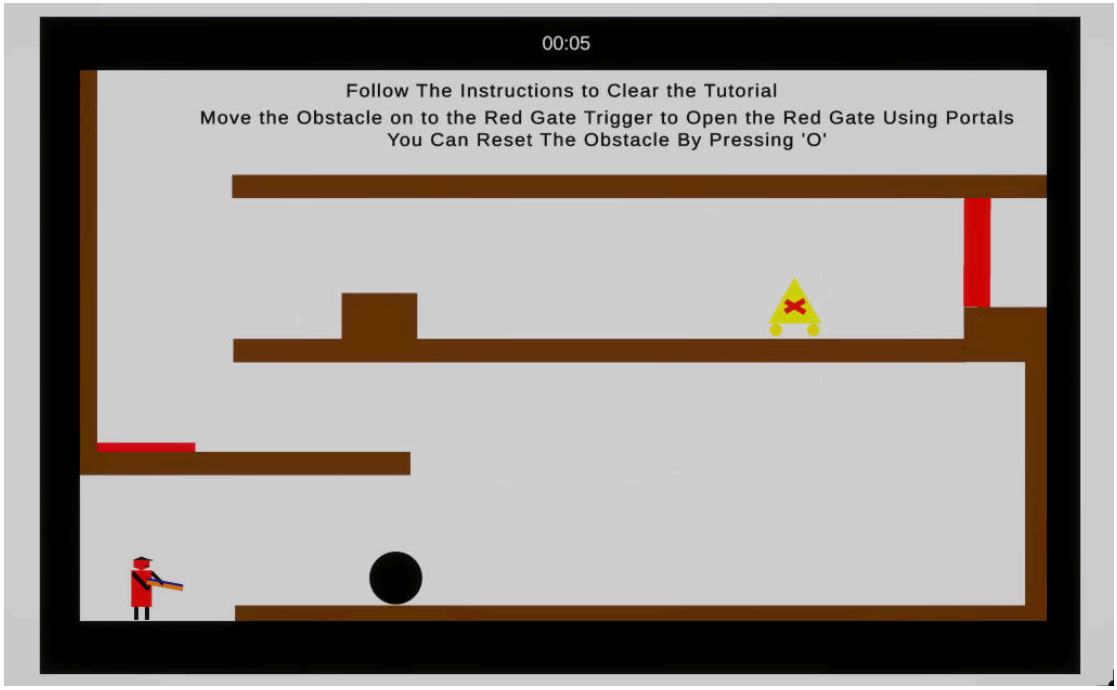

00:37
Follow The Instructions to Clear the Tutorial
Jump Into The Portals


00:44
Follow The Instructions to Clear the Tutorial
Now Reach the Exit Door
EXIT

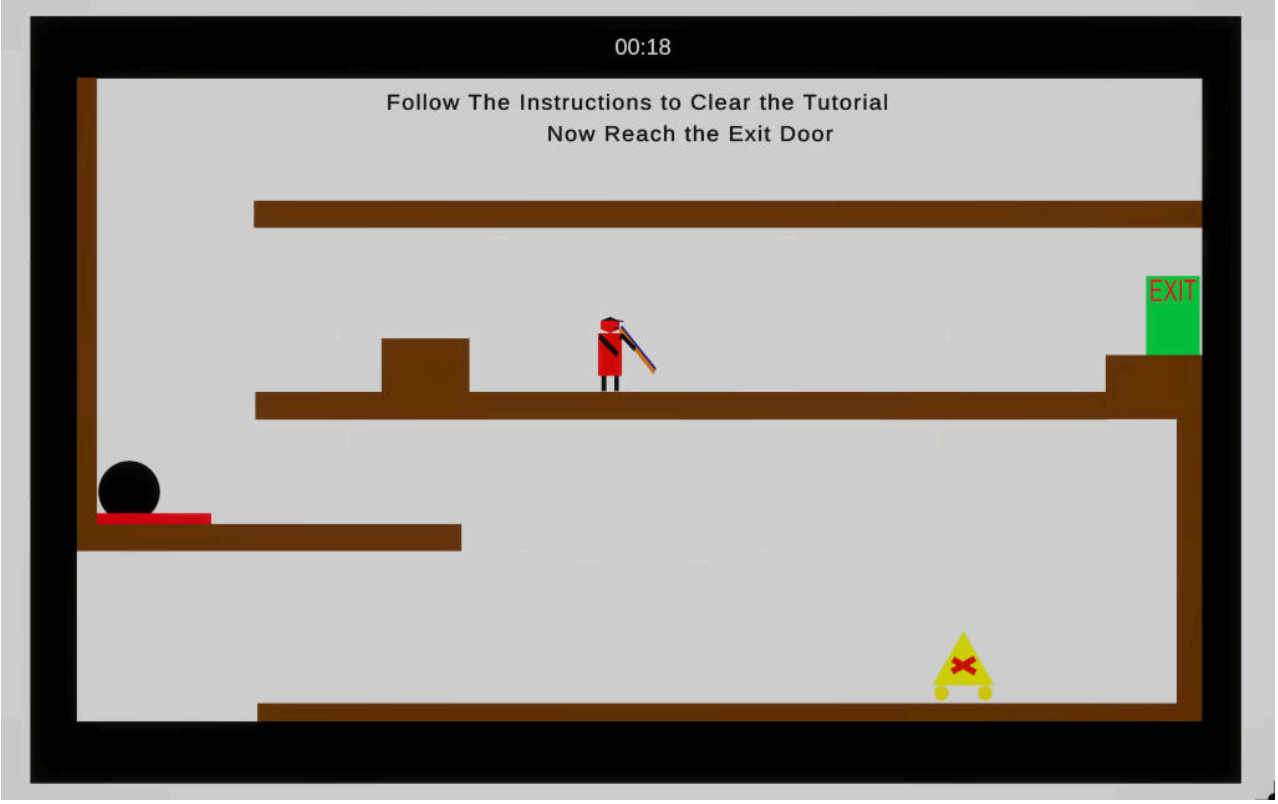
Tutorial 2:

Helps user understand how:

1. Obstacles can be teleported
2. How Gate Triggers Work
3. How Enemies are and can be teleported away







The tutorialization covers the basic mechanics of the game and gives an introduction to what to expect from the levels. Then, from Levels 1 through 10, we introduce new challenges which are derived from these basic mechanics to increase the challenge in clearing the levels.

Tutorial Gold

Tutorial 1

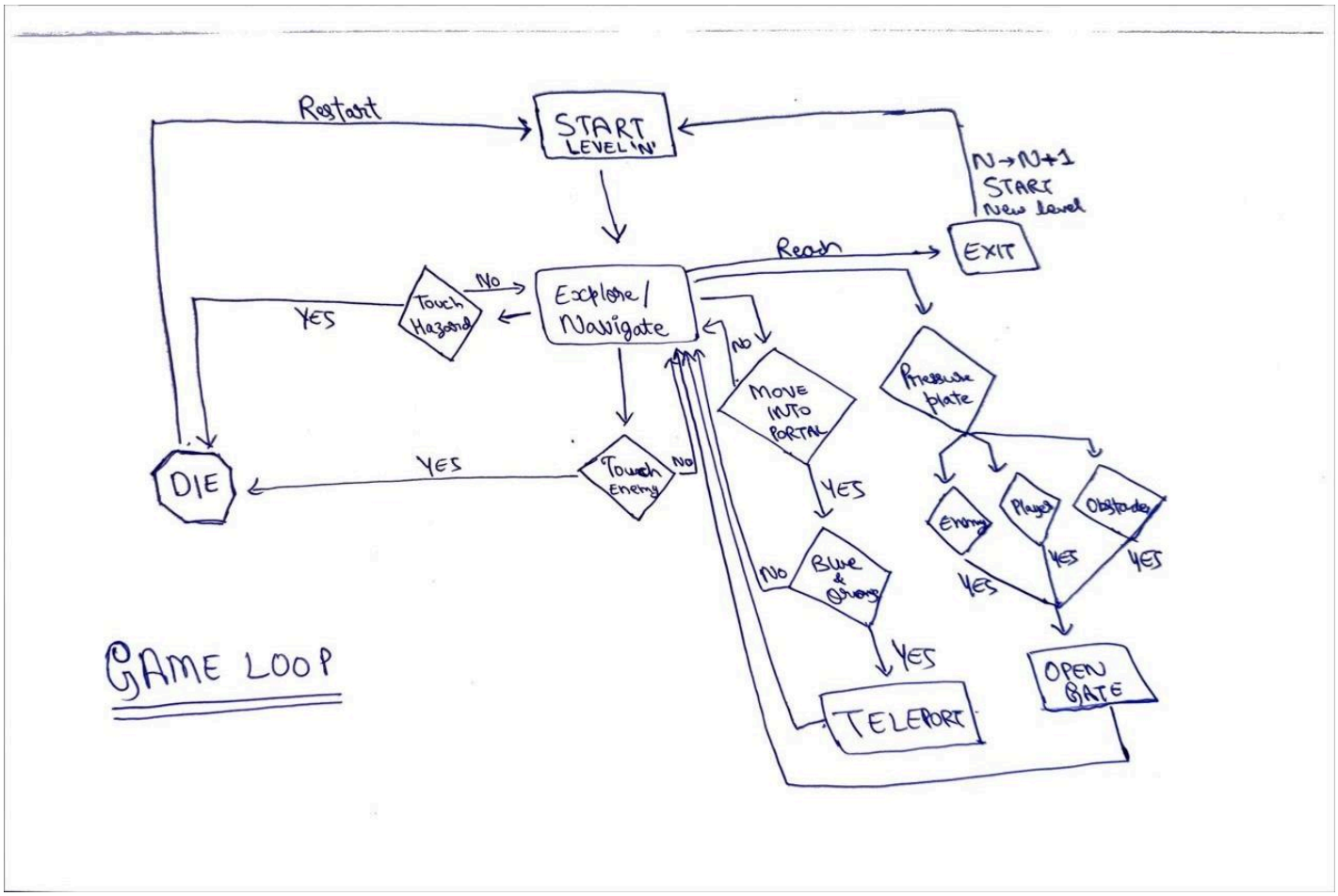


Tutorial 2

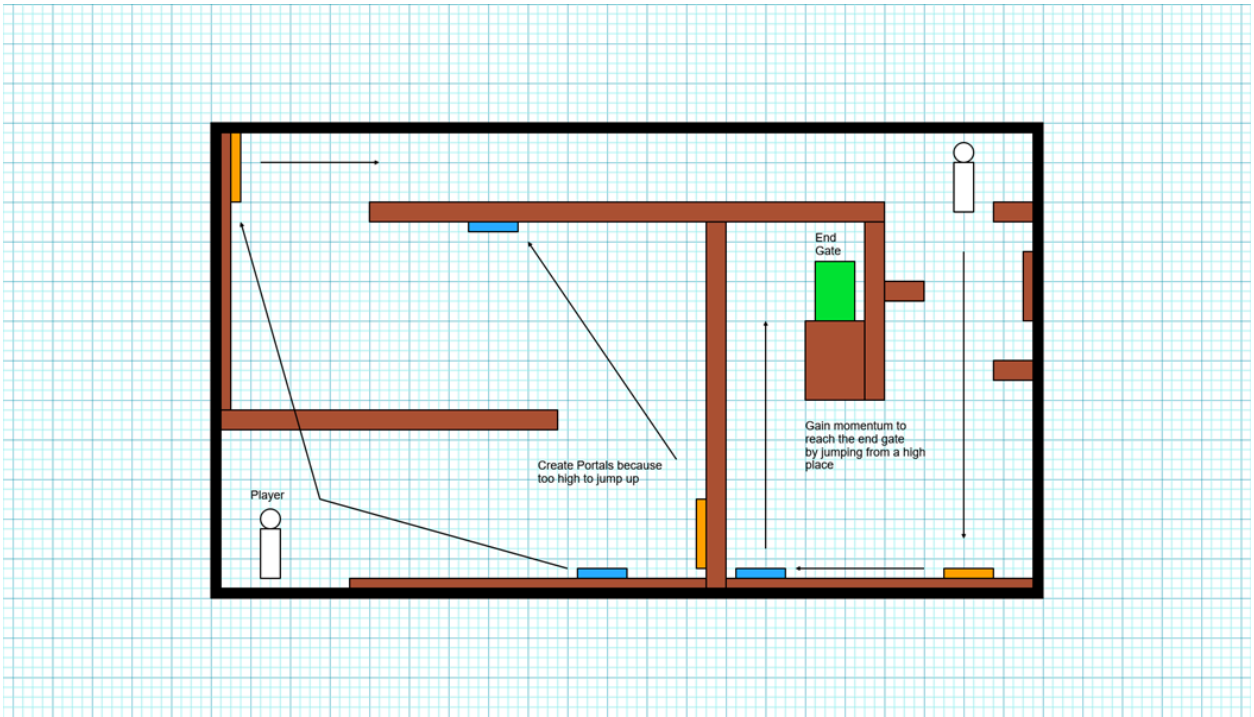


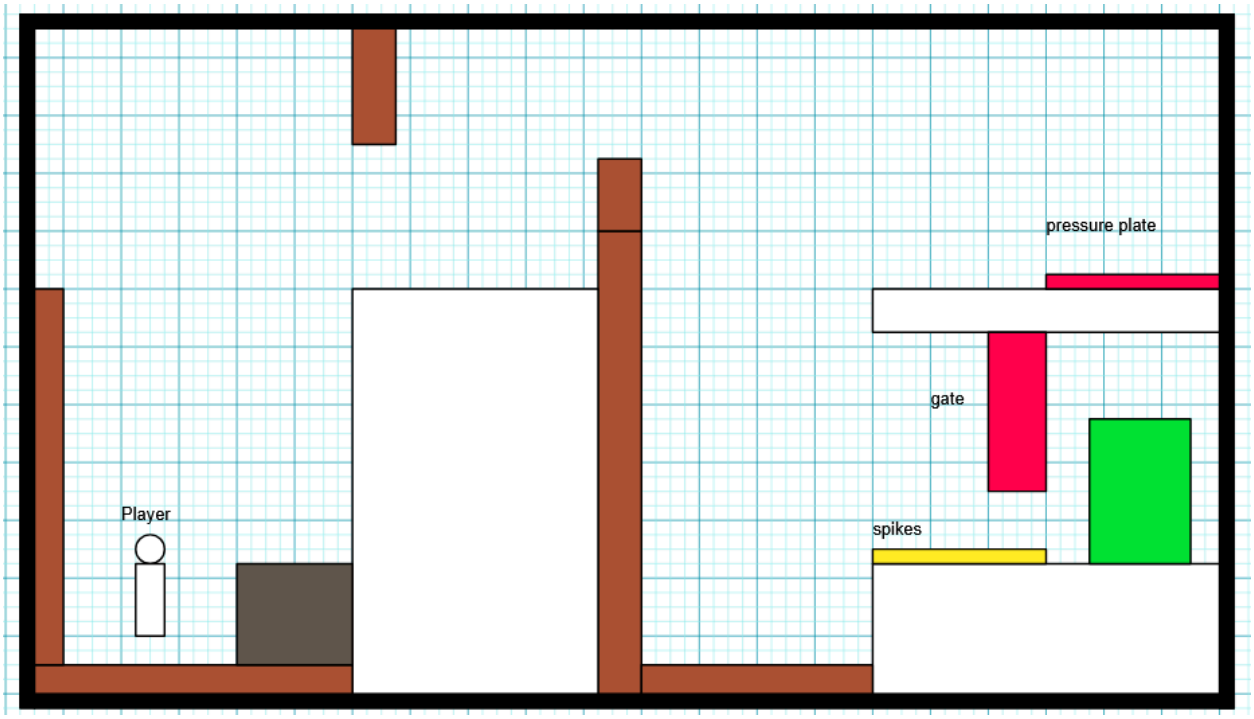
Sketches and Diagrams

Gameplay Loop



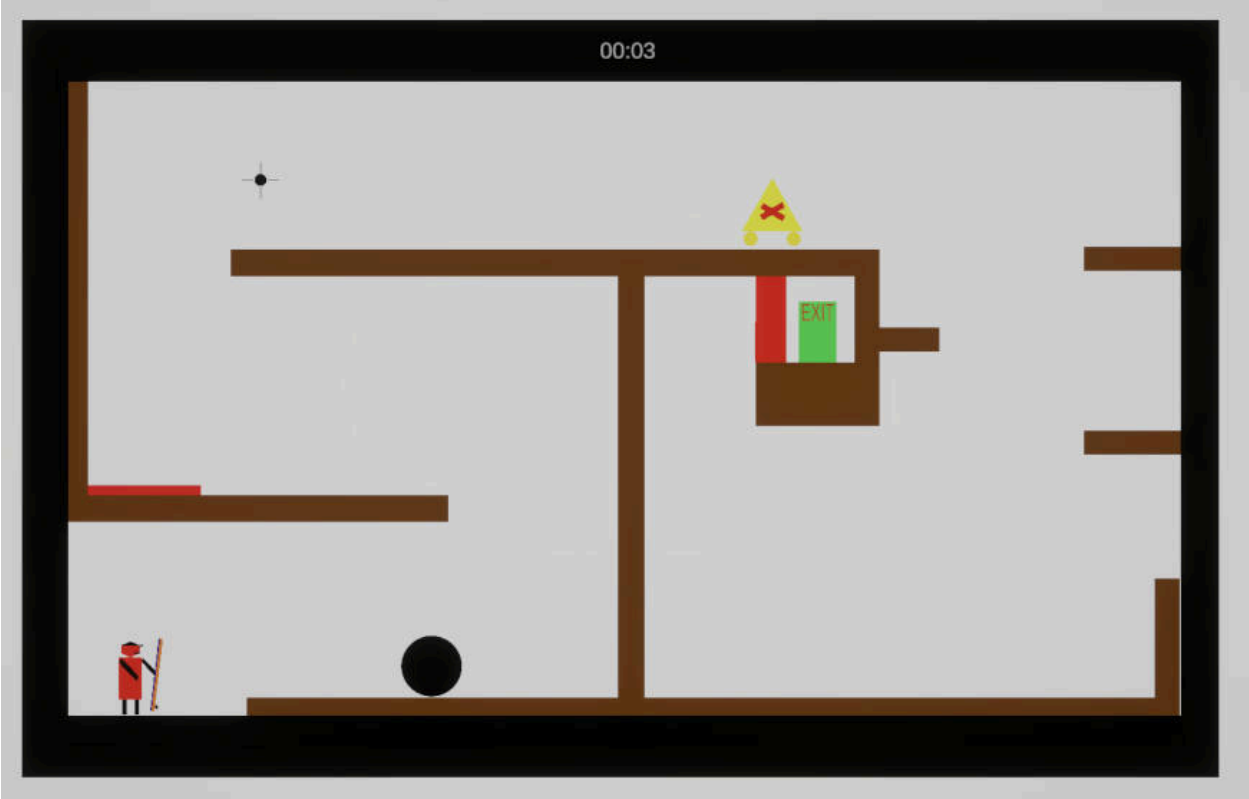
Mechanic/Level Sketches



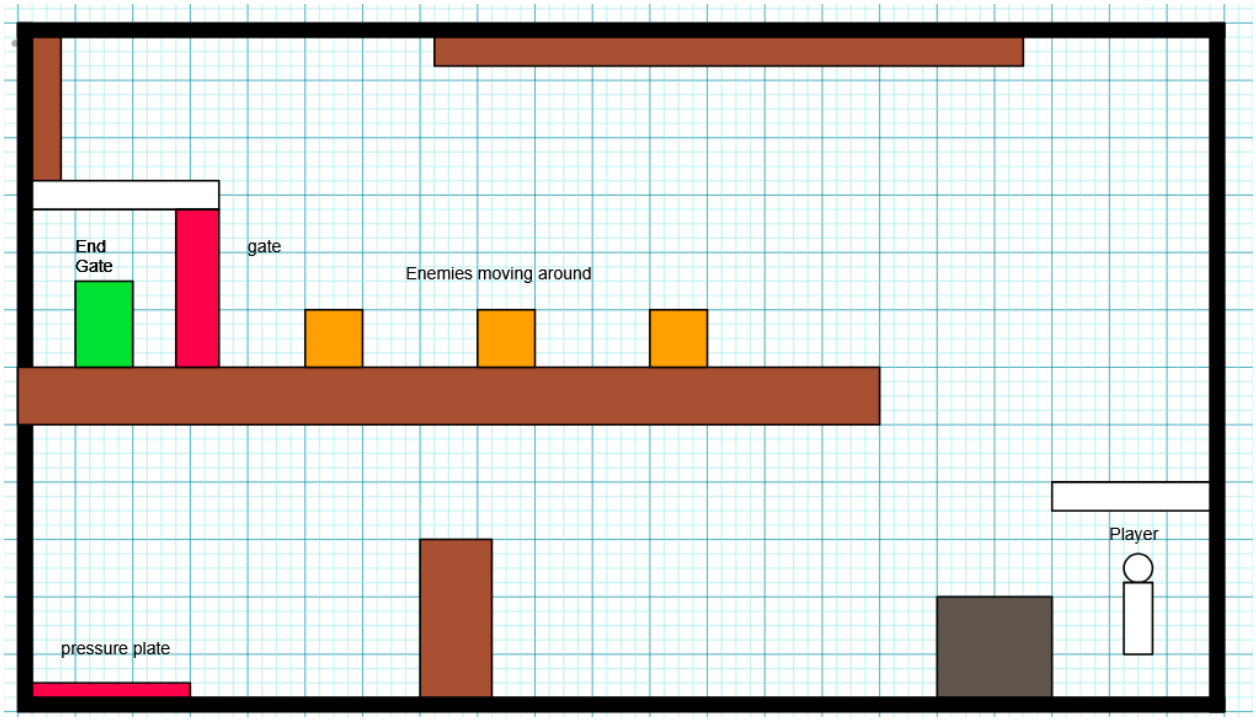


Level Designs (Beta)

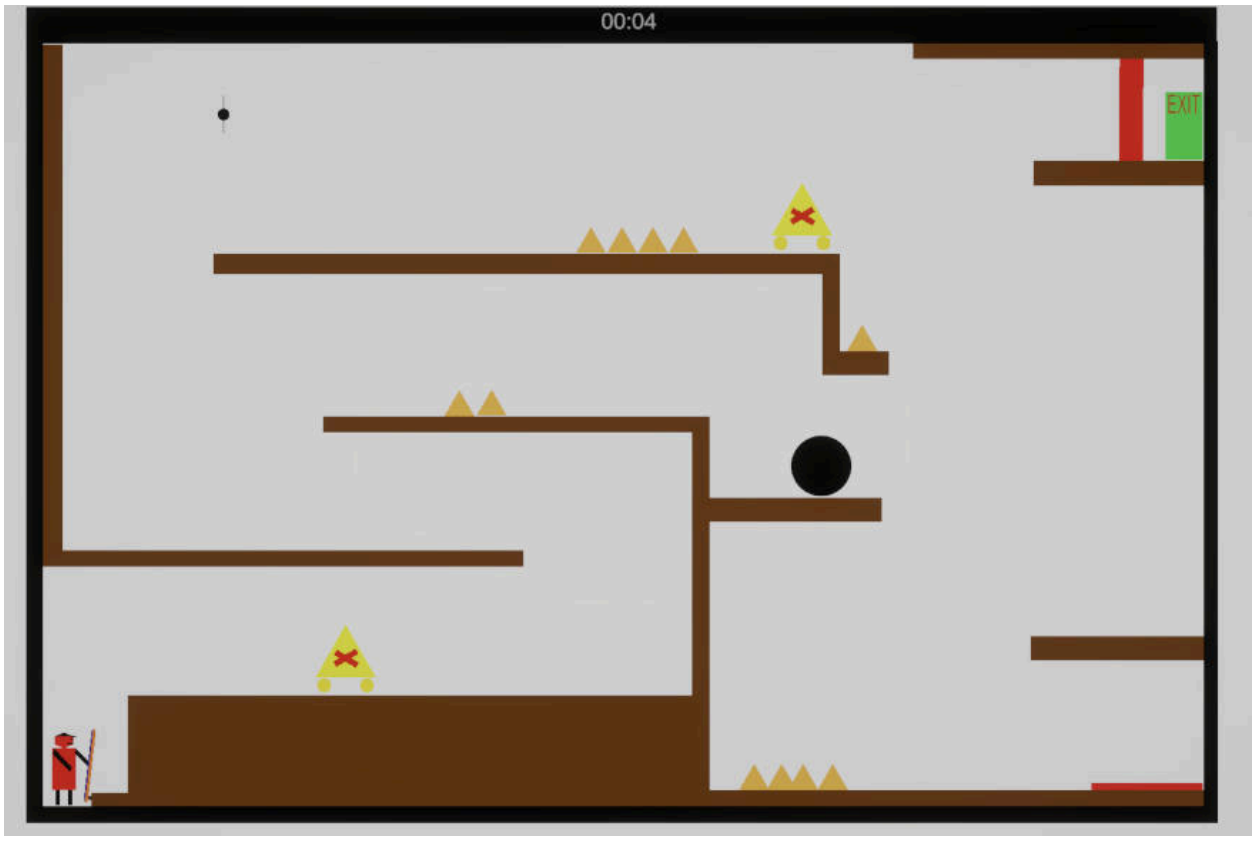
Level 1: Introduction Level with basic level design



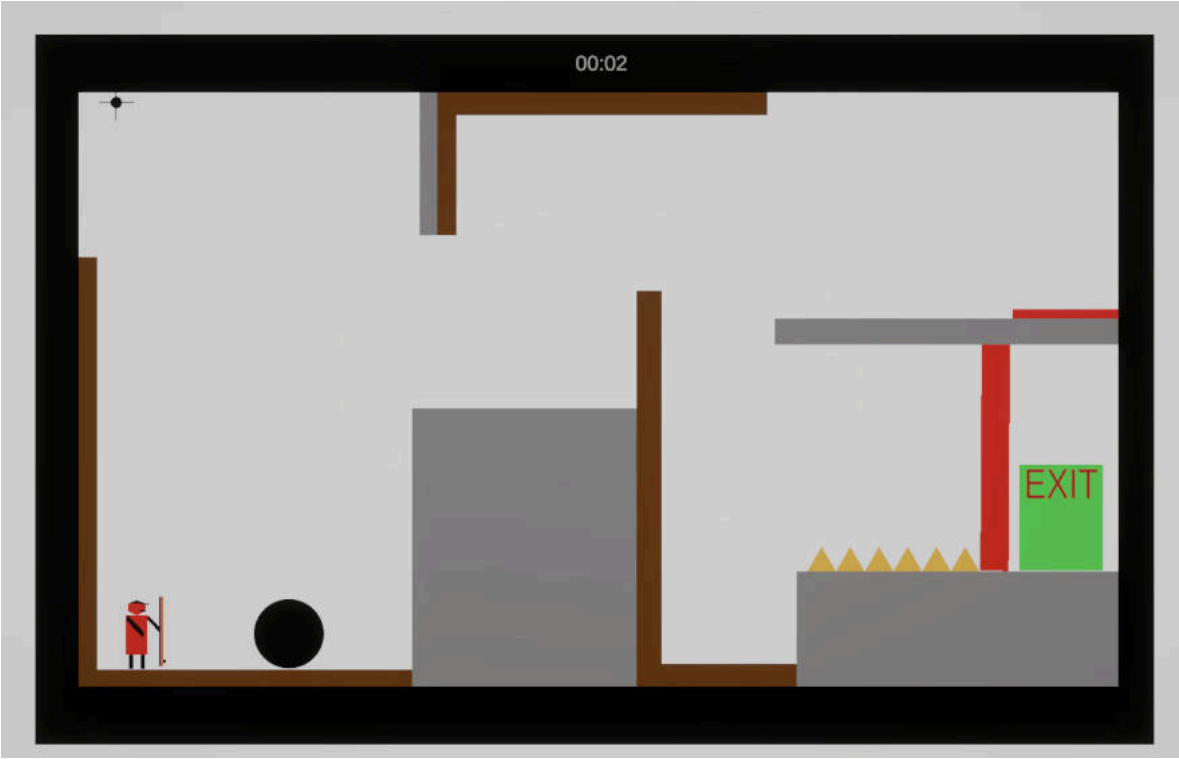
Level 2: (Multiple Enemies)



Level 3: Introduces Spikes



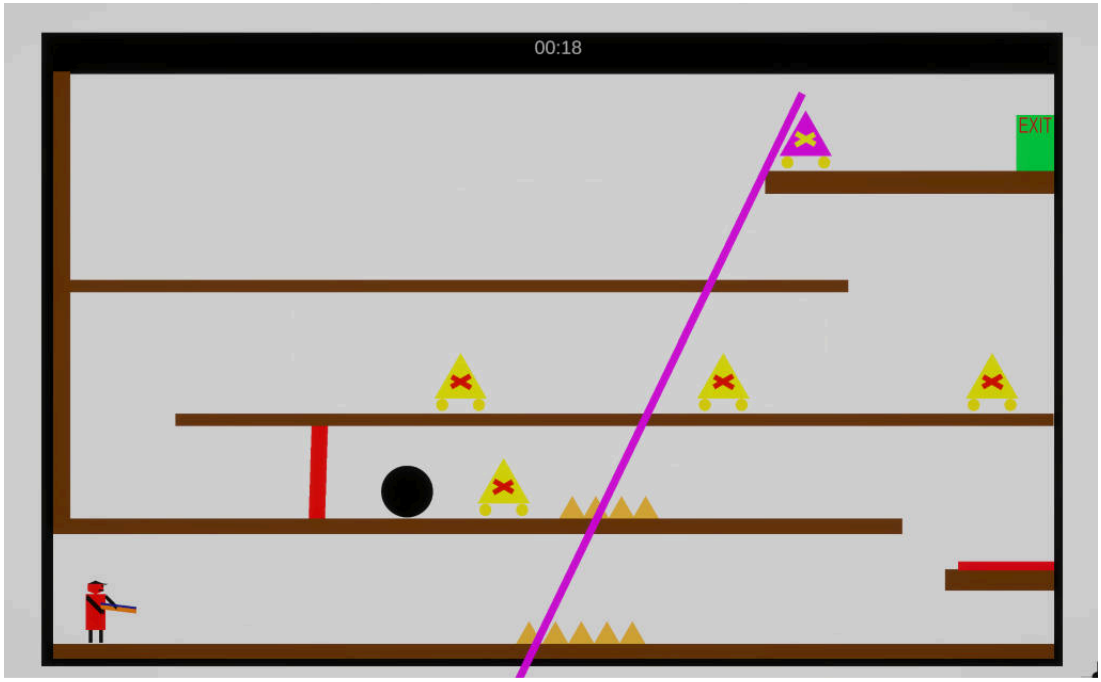
Level 4: Uses Momentum as its main focus



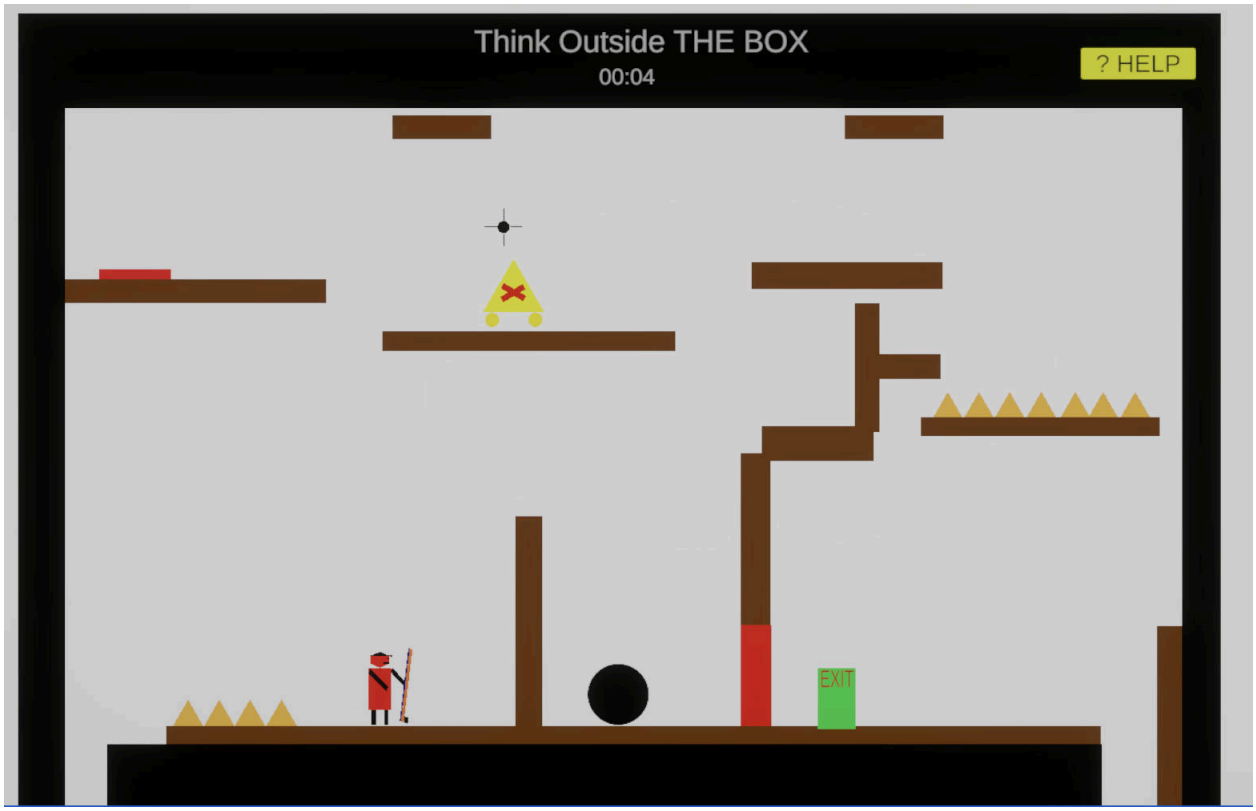
Level 5: Introduces Moving Platforms



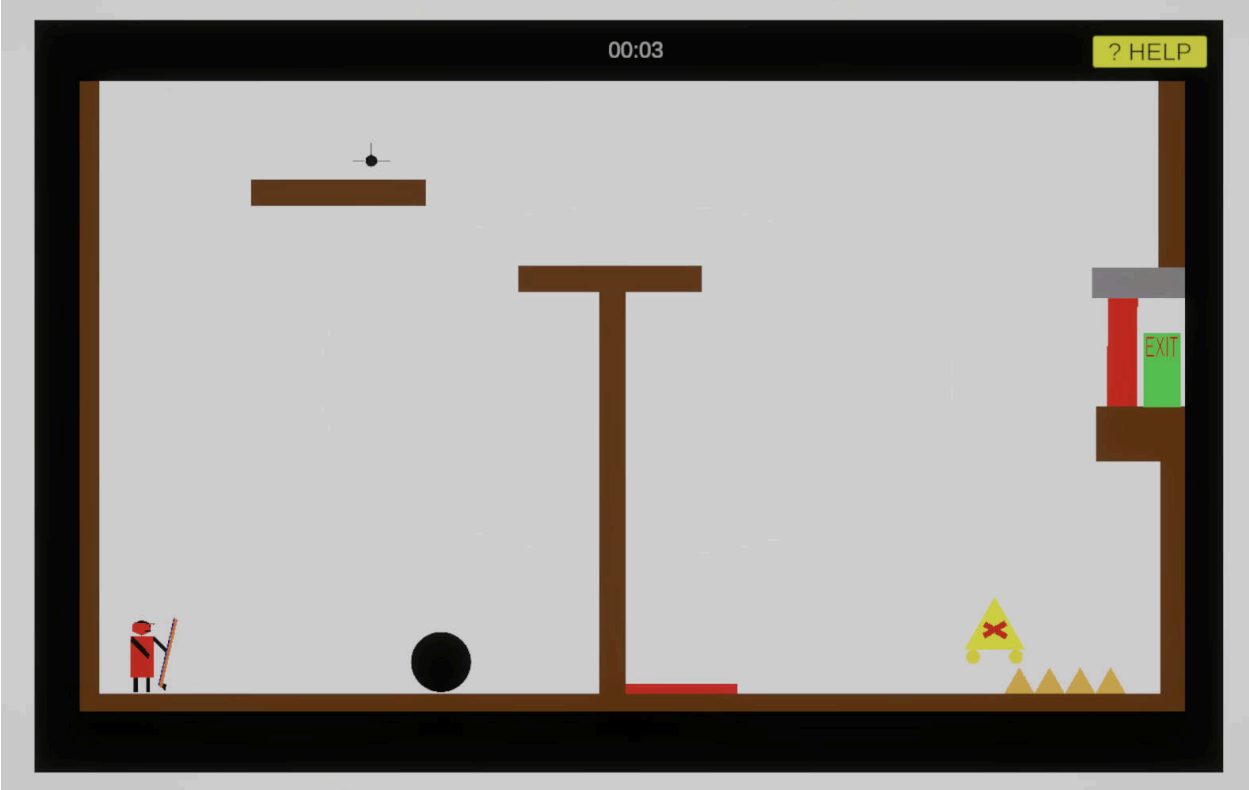
Level 6: Has Laser Shooting Enemy



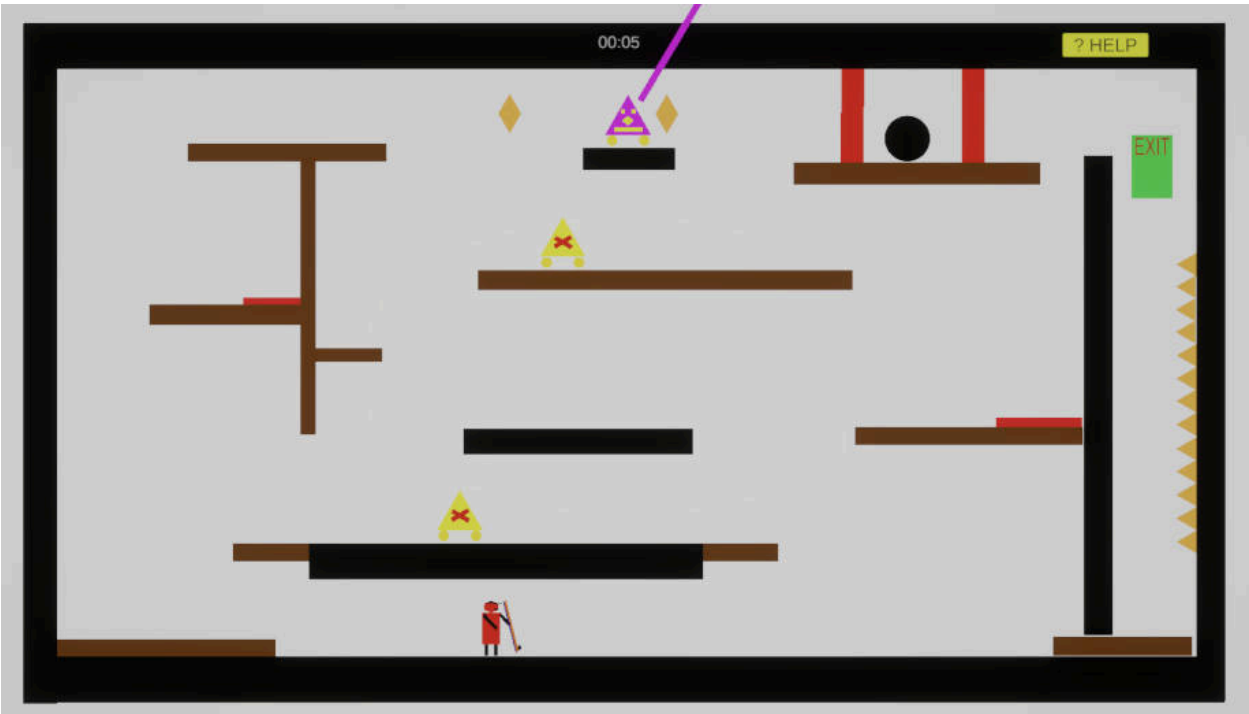
Level 7: Explores how portals can truly define the Level Design



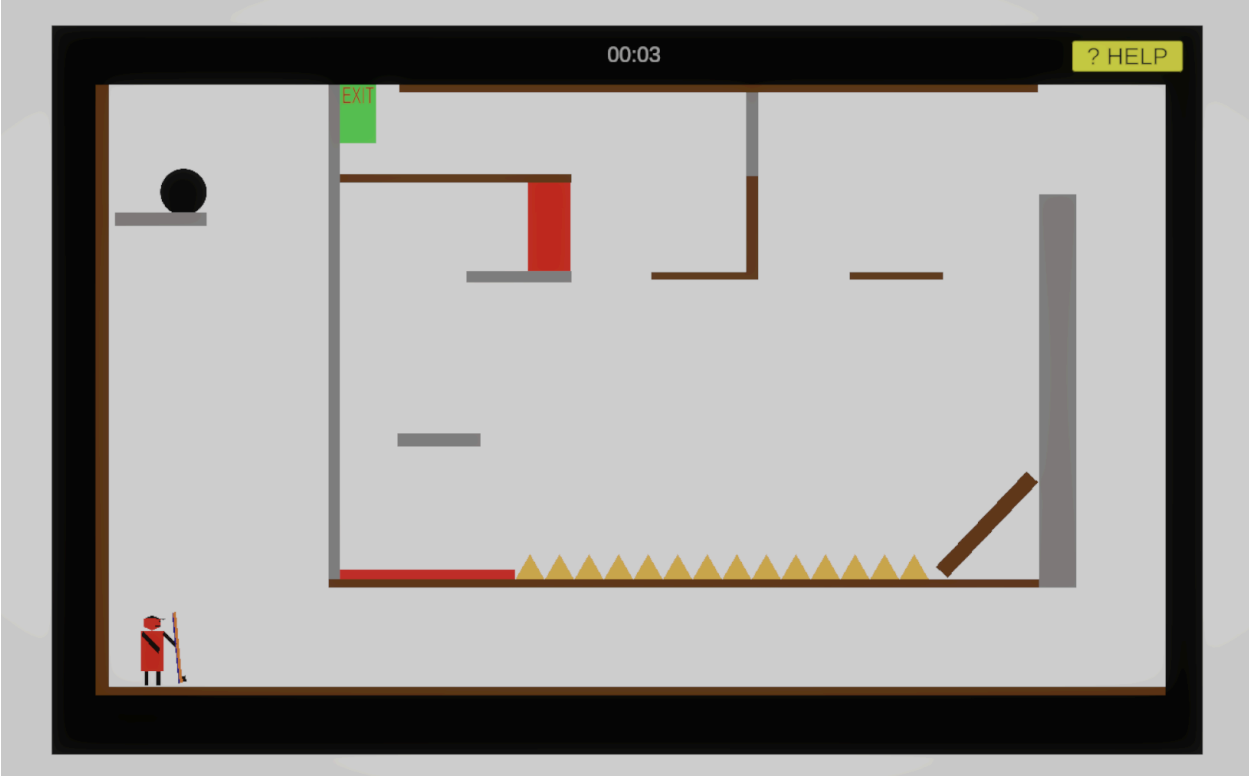
Level 8: Challenges understanding of Momentum



Level 9: Culmination of all mechanics & challenges introduced throughout level 1 to 8



Level 10: Final Level. Putting everything about DOS to Test



Level Designs (Gold)

Level 1: Introduction Level with basic level design



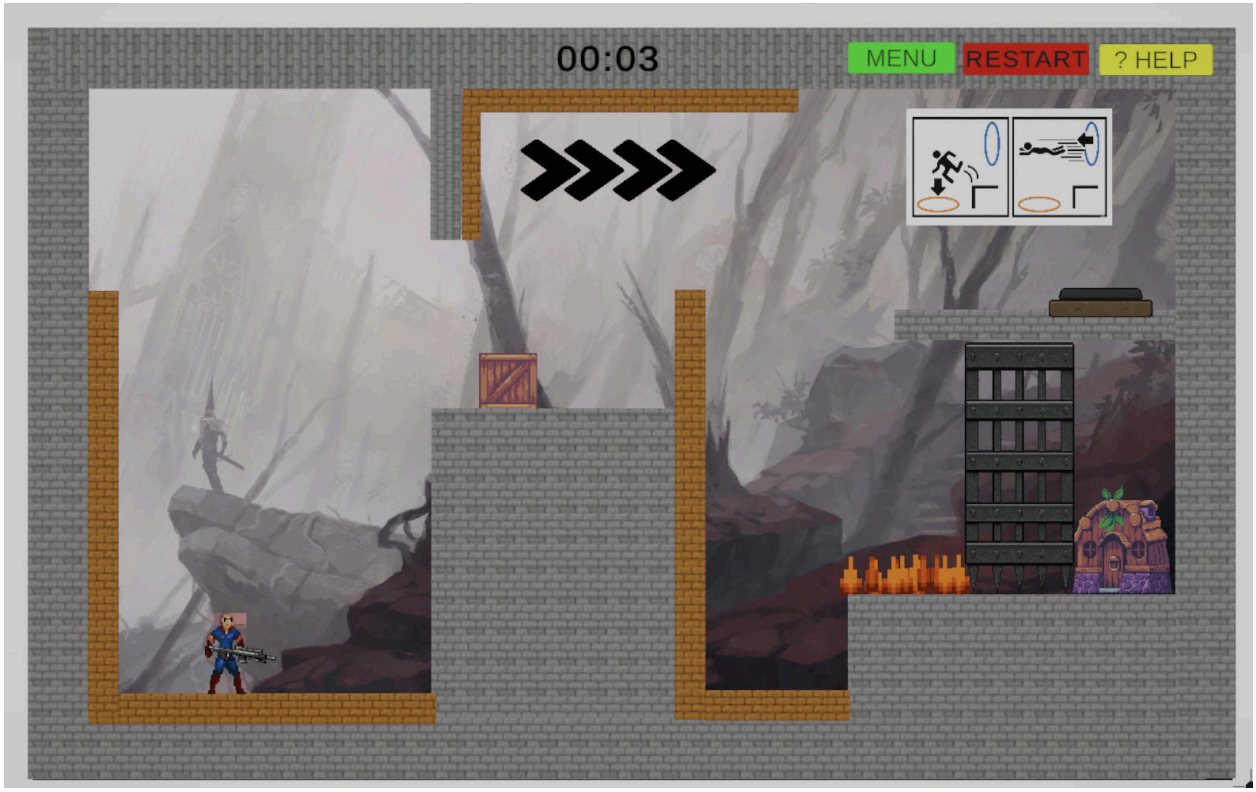
Level 2: (Multiple Enemies)



Level 3: Introduces Spikes



Level 4: Uses Momentum as its main focus



Level 5: Introduces Moving Platforms



Level 6: Has Laser Shooting Enemy



Level 7: Explores how portals can truly define the Level Design



Level 8: Challenges understanding of Momentum



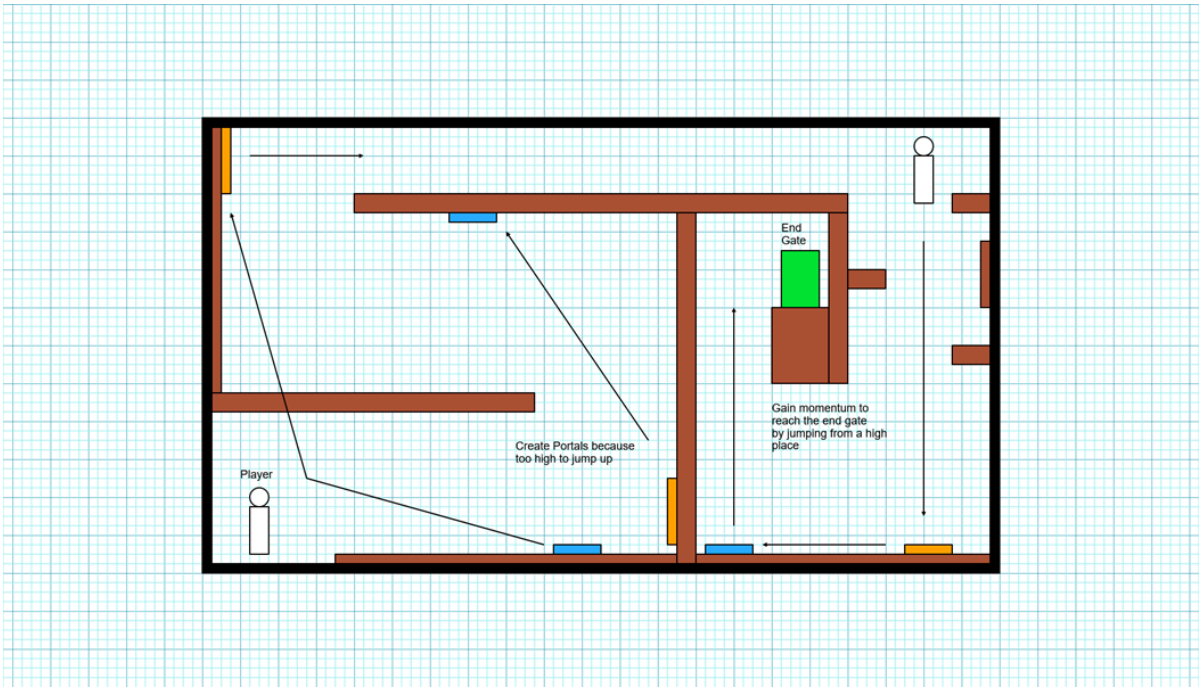
Level 9: Culmination of all mechanics & challenges introduced throughout level 1 to 8



Level 10: Final Level. Putting everything about DOS to Test



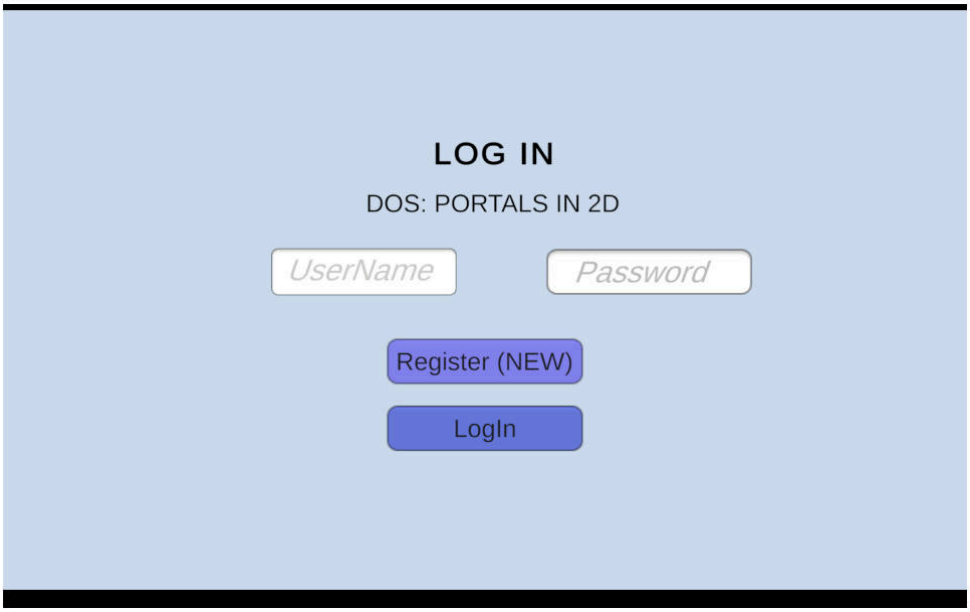
Concept Art



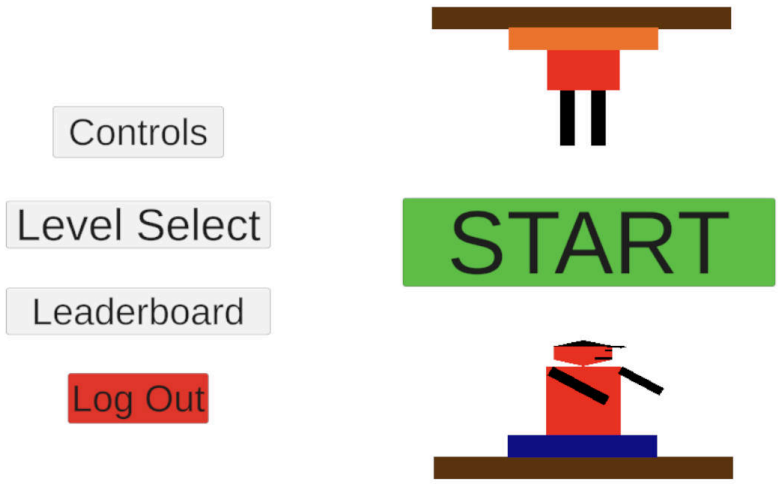
Game UI - (Beta)

Menus

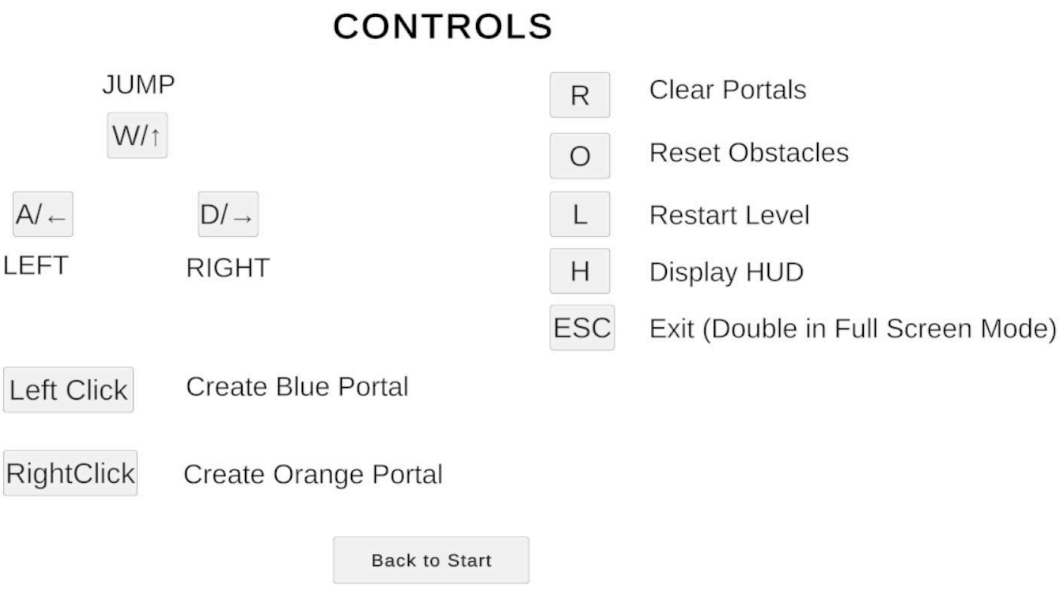
Login Screen



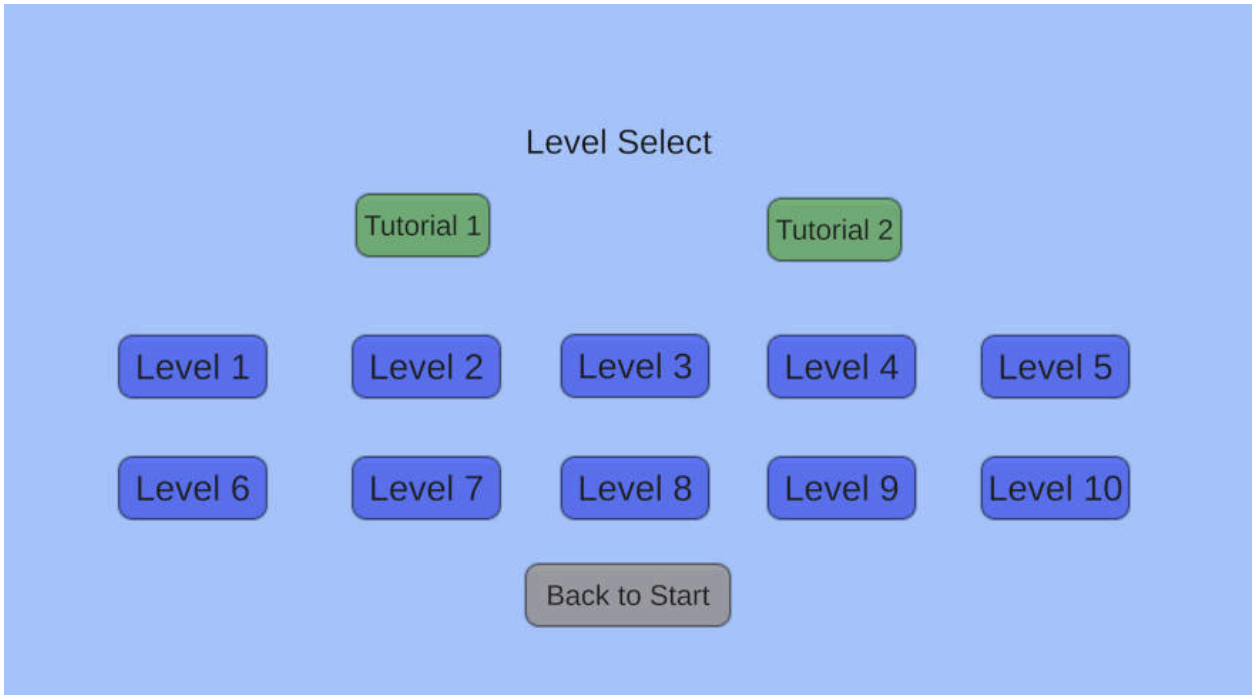
Home Page / Start Page



Controls



Level Select



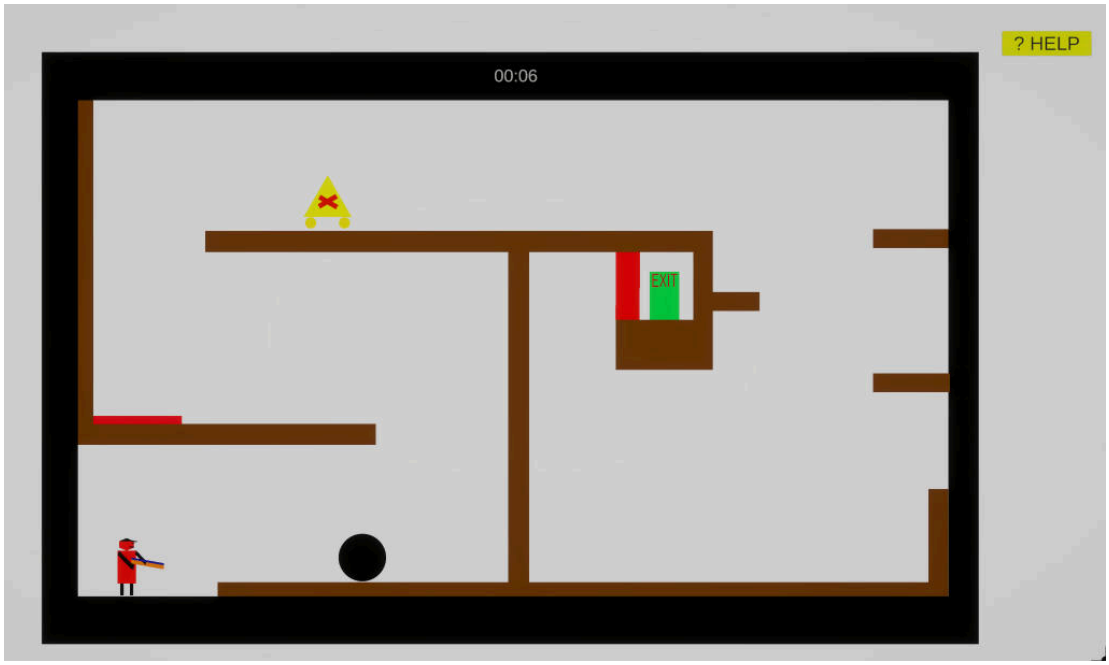
Leaderboard

You can look at your ranking and your friend's rank by typing into the Input and hitting Enter.

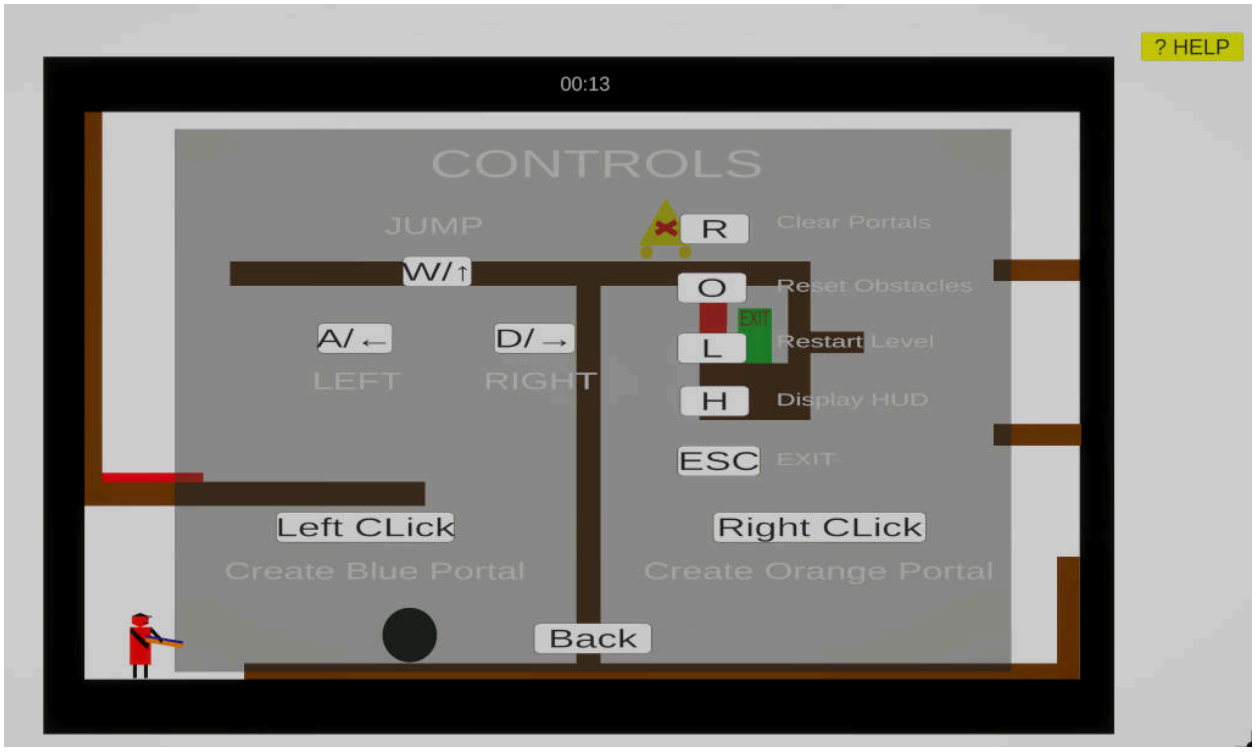


HUD

Help Button on the Top Right (Exists in Every Level)
Can Press H to bring it up as well.



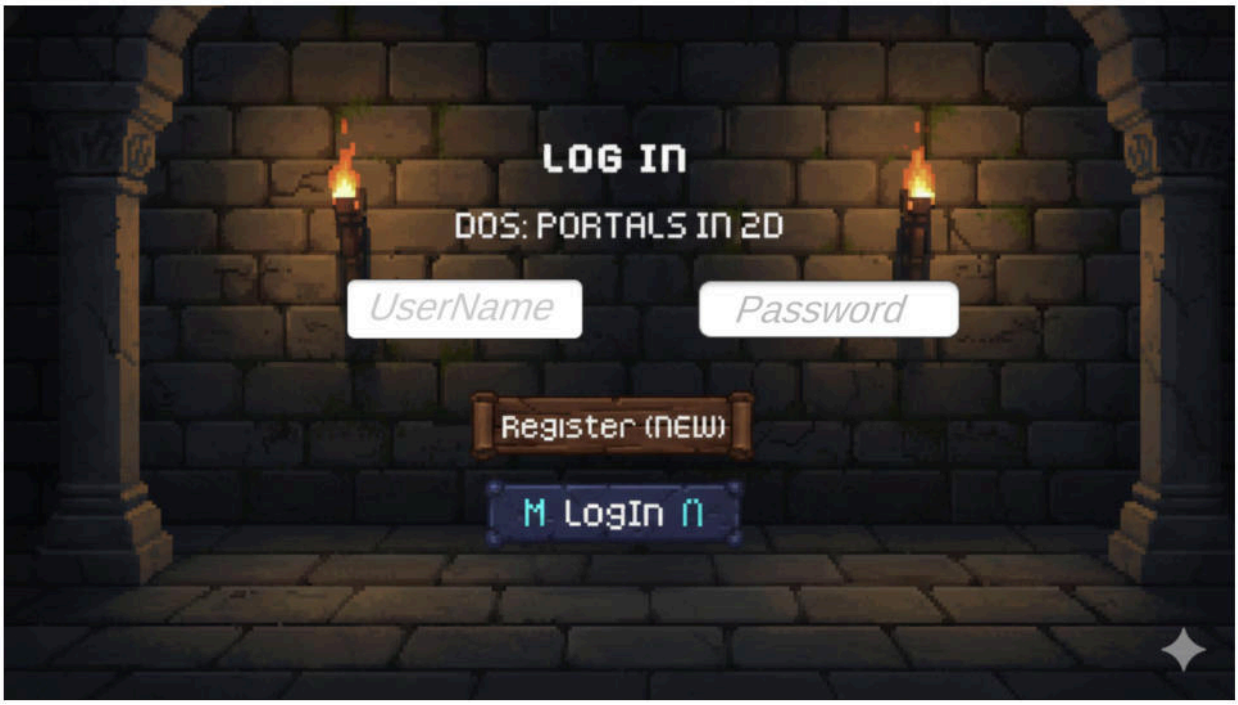
HUD (Does not obstruct the level but shows the controls in a pop up window)



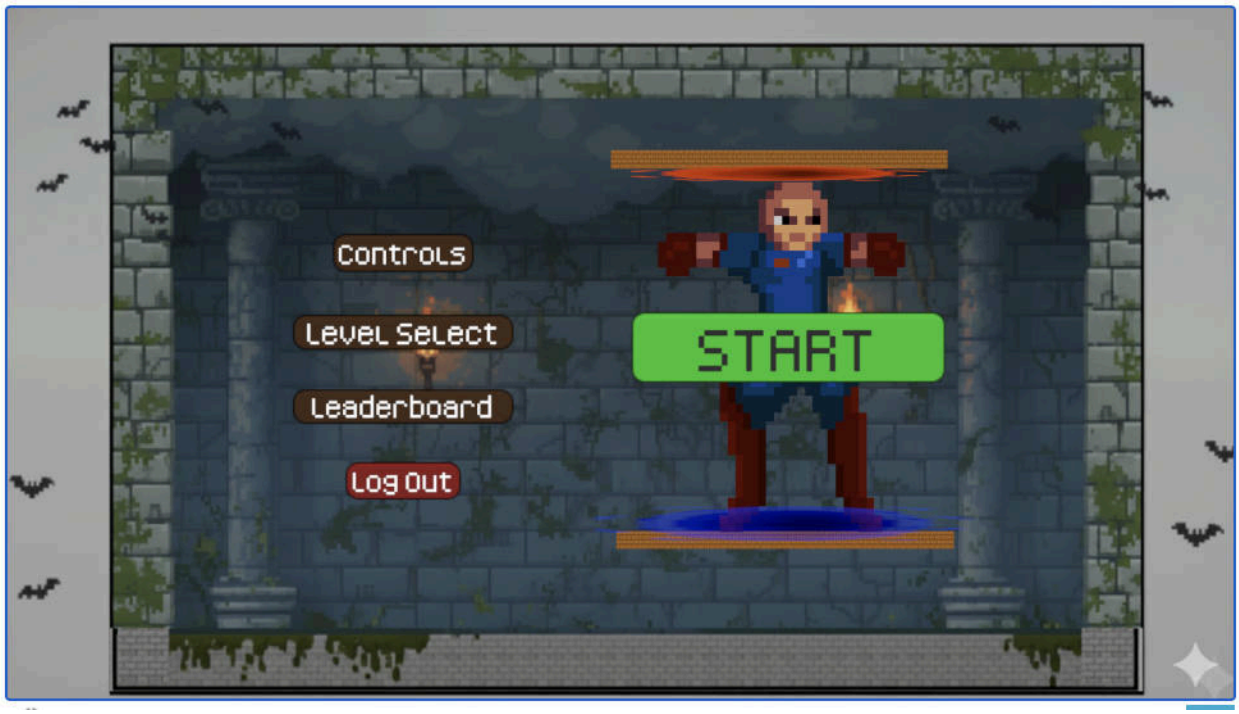
Game UI (Gold)

Menus

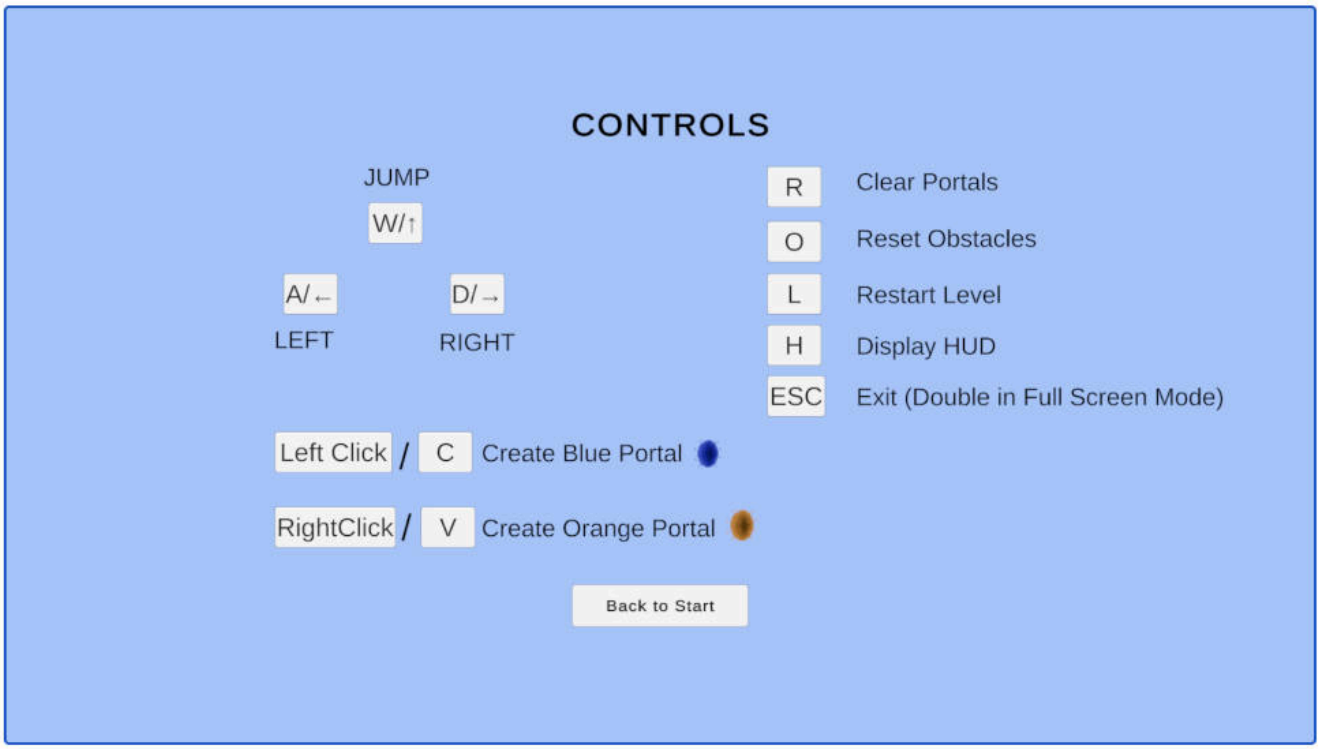
Login Screen



Home Page / Start Page



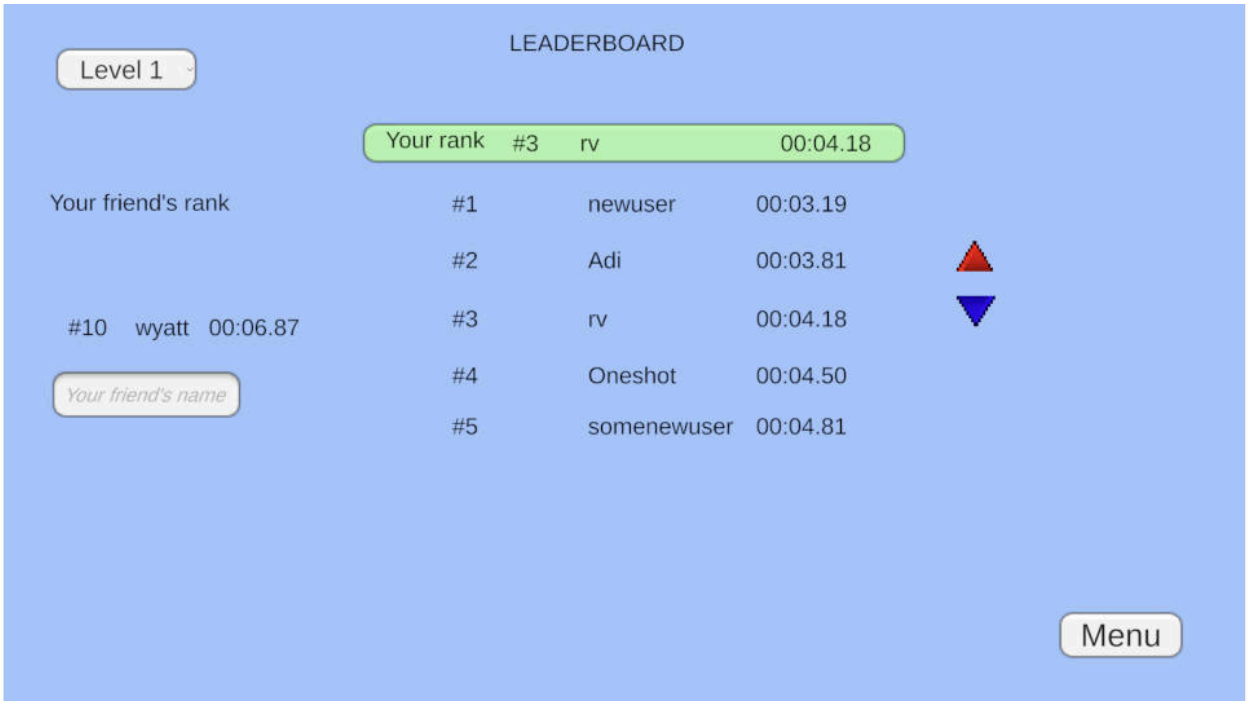
Controls



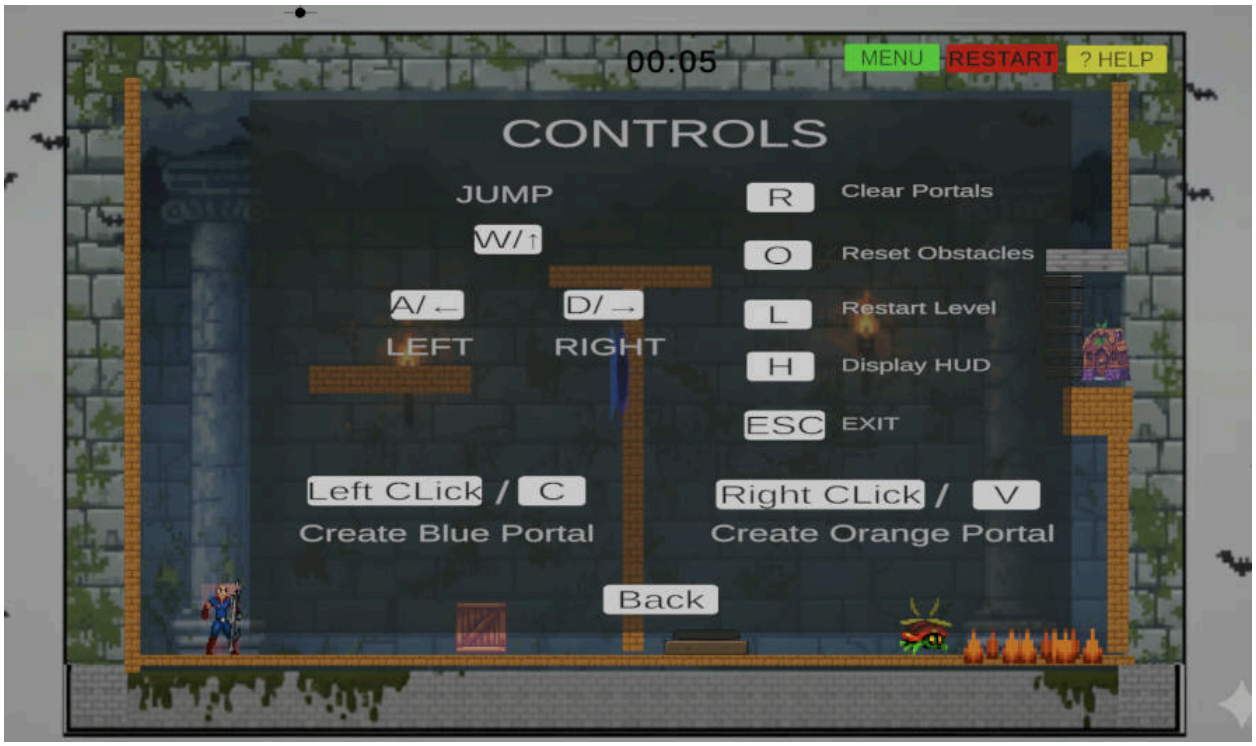
Level Select



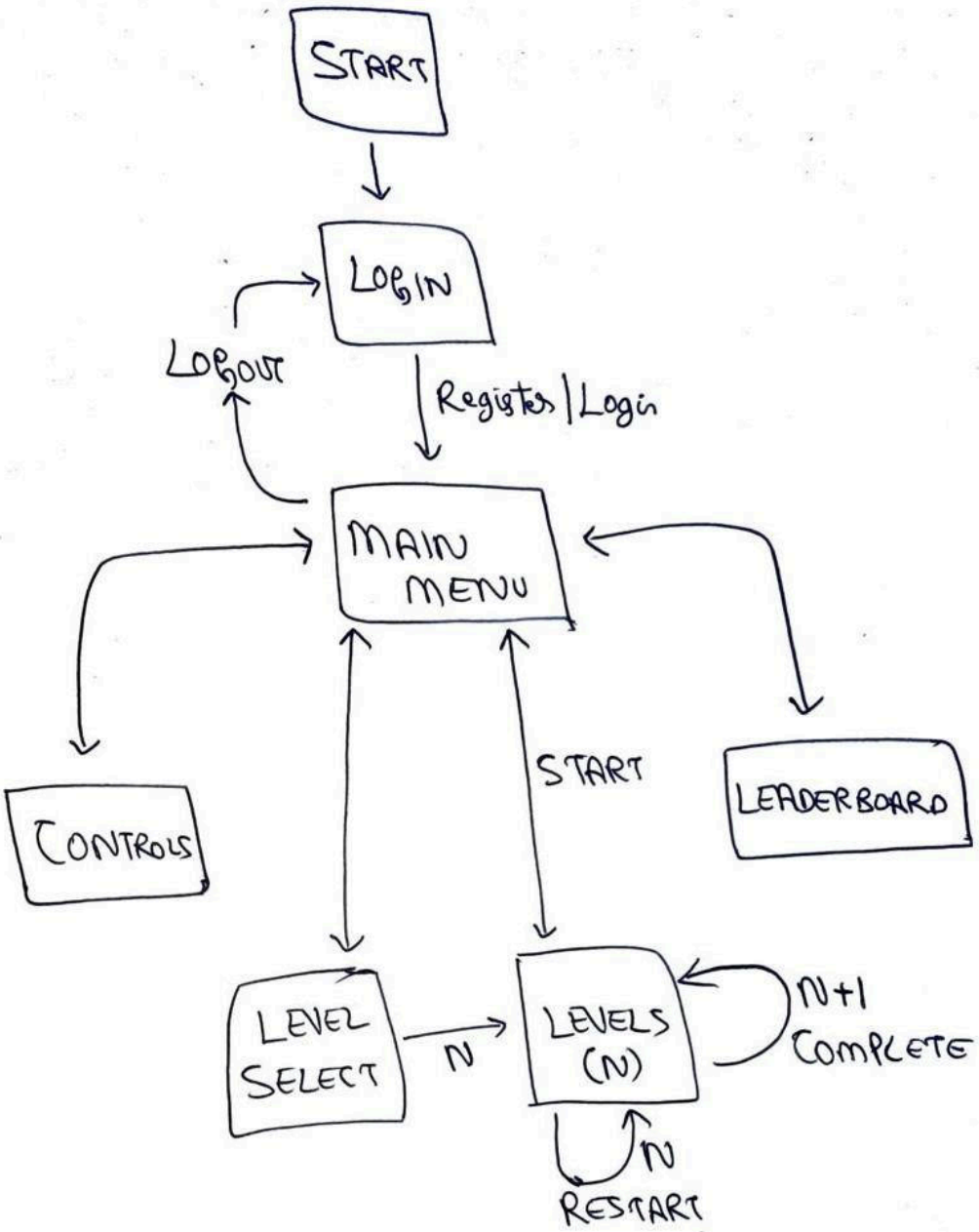
Leaderboard



HUD



UI Flowchart



Analytics

Methodology Used:

1. Data Collection:
    - Amplitude HTTP Calls
    - Amplitude API to Collect All Data
2. Data Visualization:
    - Amplitude UI
    - Unity to plot a DotPlot on the Levels

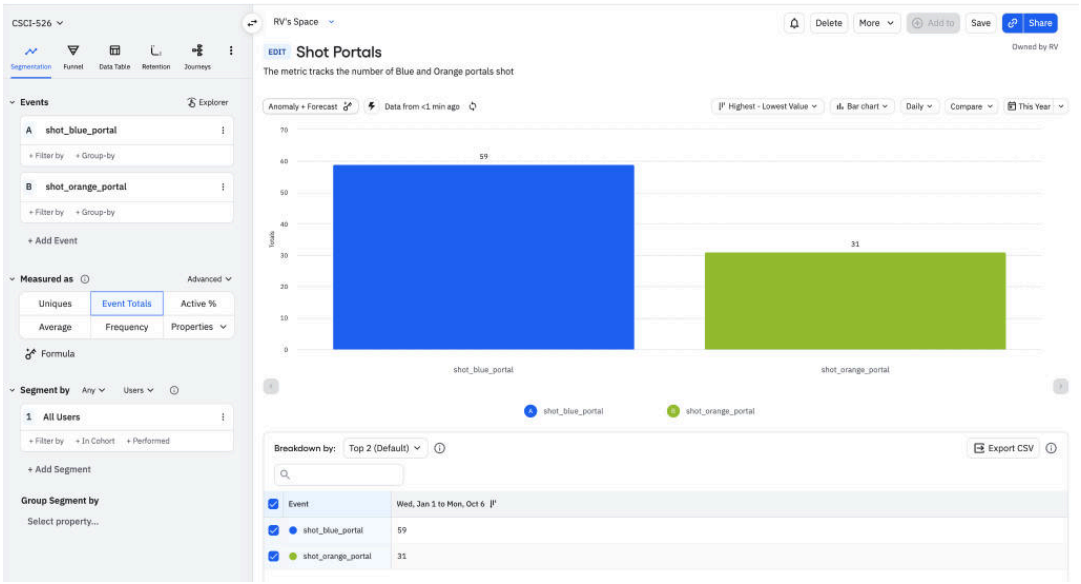
Scripts and How to get the Metrics are available in the README of the Repository.

Metrics

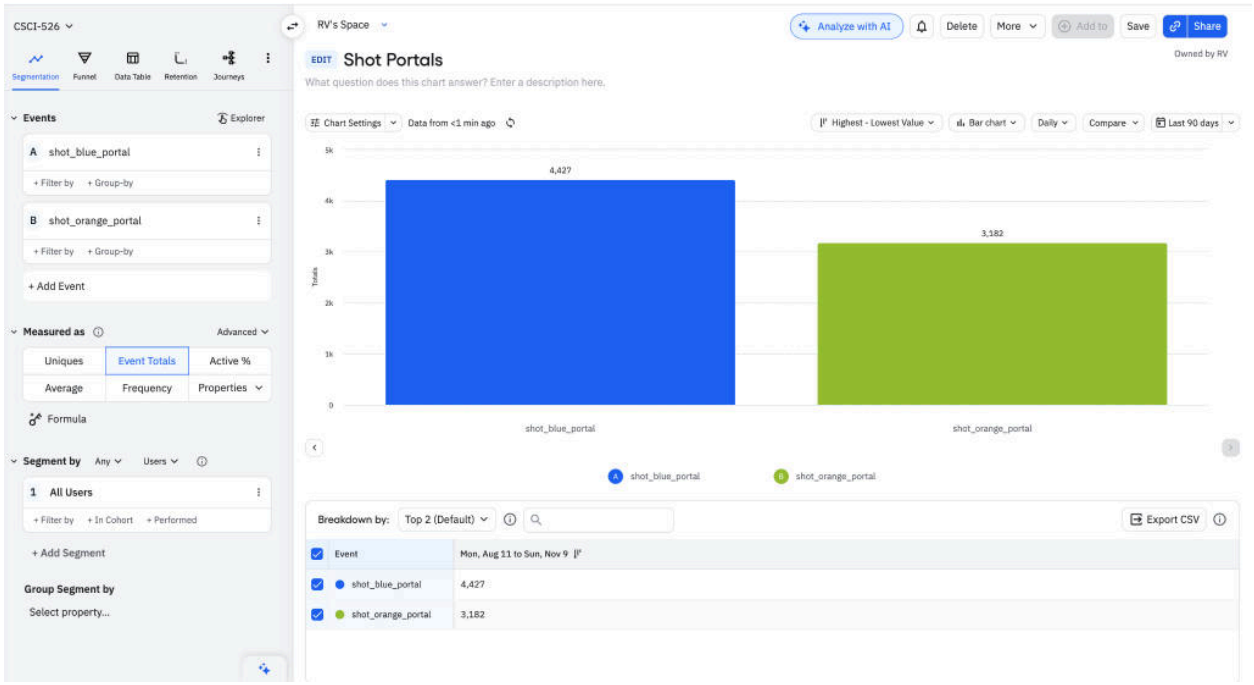
Metric #1:

Shot Portals By Level (shot_blue_portal, shot_orange_portal)

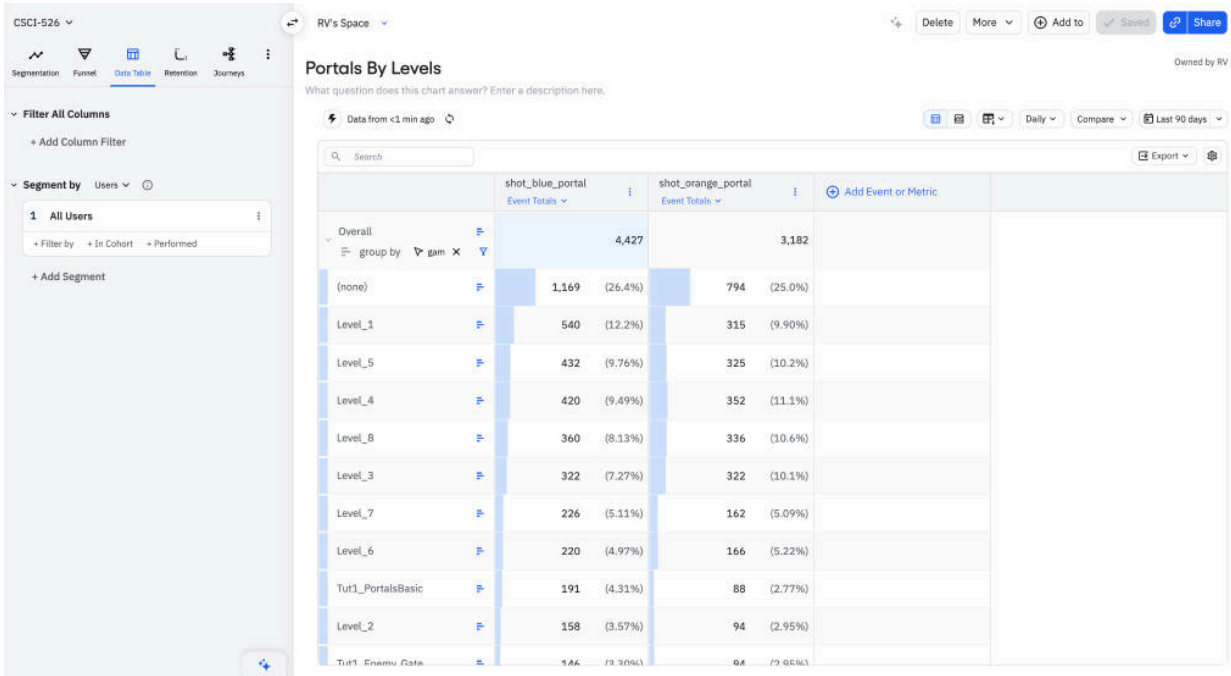
Description: *The metric tracks the number of Blue and Orange portals shot*
Link: https://app.amplitude.com/analytics/red-term-735360/chart/yfnhfcp2



Week 12



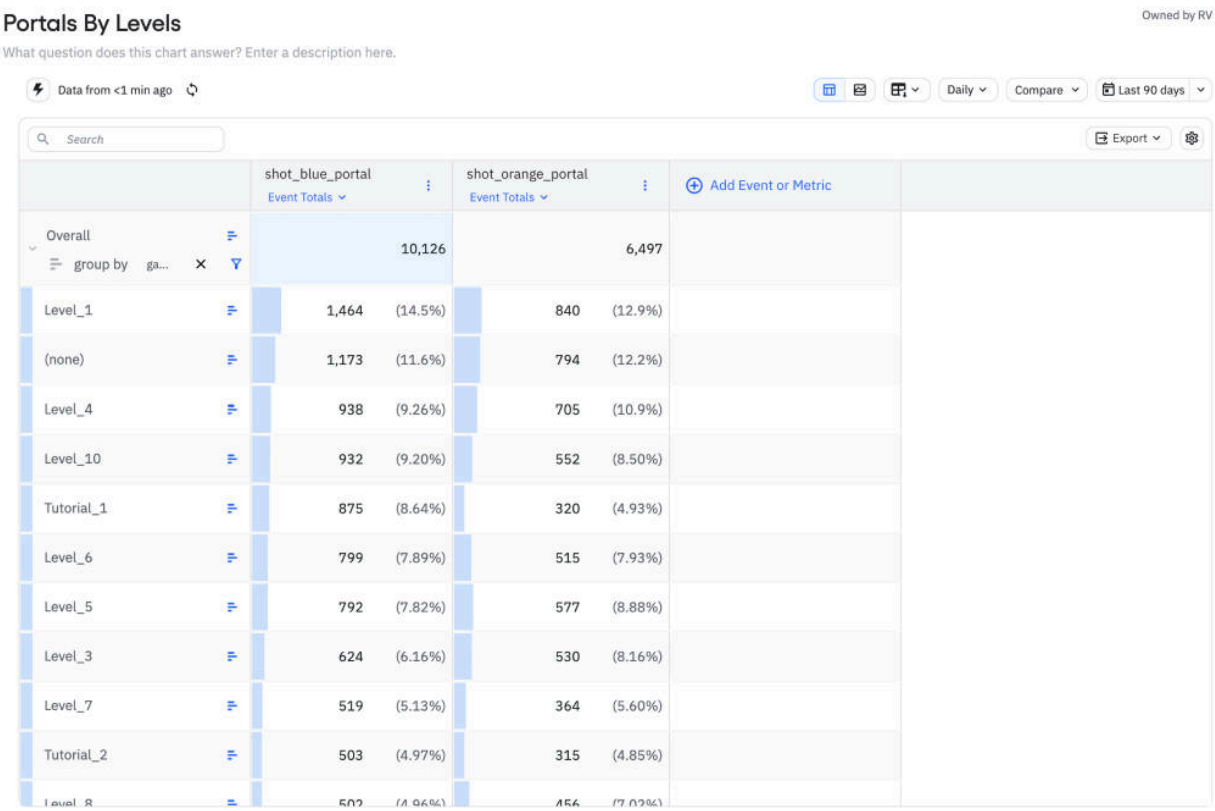
Portals By Level



Week 15



Table:



Justification:
This metric was chosen in order to track if the players are actually making use of the game mechanic or if our level design doesn't even need any use of portals. If that is the case, then we will need to change the level design. We also track what object the portal was shot on and the coordinates of the portal.
This also helps in trying to understand the preferences of people on how they shoot the portals.

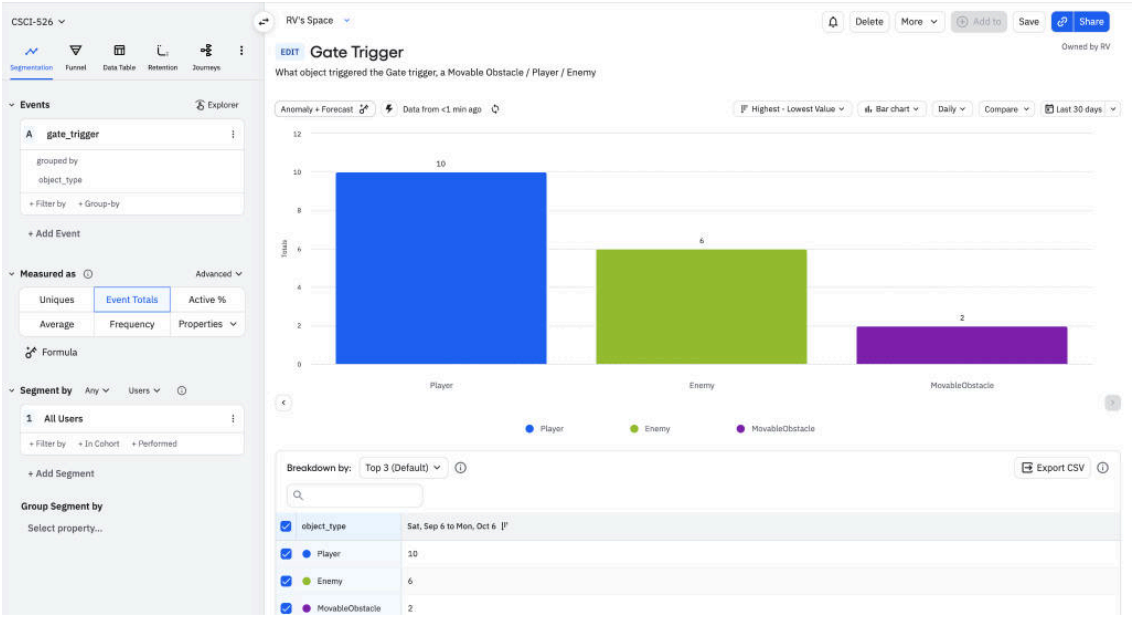
Interpretation:
Very low numbers of both metrics:
Our level design is bad and the players are able to clear the level without needing to shoot portals.

Skewed shot_blue_portal and shot_orange_portal numbers:
Our level design allows players to move around from a single pivot position, and will need to find out where and why this is happening which allows players to use one position and move around it to clear the level.

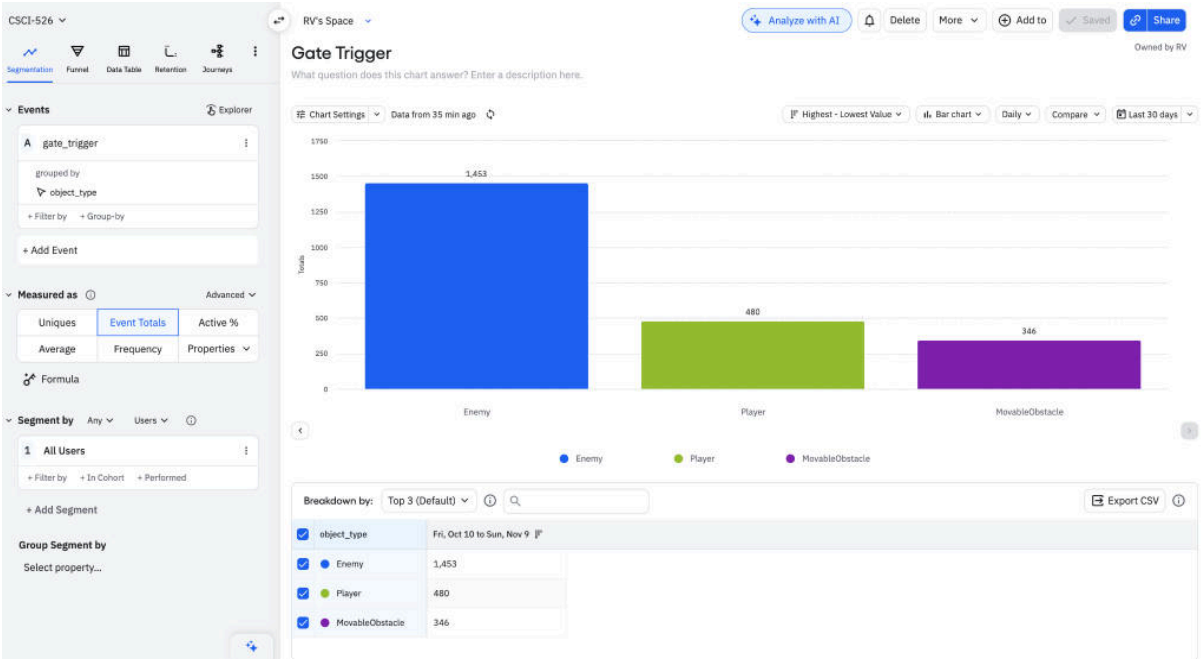
Hypothesis:
If the game starts with a tutorial and we explain how to shoot portals, the game level design should be adequate enough to see that the players are shooting enough portals for each level.

Metric #2:
Gate Trigger By Level (gate_trigger)

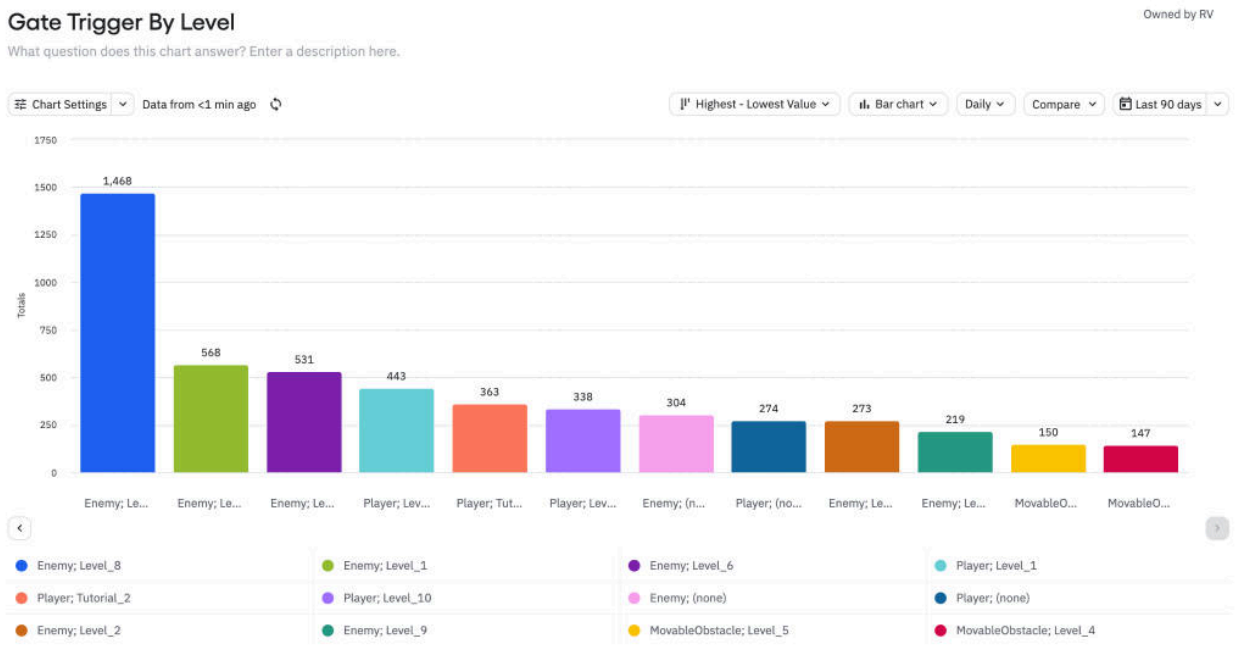
Description: What object triggered the Gate trigger, a Movable Obstacle / Player / Enemy
Link: https://app.amplitude.com/analytics/red-term-735360/chart/j8xtmbae



Week 12



Week 15



Justification:
This metric helps us understand how the players are playing the game, are they going through the linear flow that is expected of them to do, or, are they solving the levels in creative ways to gain time in the level.

Interpretation:
Very low number of enemies being used to trigger the Gate:
Shows that the players are not using enemies to trigger the gate. This means either:

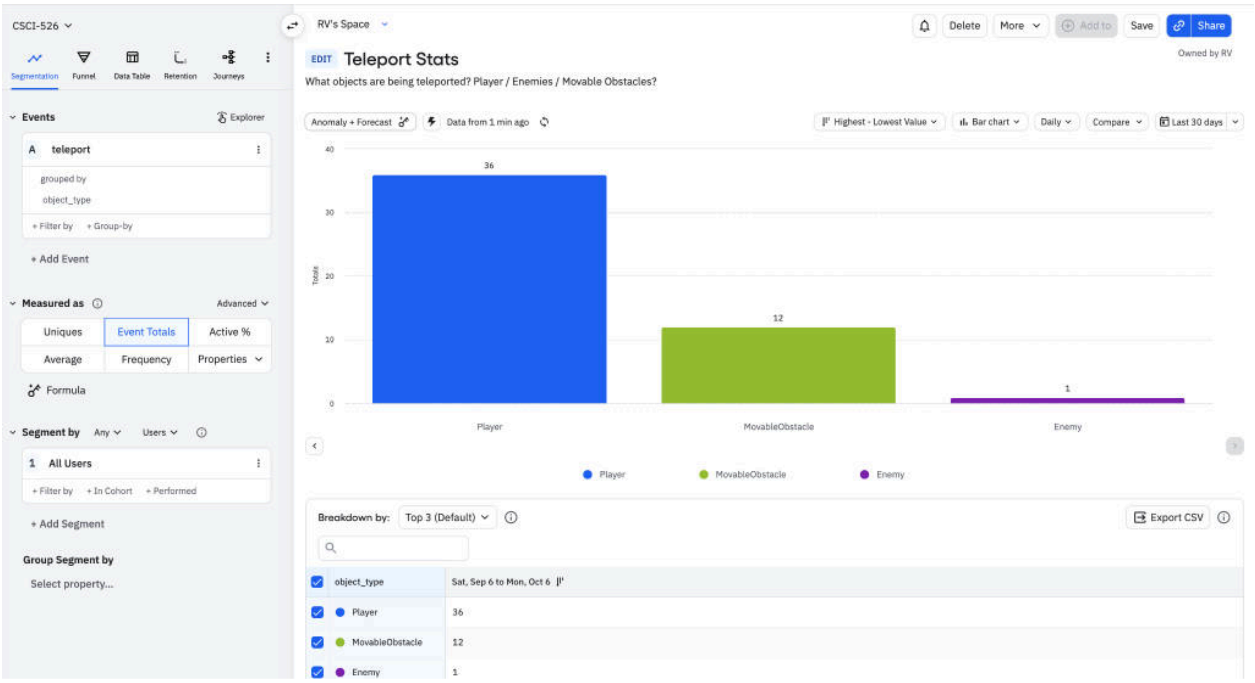
1. The players do not understand that the enemies could be used to trigger the gate.
2. The players are not using the enemies as it is not saving them enough time.

Case 1 would be a communication problem we will need to solve in the Tutorial but Case 2 would be a level design issue.

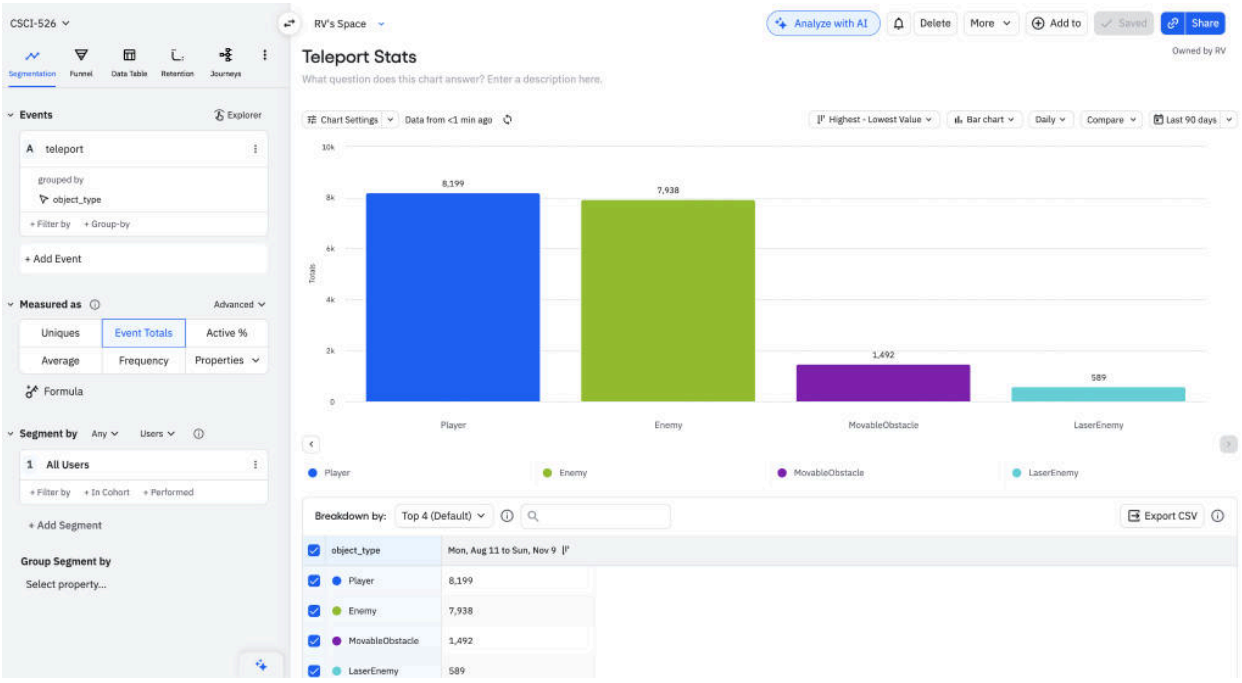
Hypothesis:
If we place the gate triggers near the enemies, we should see an increase in the players using not just the movable obstacles but the enemies as well to trigger the gates.

Metric #3:
Teleport Stats By Level (teleport)

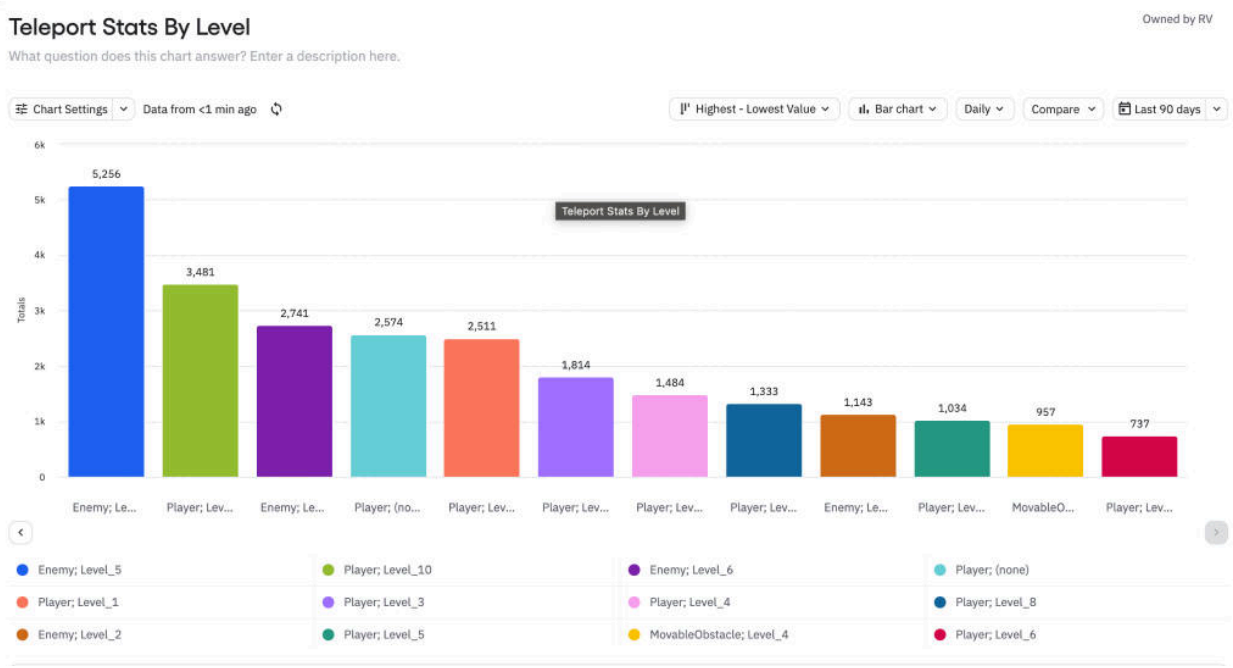
Description: What objects are being teleported? Player / Enemies / Movable Obstacles?
Link: https://app.amplitude.com/analytics/red-term-735360/chart/68bfsf8x



Week 12



Week 15



Justification:
This metric helps us analyze how the game is being played and what is being teleported by the players. Knowing if the player only teleports themselves or uses the movable obstacles and enemies to clear the level will help us design better levels and know where we might be going wrong with our expectations on how one must clear the level.

Interpretation:
Only the player is being teleported and just a marginal amount of obstacles and enemies are teleported:
Either the player does not understand that there are more things that can teleport except themselves or our level design doesn't need them to teleport anything else.

There is a good amount of moveable obstacles and enemies being teleported:
This would mean that the communication to the player about what can be teleported was done well in the tutorial and the level design is interesting and complex enough to justify using movable obstacles and enemies. This will also let us know that the game is engaging enough for players to try and find clever ways to clear the level.

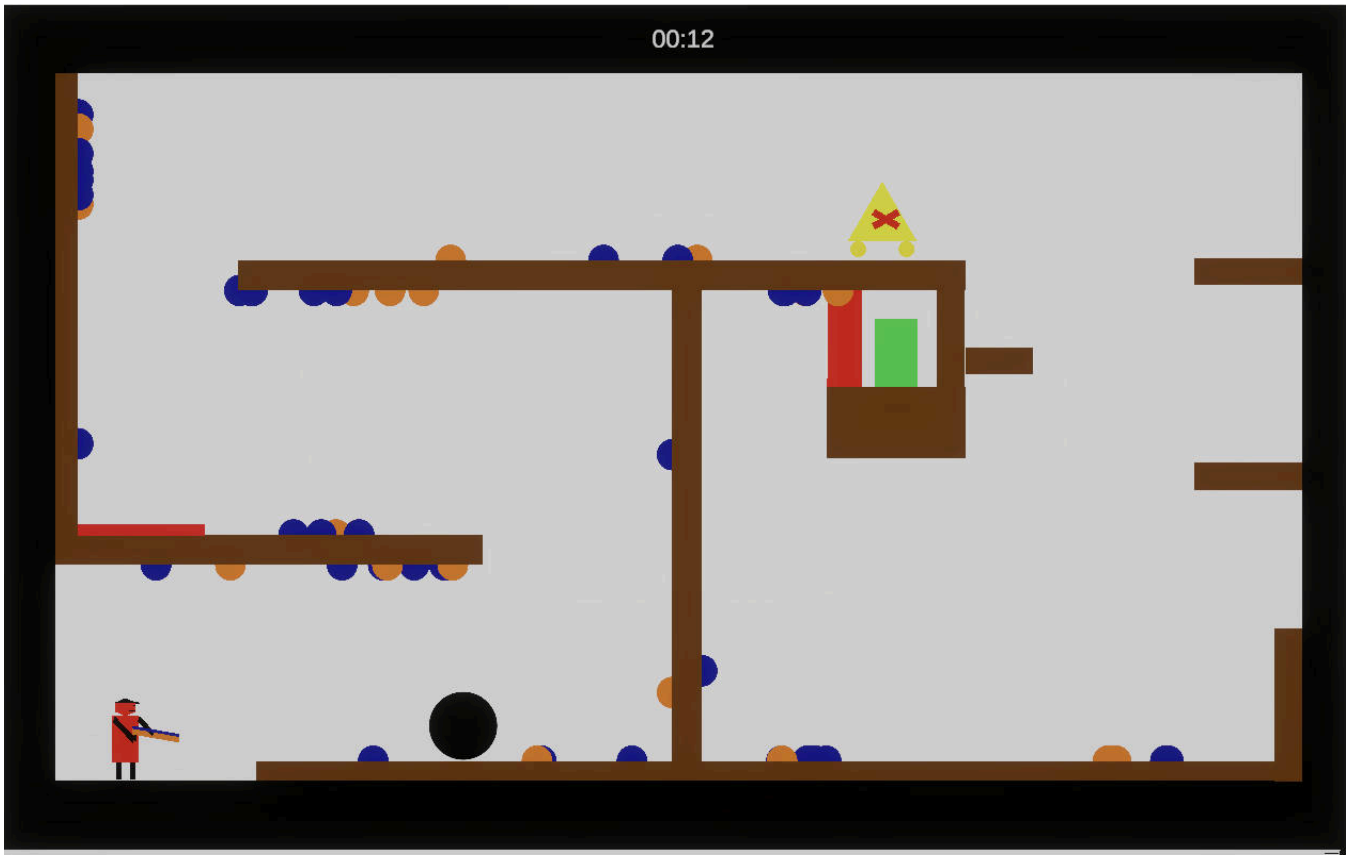
Hypothesis:
A better tutorial level which explains that Enemies and Movable Obstacles can be teleported will result in an increase in more enemies and movable obstacles being teleported.

Metric #4:
Portals Dot Plot

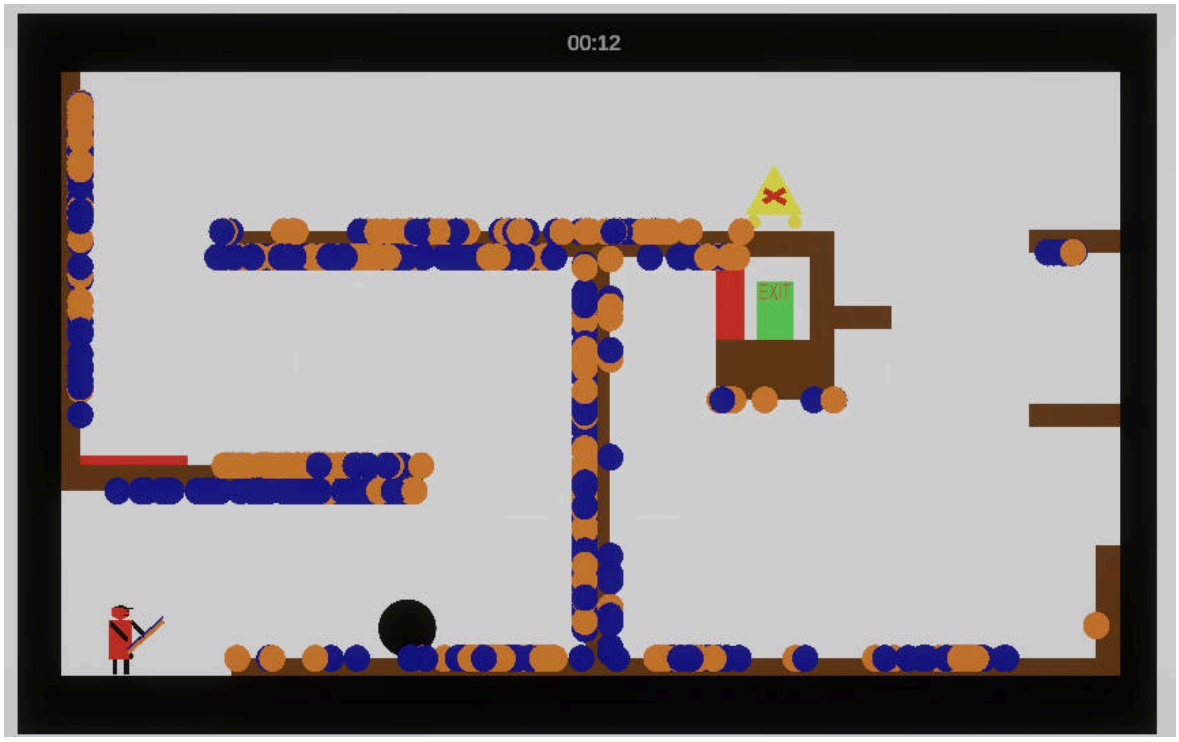
Description: The dot plot of the orange and blue portals that are being shot at each level.
How to access ? The scripts are available in the Repository and the README.md has the exact description of how to get this:

```
1. Run AmplitudeData.py to get amplitude_data/all_events.csv
2. all_events.csv file has ALL the events from Amplitude in the time period that we wanted
3. Then use FilterEventData.py to get the CSV for the shot_blue_portal and shot_orange_portal events with X,Y,gameLevel values
4. The FilterEventData.py will create a portal_events.csv file
5. Copy portal_events.csv as portals.csv in the Resources folder in Unity
6. There exists an Empty GameObject called DotPlotter in All Levels, Activate it and run it with the proper prefabs
7. The script should plot all the blue and orange portals that were shot
```

Level 1



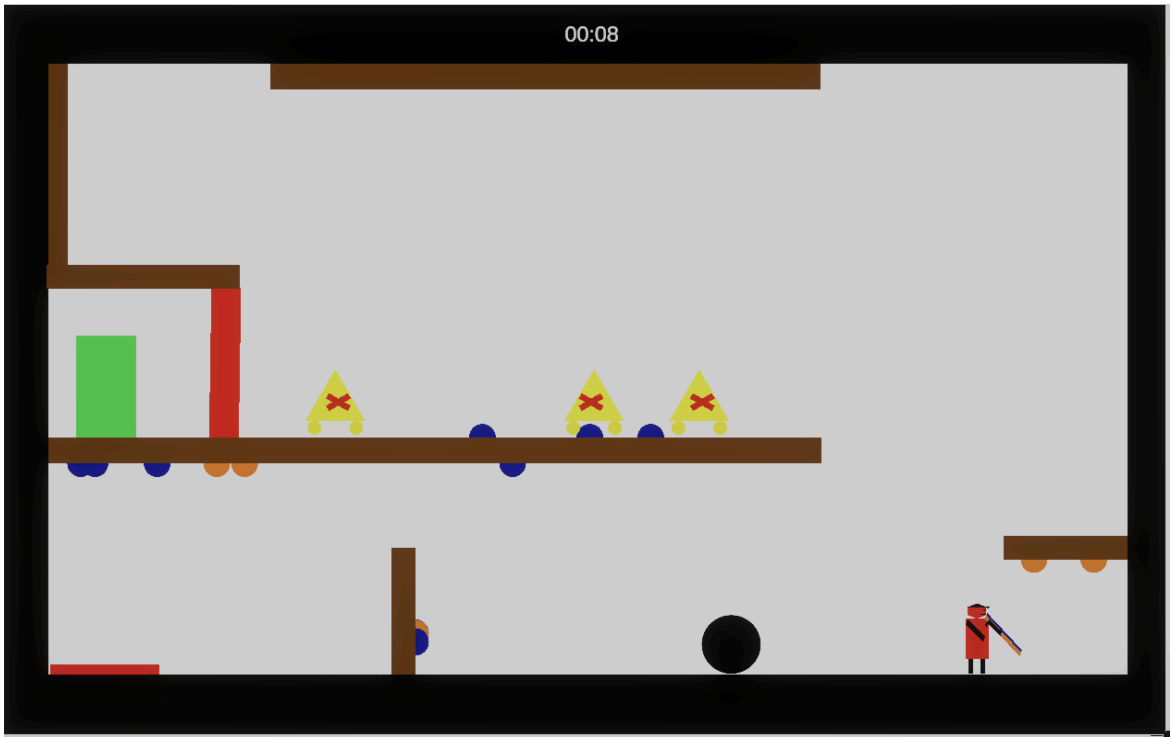
Week 12



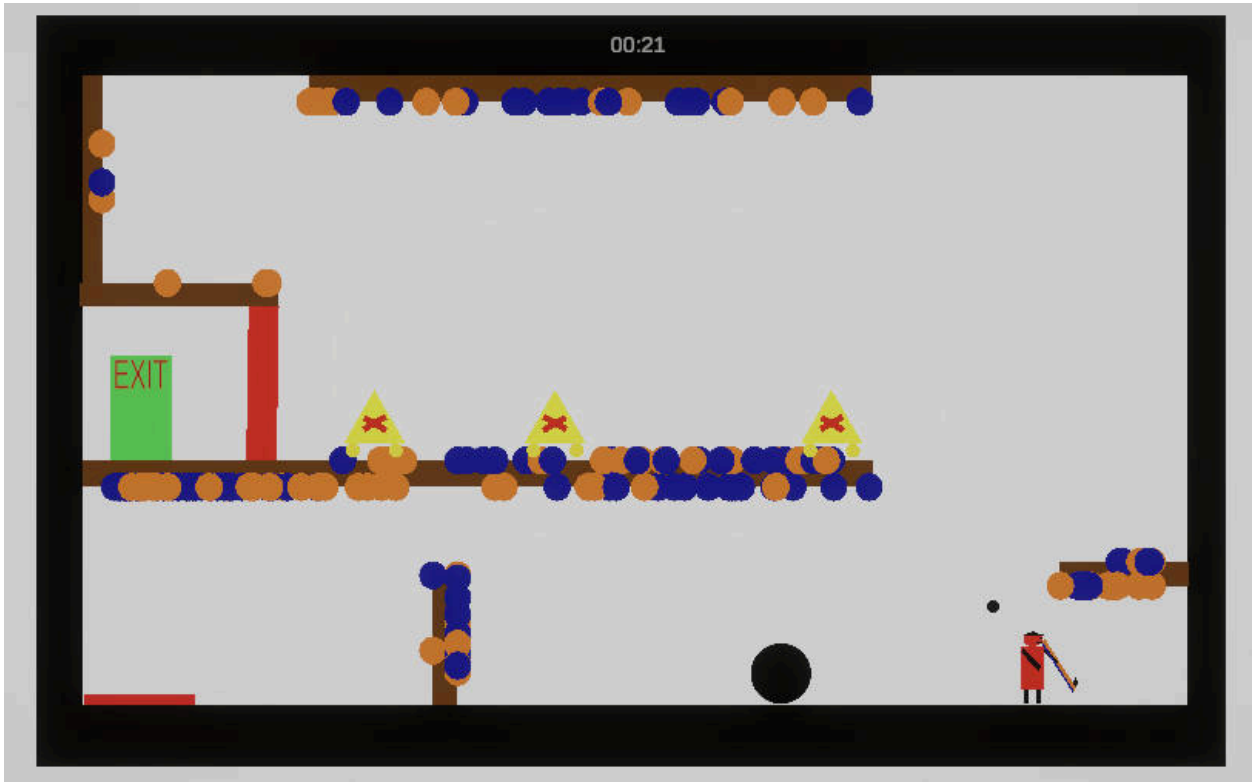
Week 15



Level 2



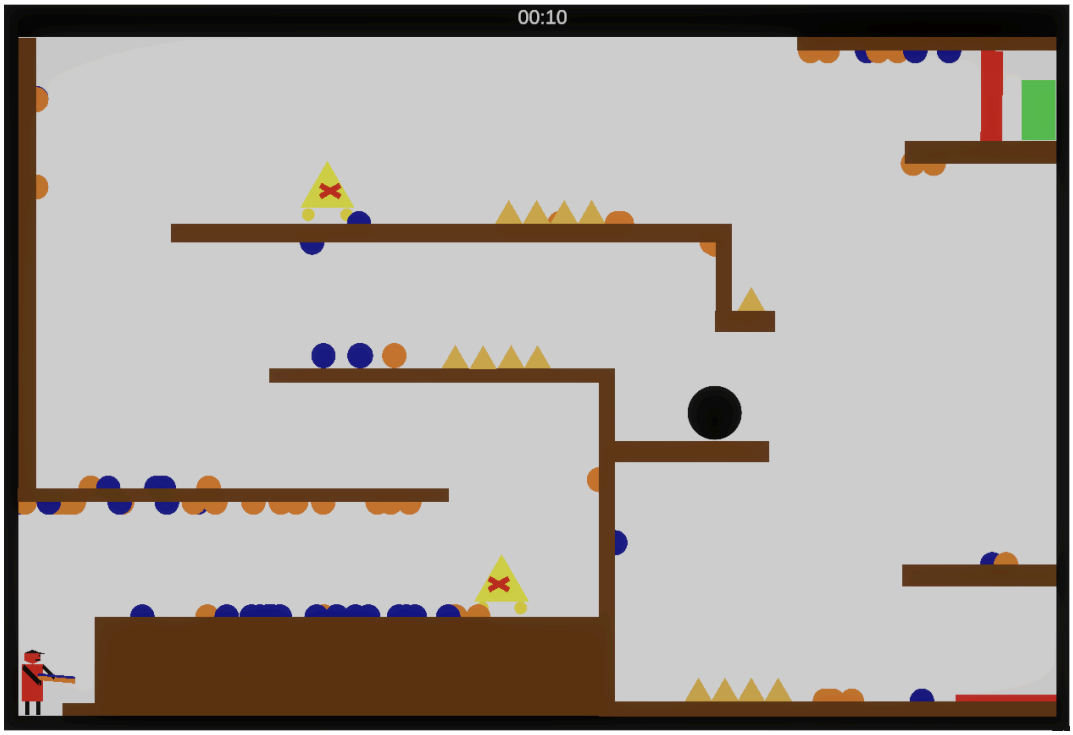
Week 12



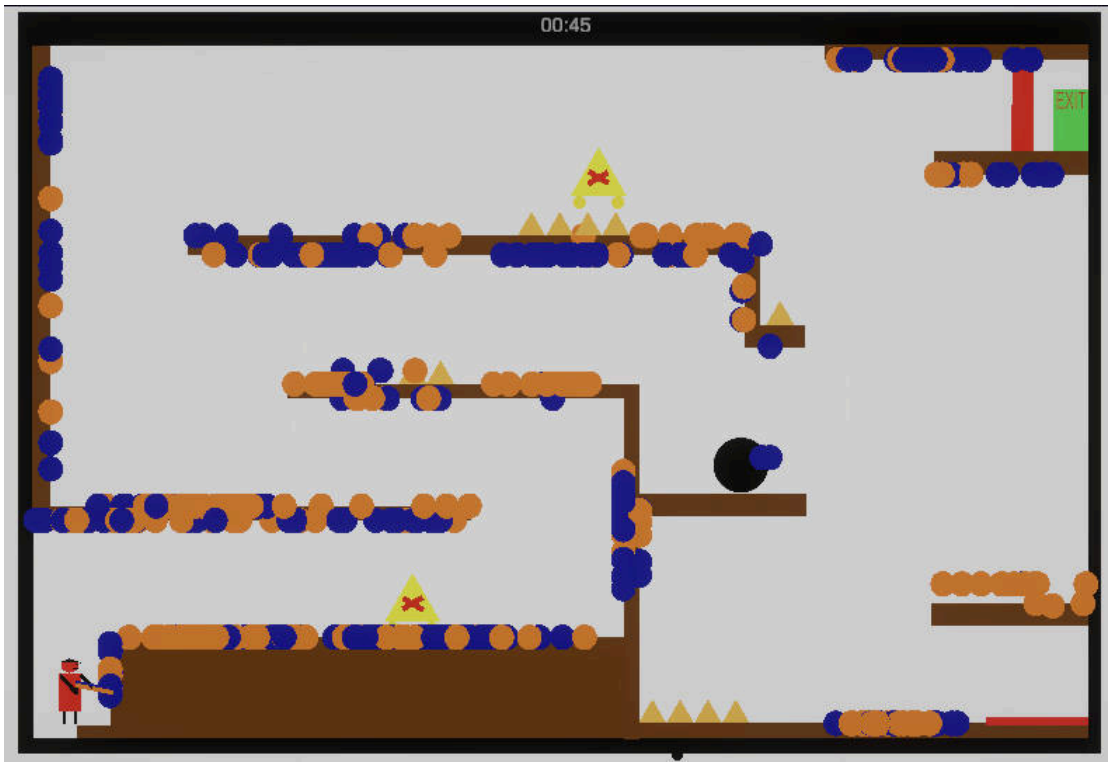
Week 15



Level 3



Week 12

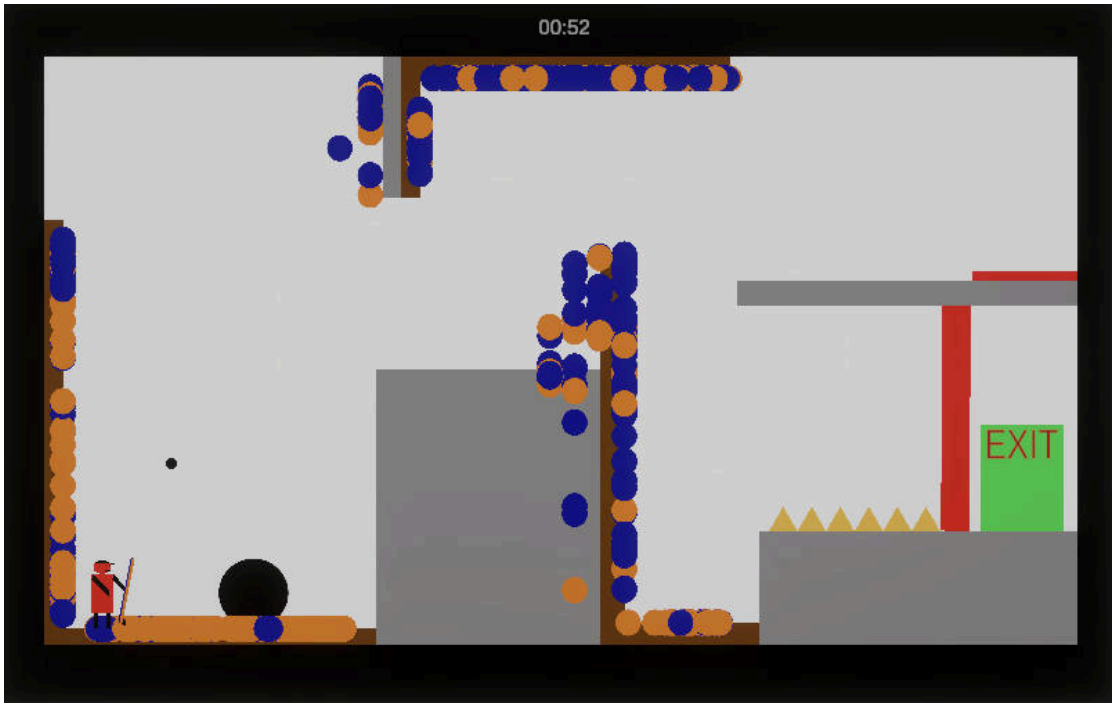


Week 15

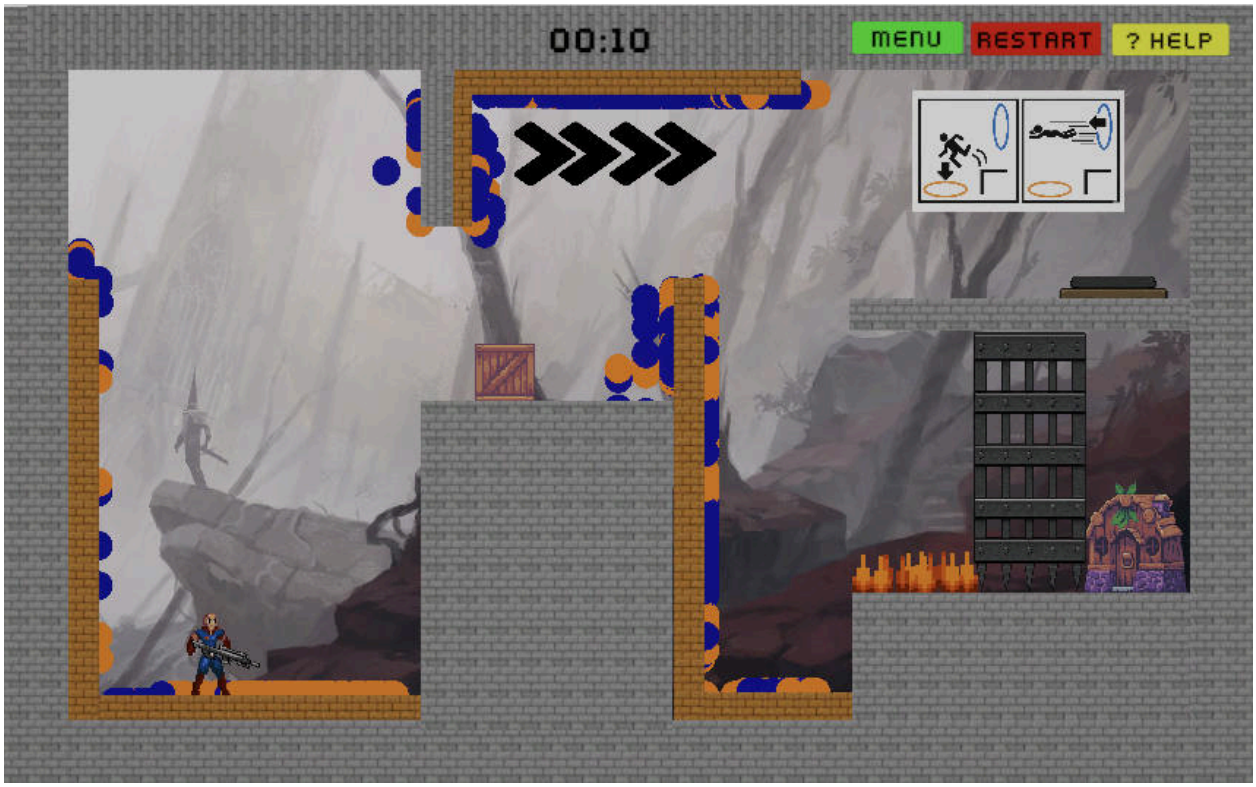


Level 4

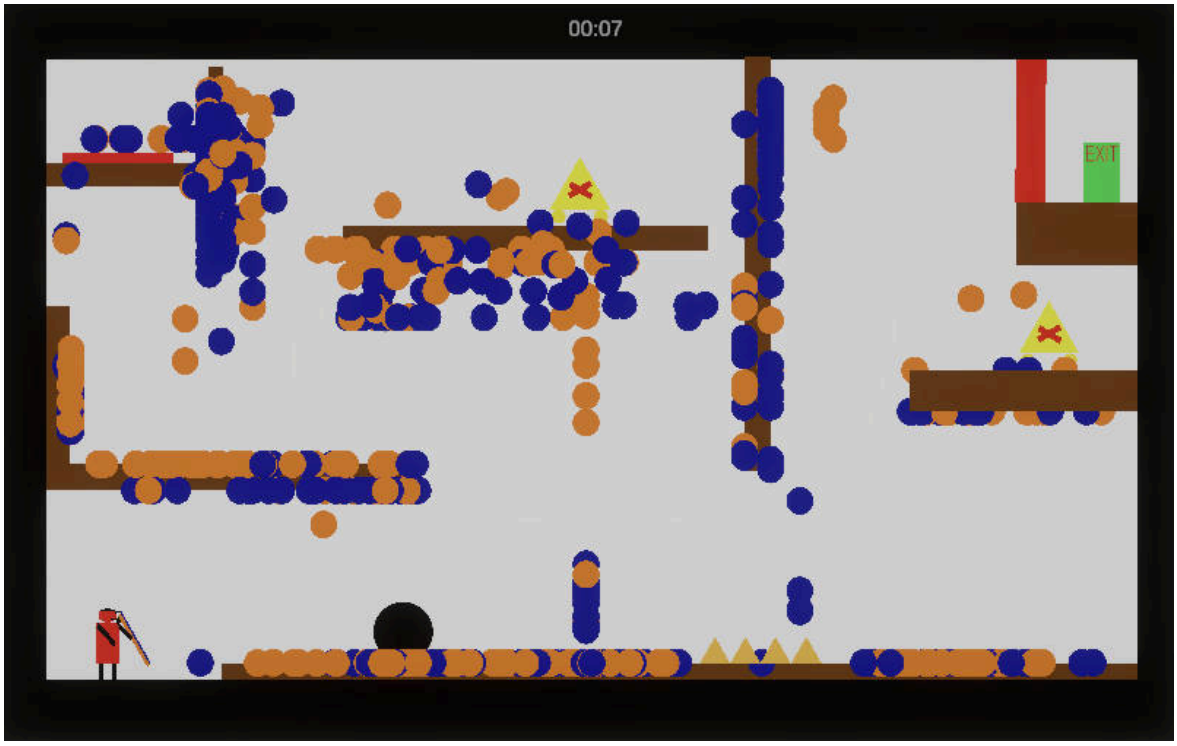
Week 12



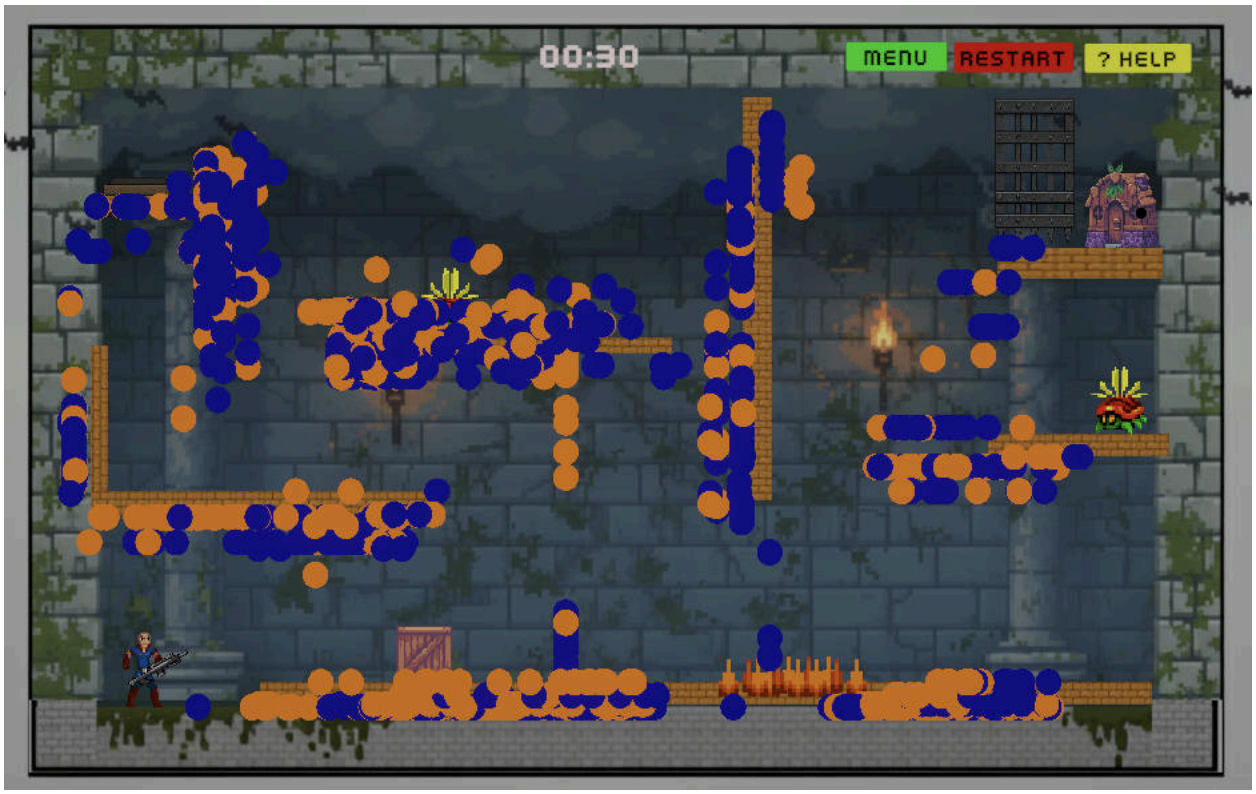
Week 15



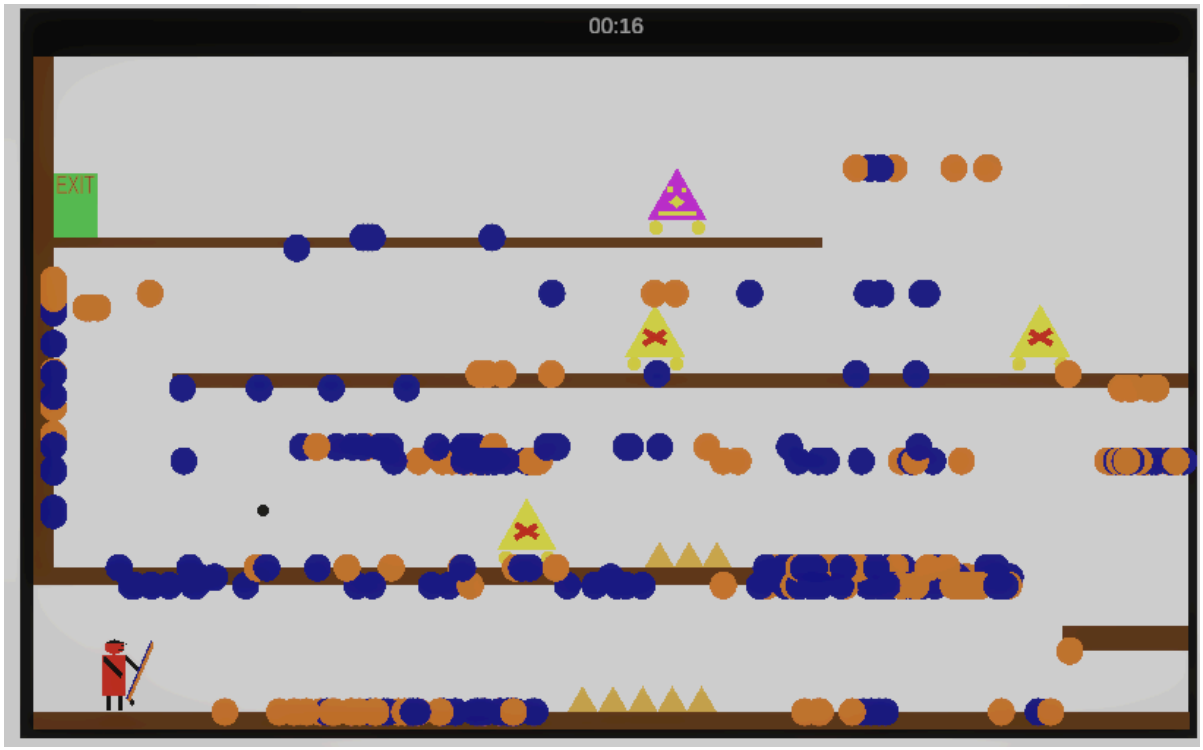
Level 5



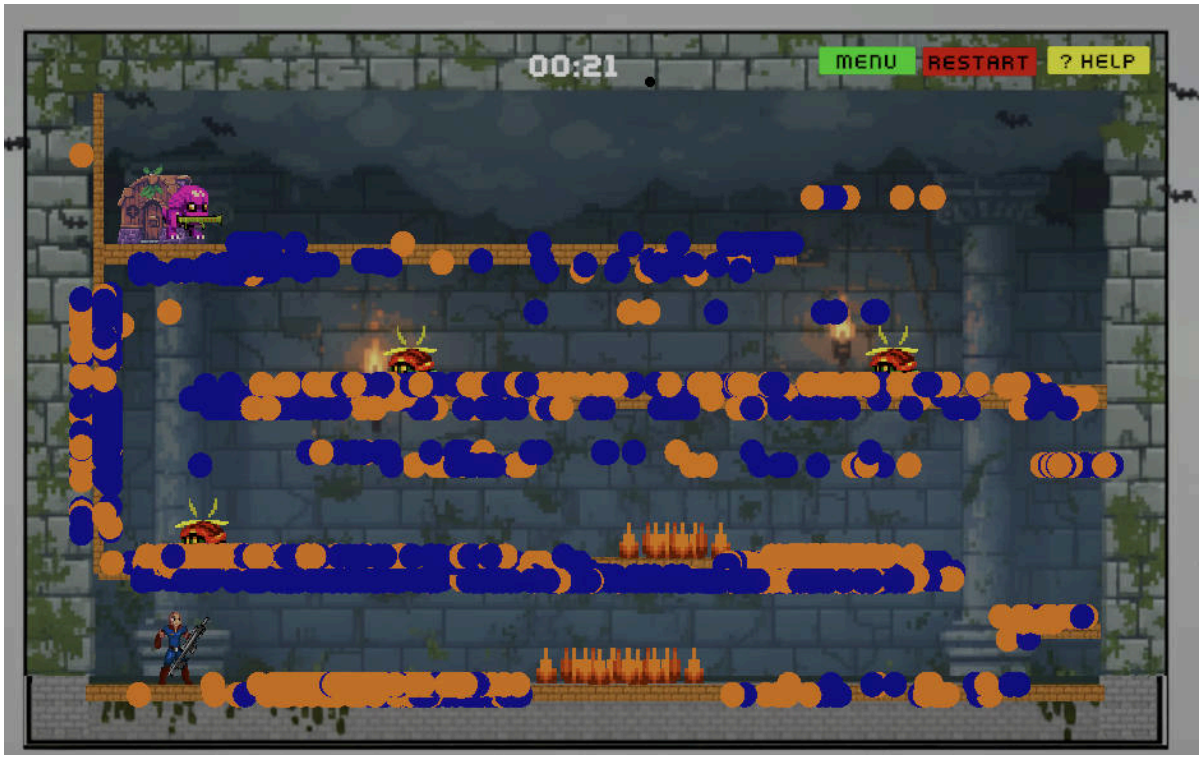
Week 15



Level 6



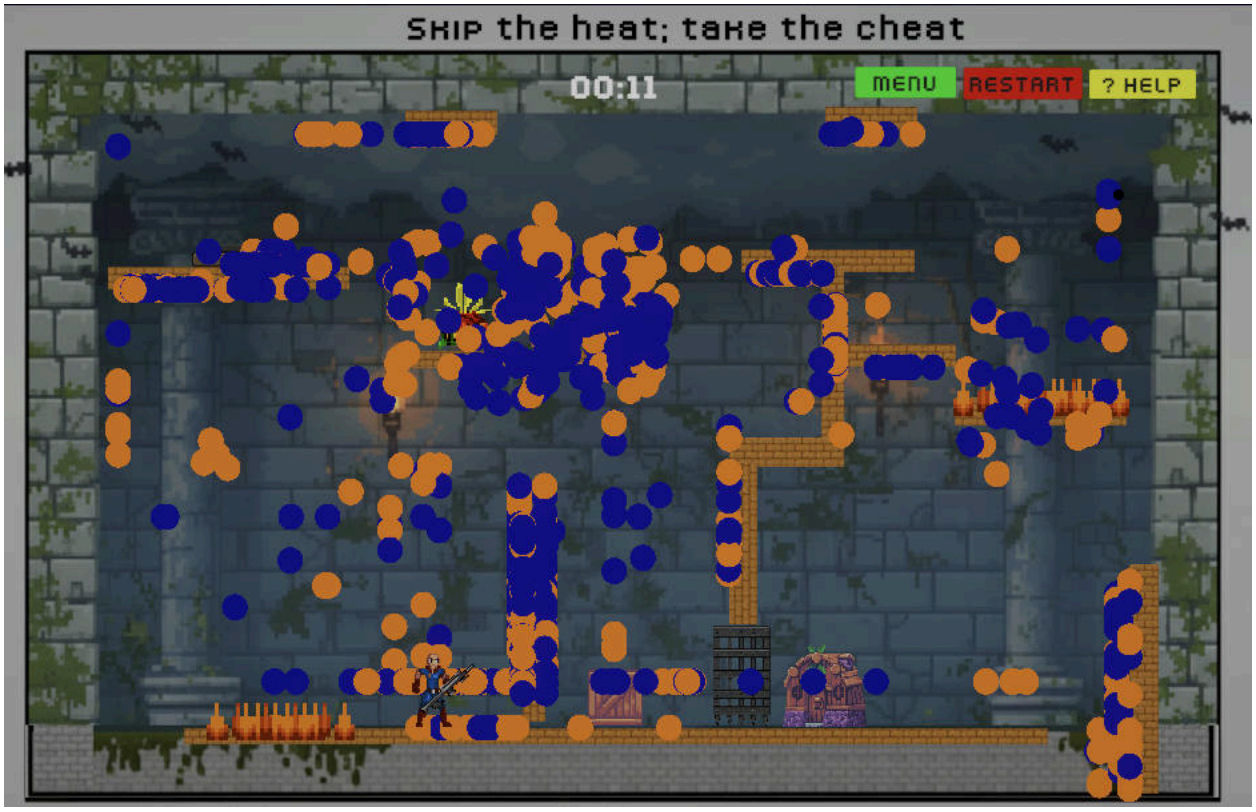
Week 15



Level 7



Week 15

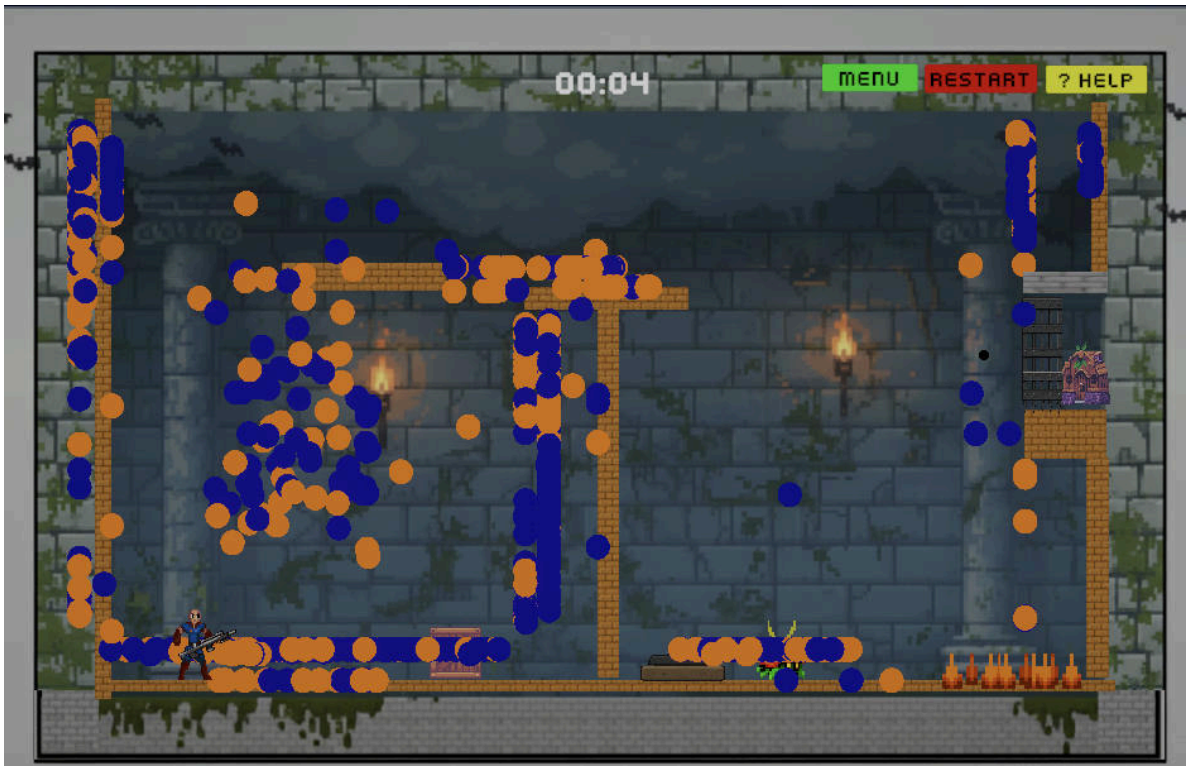


Level 8

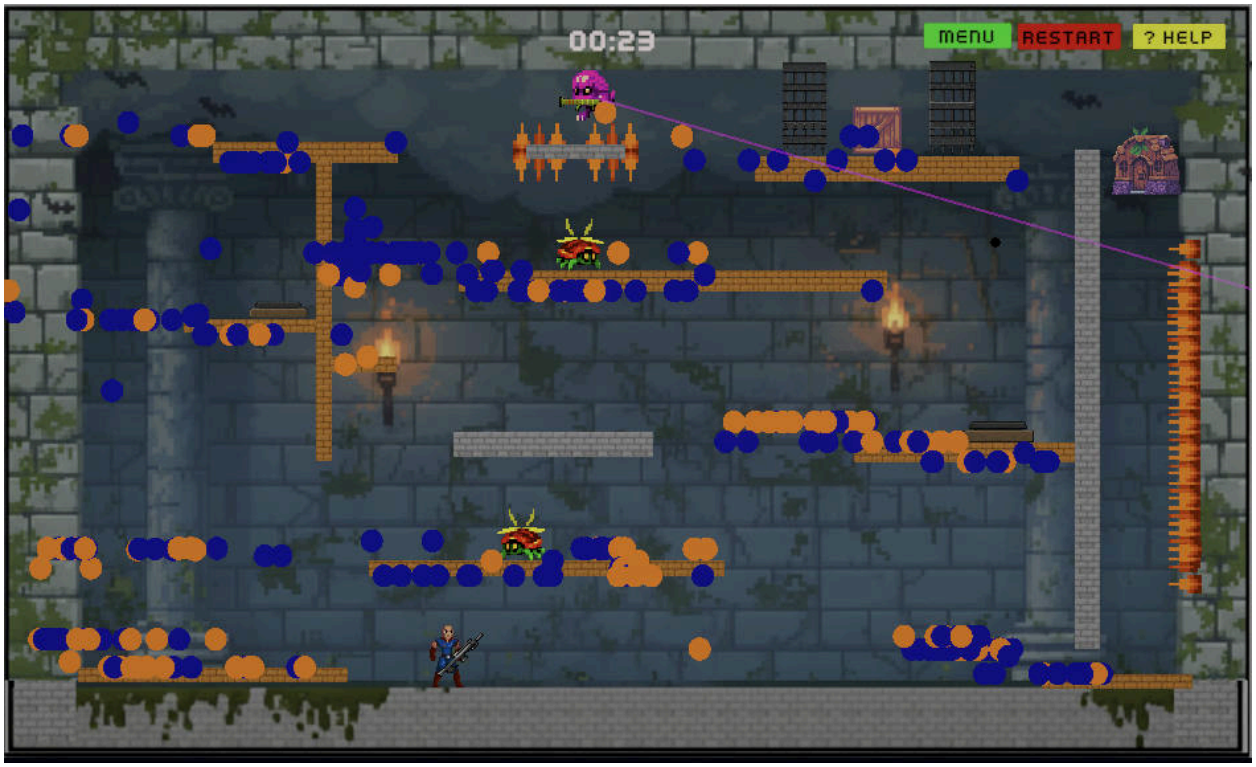
Week 12



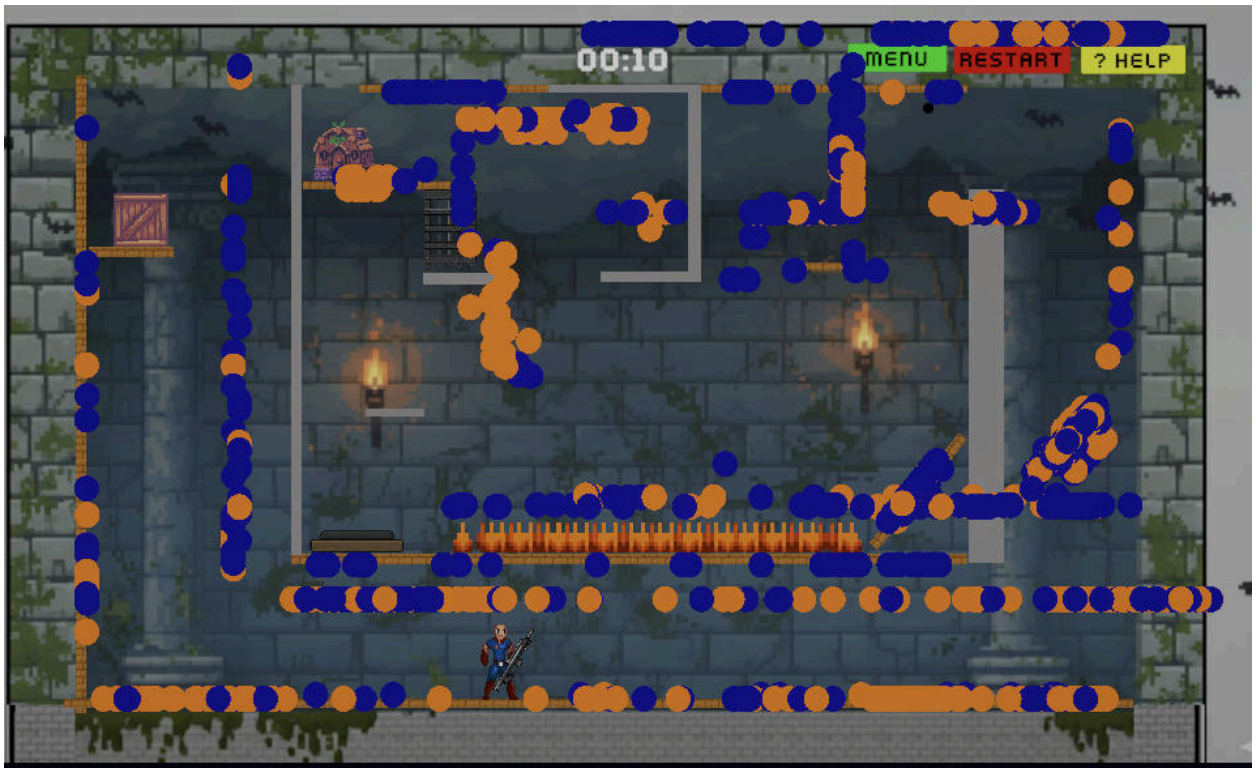
Week 15



Level 9



Level 10



Justification:
We will need to not just check how many portals are being shot at each level but also WHERE the portals are being shot. This will provide us with valuable insight to figure out how players are playing the game. If the players are getting stuck, if the players are using a section of the level a lot more than others and will also get to know where the players are not going.
This will give us a lot of great pointers on how to design the next levels.

Interpretation:
Overly Concentrated Dot Plots in all levels:
This would mean that the game is very linear and we are not providing the players enough opportunities or we are not educating them about alternative ways to clear the levels.

A good spread of where the portals are being placed:
This would conclude good level design and also good tutorialization of the game. The players are able to use the whole map and the game provides enough alternatives for each player to progress in their own unique way.

Hypothesis:
If the tutorial does not help the player understand the game, there will be very less number of portals being shot near the gate to try and get through the game. There will be less number of portals near enemies to try and teleport them above. If we fail to explain the game well, we will see more players try to avoid the enemies and obstacles instead of using them.

An increased usage of portals near enemies, gates, obstacles and the exit in corollary would mean that the players have the required knowledge and the game is interesting enough for them to try and outsmart the levels to get through.

Weekly Prototype Descriptions

NOTE: Images of the changes are available in the Table below

Week-7 & Week-8:

- Analytics from Amplitude
- Player and Gun fixes
- Tutorial levels
- Level 2 and Level 3
- Global leaderboard
- Progression

Week-9:

- Spikes & Player Prefab changes
- Level 3 changes
- Attribute “game_level” added in the Analytics
- Heat-map analytics
- Changes in Game flow **(Change due to feedback from PlayTesting)**
- Leaderboard with pages and the current player’s score
- Resetting Obstacles **(Change due to feedback from PlayTesting)**
- Restarting Levels **(Change due to feedback from PlayTesting)**

Week-10:

- Tutorial 1 and Tutorial 2 (Introduces the mechanic to the users who do not understand how portals work. Change due to the feedback during play testing.)
- Level 4, Level 5 and Level 6
- New Mechanics with the Levels
- Logging all game complete times
- UI updates
- Bug Fixes

Week-11:

- In game HUD
- Level 7
- Level 8
- Laser Gun Improvements
- Bug Fixes and Level Improvements
- UI updates
- 360 Aim **(Changed due to consistent PlayTest feedback)**

Week-12 (BETA):

- Level 9
- Level 10
- Analytics + Youtube Vide
- Leaderboard UI Changes
- Mission Complete and Mission Failed Popup changes
- Quality of Life changes with Tutorial

Week-13:

- Gate and Gate Trigger Prefab changes
- Wall prefab changes
- Portal Prefab changes
- Momentum Fix when jumping through portals **(Changes from Feedback from Beta from Graders and PlayTests)**
- Tutorial changes to make it more engaging **(Changes from Feedback from Beta from Graders and PlayTests)**
- Adding Restart and Obstacle Reset Button on Screen **(Changes from Feedback from Beta from Graders and PlayTests)**
- Laptop Friendly Key Bindings. Portals can now be shot by Pressing C (Blue Portal) and V (Orange Portal) **(Changes from Feedback from Beta from Graders and PlayTests)**

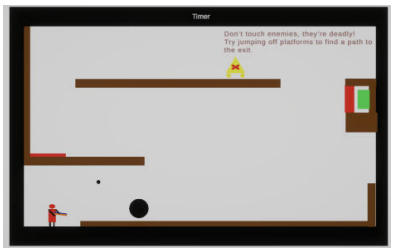
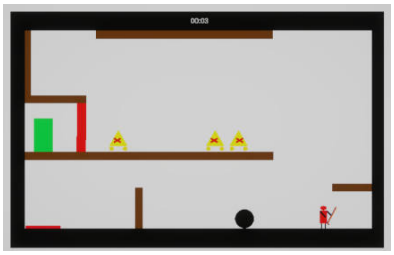
Week-14:

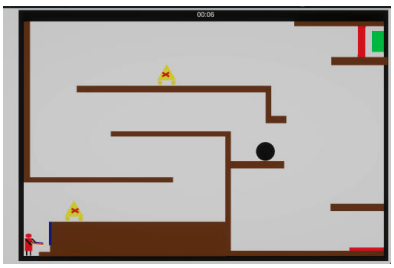
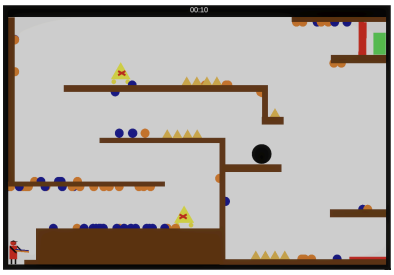
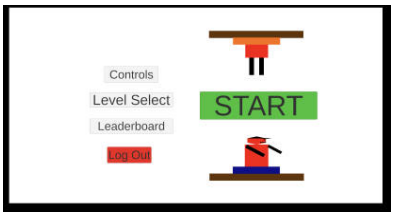
- Player, Enemy, Exit Door assets and Animations
- Background changes for all levels
- Popup for obstacle when it gets stuck next to a wall
- Sound effects and BGM
- Quality of life fixes for gameplay

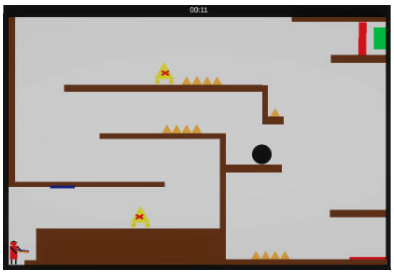
Week-15 (GOLD):

- Start page revamp
- Controls page updates
- QoL changes
- QA testing for Gold release

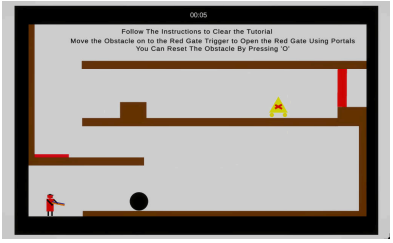
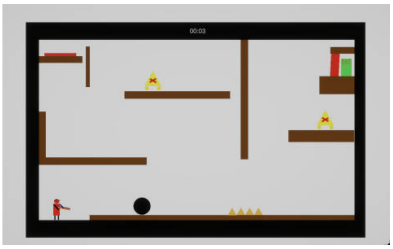
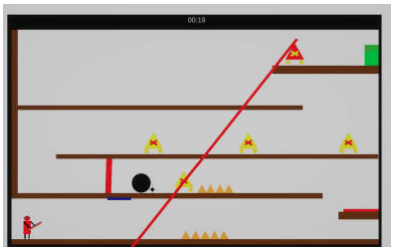
Weekly Progress

Week Number	Team Member	
Week 7 & 8   	**Anirudh Ramadurgam Venkat**	**Wyatt(Yeefong) Wu**
	Analytics, Player & Gun turning, Progression *Jira: DOS-21, DOS-12* *Github:* *https://github.com/CSCI-526/paired-prototype-dos/commit/7042a82f8e641dfbba0665b916f4c066e2b5cdd8,* *https://github.com/CSCI-526/paired-prototype-dos/commit/b7444b121b6ce1fd5be4f8dd9e6ad1cfe4b5fc7a*	*Cloud, DB for global leaderboard & login / register* *Jira: DOS-10* *Github:* *https://github.com/CSCI-526/main-team-dos/commit/b7d78a2a6ab4a550e6fd057bf067d00e5997dd27*
	Aditya Prasanna Mogare	**Saheel Mayekar**
	Created two tutorial levels to improve player progression and ensure players are well-prepared for advanced stages; refined environment flow and level sequencing. *Refined the menu page with redirected links and fixed the start button bug.* *Github Commit:* *https://github.com/CSCI-526/main-team-dos/commit/9f4294e84d09fc985d79eb6a7576509d1b555e65*	*Fixed enemy collision bug when teleporting* *Fixed corner portal adjustments* *Added raycast effect when shooting portals* *Added the ability to make micro adjustments while creating portal of the same color* *Level 2 design and implementation* *Enemy logic rework* *https://github.com/CSCI-526/main-team-dos/pull/6/commits* *https://github.com/CSCI-526/main-team-dos/pull/8/commits*

	**Vedant Jayesh Oza**	**-**
	• *Designed and implemented Level_3_Final scene with vertical multi-level platforming* • *Created 4-tier vertical progression system requiring strategic portal usage* • *Positioned 2 enemies across different platform heights* • *Placed movable obstacles and triggers for puzzle mechanics* • *Github Commits - https://github.com/CSCI-526/main-team-dos/tree/vedant-level-3-game-alpha*	
Week 9 	**Anirudh Ramadurgam Venkat**	**Wyatt(Yeefong) Wu**
	• Analytics for Dot Plot of Portals • Analytics for “game_level” added to all events being to Amplitude • UI changes • GDD Updates	• Leaderboard with pages and the current player’s score
	**Aditya Prasanna Mogare**	**Saheel Mayekar**
	• Implemented a keybind (O) to instantly reset movable obstacles to their original position, restoring transform and physics state.	- Added spike prefabs - Updated enemy ground check

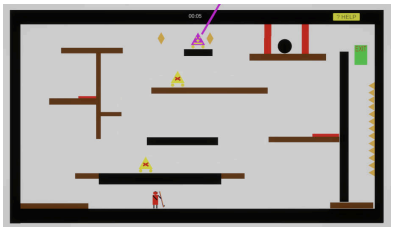
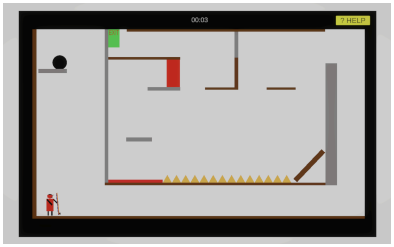
	• Added a universal scene restart (R key) to reload levels seamlessly for quick testing and replay. • https://github.com/CSCI-526/main-team-dos/activity?ref=DOS-29 • https://github.com/CSCI-526/main-team-dos/activity?ref=DOS-32	
	Vedant Jayesh Oza	**-**
	• Alpha Build presentation covering game mechanics, research, and analytics. • Gameplay video for Alpha Build submission • Implemented spike hazards in Level 3 to increase difficulty (PR submitted to alpha branch)	

Week 10	**Anirudh Ramadurgam Venkat**	**Wyatt Wu**
	• Tutorial 1 • Tutorial 2 • Issue Fix with Tutorial	• setup DB/script to log all attempt • create mission fail card

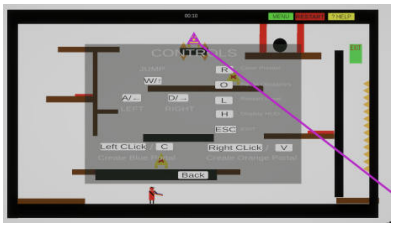
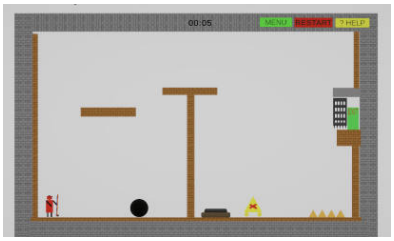
  	times being added to Level 1 Leaderboard times	• Questionnaire • UI clean up
	Aditya Prasanna Mogare	**Saheel Mayekar**
	Design and implementation of Level 5	Design and implementation of level 4
	Vedant Jayesh Oza	**-**
	- Design and implementation of Level 6 - Introduced Laser shooting enemy mechanics at Level 6	

Week 11	**Anirudh Ramadurgam Venkat**	**Wyatt Wu**
 	• In Game HUD Button and Keyboard Click • HUD Integration in all levels • Revamp of Controls Screen • https://github.com/CSCI-526/main-team-dos/pull/31 • Documentation for UI • Tutorial Fix • 360 Aim Fix	Design and implementation of level 7
	Aditya Prasanna Mogare	**Saheel Mayekar**
	Design and implementation of level 8 https://github.com/CSCI-526/main-team-dos/pull/37	Momentum rework Bug fixes with moving entities Added momentum art

	Vedant Jayesh Oza	**-**
	Fixed LaserGunEnemy bugs in Level 6: corrected laser alignment, implemented right directional shooting constraints, and increased warning duration from 1.5s to 3.0s for better player reaction time. Also simplified the level layout. https://github.com/CSCI-526/main-team-dos/pull/39	

Week 12	**Anirudh Ramadurgam Venkat**	**Wyatt Wu**
 	• Level 9 • Tutorial Change • Level Select Page now has Level 9 and Level 10 • Dot plot update for all the levels • Analytics and Presentation prep	• fix the leaderboard UI • Fix the camera sizing issue • Design questionnaire
	Aditya Prasanna Mogare	**Saheel Mayekar**
	• Extracted and filtered portal event data from Amplitude using AmplitudeData.py and FilterEventData.py, generating a clean portals.csv with X, Y, and level info. • Integrated a DotPlotter system in Unity to visualize all blue and orange portal shots across levels using real gameplay data. • Configured prefabs,	- Level 10 - Bug Fixes and Improvements

	scaling, and scene-based plotting to accurately display player portal placements for analysis and debugging.	
	Vedant Jayesh Oza	**-**
	Improved Mission Complete and Mission Failed UI screens. Changed Mission Failed panel to. Implemented changes universally through PlayerCollision.cs to tackle the background duplicate Mission failed Text Pop Up https://github.com/CSCI-526/main-team-dos/pull/42	

Week 13  	**Anirudh Ramadurgam Venkat**	**Wyatt Wu**
	- Laptop Friendly Keybindings - Tutorial Changes - Menu and Restart Button Added to Each level next to the Help Button - Changes in the UI	- Overhaul all wall design - Create art for all wall
	Aditya PrasannaMogare	**Saheel Mayekar**
	-Gate and Gate Trigger Overhauls - Animation for Gate Triggers	• Fixed enemy bug • Reworked momentum • Added art to lvl 4 • Fixed bug/feature
	Vedant Jayesh Oza	**-**

	- Blue and Orange Portal Overhauls	

Week 14 	**Anirudh Ramadurgam Venkat**	**Wyatt Wu**
	- Player, Enemy, Exit Door asset changes and animations	Background design from lv5-lv10 Bug fix
	Aditya Prasanna MOgare	**Saheel Mayekar**
	Background design from tut1-lvl4 Bug fix	Added background music and SFX
	Vedant Jayesh Oza	**-**
	- Portal Glow - Popup on the Movable Obstacle when it touches a Wall saying "You can reset me by Pressing O"	

Week 15 	**Anirudh Ramadurgam Venkat**	**Wyatt Wu**
	- Level 4 design changes - QA - Updates of QoL improvements	- Overhaul login, starter, level select page - Create new font for the game
	Aditya Prasanna Mogare	**Saheel Mayekar**
	- Analytics through Dot Plotter Update - Creation of YouTube Video	- QA Testing - Bug Fixes - Speed Run Documentation
	Vedant Jayesh Oza	**-**
	Overhaul Controls page UI: add blue background, portal	

	icons, and visual consistency	

Note: This is to be followed for the rest of the semester. Make sure it is updated each week.

Review Notes

Notes/feedback that you receive from staff and playtesters, as well as possible improvements you could make based on it. It is important that the stenographer makes note of these each week, but other team members should also keep track of incoming feedback.

Playtest 1: 10/17/2025

- Player feedback during live playtesting:
    - Player thought they soft-locked themselves by putting the ball in the corner
    - Player didn't know how to restart the game, tried to restart the webpage
    - Player directly went to the start, didn't know there was a tutorial
    - Player didn't figure out about the momentum aspect
    - Player didn’t know you could teleport the enemy using portals

Improvements based on Playtest 1: 10/17/2025 (Alpha build)

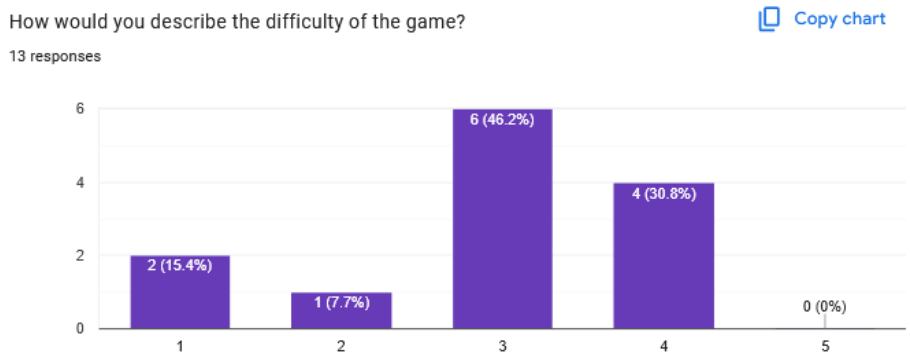
- Soft-locking through Obstacles is mitigated as obstacles can be reset by pressing “O”
- The START goes directly to the tutorial
- Restarting levels can be done by pressing “L”
- Teleporting enemies is a part of tutorial progression

Survey Based on Alpha Build:

PlayTest 1 / Alpha Playtest results

- Analytics:
    - Added in the Analytics section.

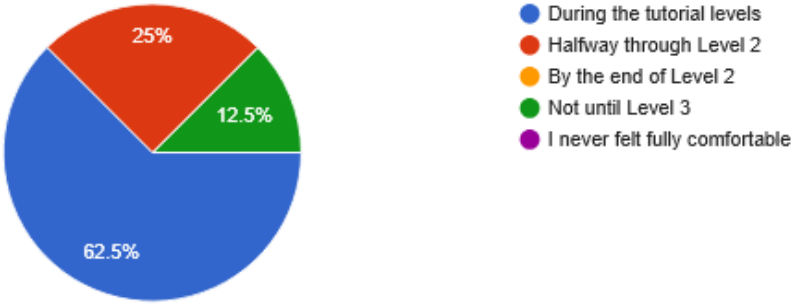
- Survey:



At what point did you feel comfortable with the portal mechanics?

Copy chart

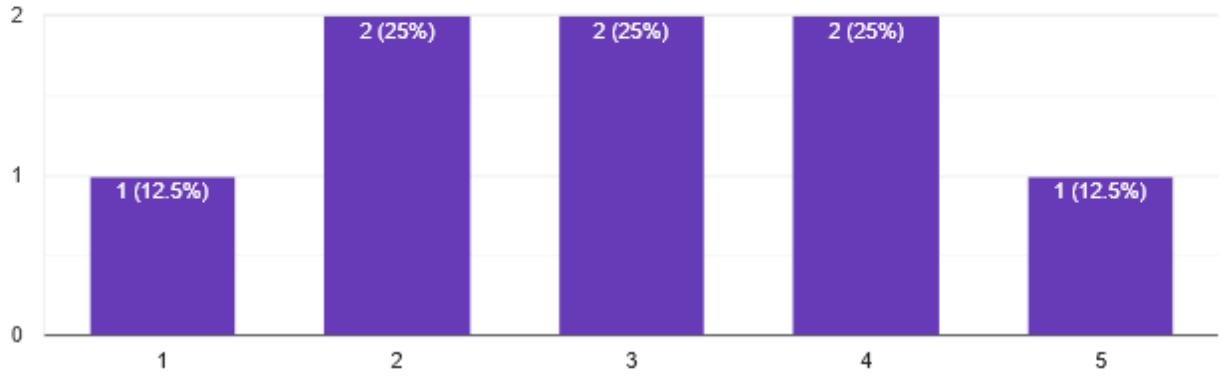
8 responses



How often did you intentionally use falling momentum through portal to reach higher platforms per game

Copy chart

8 responses



Playtest 2 10/31/2025

- Player feedback during live playtesting:
  - Could figure out how they died in the portal in tutorial 2 - death not clear, could make death card smaller/transparent
  - Players don't know about the portal reset button

Playtest 3 11/07/2025

- Player feedback during live playtesting:
    - Did not realise what the pressure plate does in the tutorial
    - Need to include in the tutorial that you can reset portals and reset the obstacle
    - Not clear on how momentum mechanics work, it does not work when walking into opposite-facing portals
    - Players had a confusion that blue portal was to go in and orange was to come out (Why do the colors matter?)

Beta PlayTest 11/14/2025

- https://docs.google.com/spreadsheets/d/16UeaoaVMz5aIT3KDX-lUbYRM8_7J8VcGPVDY1-cX_4w/edit

Staff Feedback

- Tutorial text placement makes it harder to focus on. **[Fixed]**
- Momentum Issue while jumping **[Fixed]**

Playtest 4 11/21/2025:

- Player feedback during live playtesting:
    - Gun sometimes does not work when close to wall, no gun feedback - not a big fan
    - Portal bug - tp from the other side in vedant's lvl
    - Players took a lot of time on the 3 story lvl
    - Reset portal and obstacle bindings not that evident, people forget about it
    - Hidden stage in the think outside the box level
    - Guidance in what to do - the main way to complete the level
    - Invincible against the finish gate
    - Lvl 2 a lot of bugs, collider of pressure plate, collider of gate

Hypothesized Issues from Feedback (Gold)

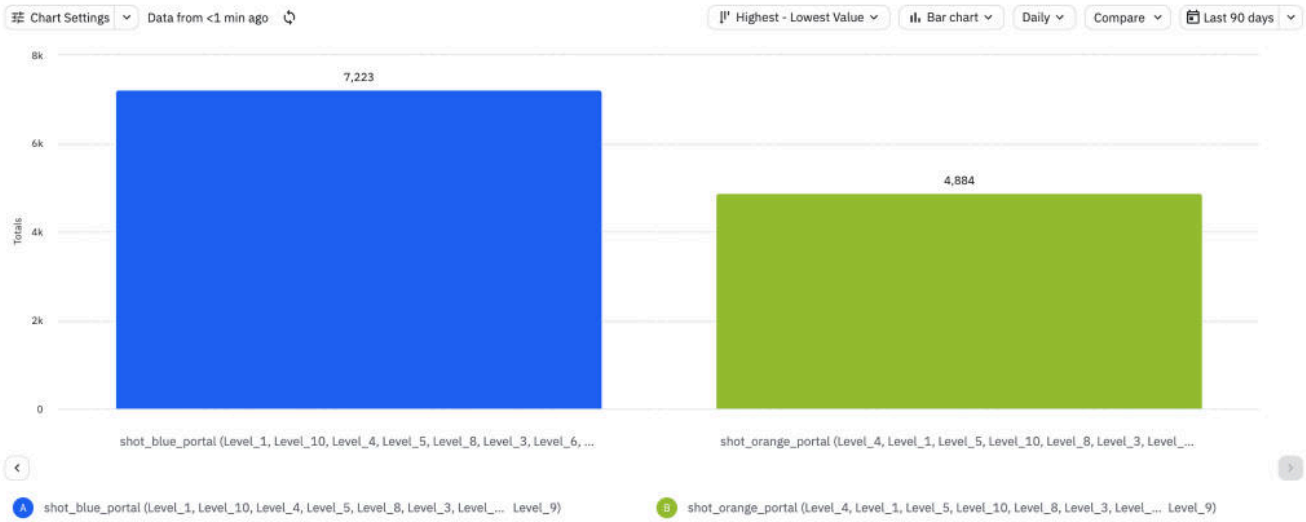
Hypothesized Issue #1

Shooting of Portals on Laptops or Devices without a Mouse is difficult.

Significance
Since shooting of the Portals, especially the Orange portal is difficult on laptop devices, it alienates a large portion of the users we were focusing on and makes the game unplayable. Having alternate shooting controls will help bring in a larger audience.

Feedback/Data points

- From the feedback we received on the beta playtest:
  - “Would be better if there were options to shoot portal(not only mouse buttons, maybe some dedicated keyboard buttons too).”
  - “Find a way to allow players to play on Mac laptop.”
  - “May be bound left click and right click to keyboard(optional)”
  - “Without a physical mouse, i was confused about how to place the orange portal initially”
  - “maybe different ways to set down the portals?”
- From the Metrics #1 from the Analytics:
  - There is a significant skew in the amount of blue and orange portals being shot. We would likely assume that for every blue portal shot, there must be an accompanying orange portal. A small skew would be negligible based on the player’s play styles but a 32% decrease from the blue portal's number hints at a problem.



- From play testing and Grader Feedback:
    - During the live play with the Grader and other pods, we saw people really struggle with the right click on Mac devices and even on other laptops it was difficult to get accustomed to the game without a mouse.

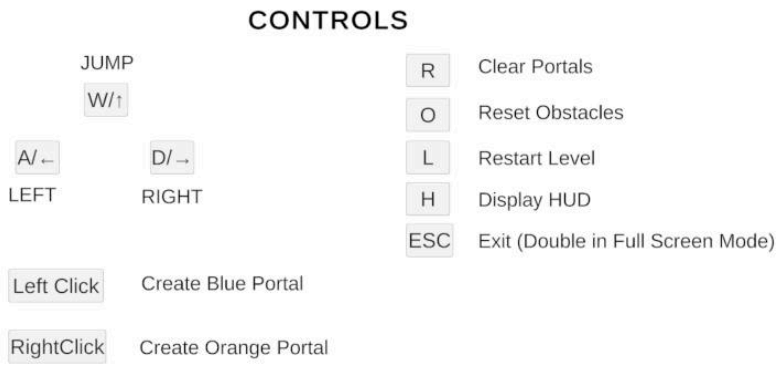
Potential Fixes

1. Add alternate controls with some keyboard key options for shooting the portals and not rely solely on the clicks of a mouse/trackpad.
2. Make the gun shoot blue and orange portals alternatively with just the left click and remove the need for 2 buttons to shoot the portals.

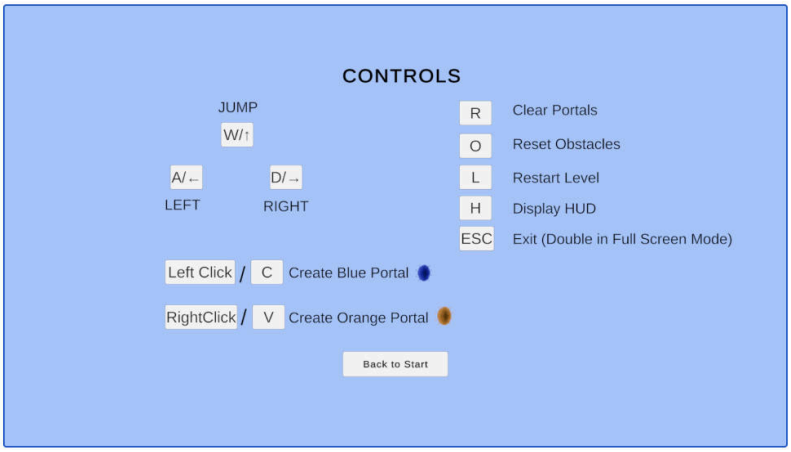
Implemented Solution

We binded the keys “C” for shooting the blue portal and “V” for shooting the orange portal. The specific keys C and V were chosen because when the players use their left hand on WAD to move around, the Thumb is really close to “C” and “V” and it becomes intuitive very quickly to use these buttons to shoot and just use your right hand for aiming.

Before:



After:



Hypothesized Issue #2

Too many buttons to memorize and difficulty in restarting levels or going back to the Menu

Significance

Since the players found it increasingly frustrating to remember different buttons for really important controls of the game, we found that a lot of players were getting stuck at really mundane places where we did not expect them to. Players were having difficulties with restarting the level, finding the controls or going back to the Menu to select another level to play even though there were keyboard controls to take all these actions. This made the user experience frustrating even though the players enjoyed playing the game.

Feedback/Data points

- Feedback from the Beta playtest:
    - “Frustrating: There are so many controls, maybe for stuff like restart level and all that, put it in the menu rather than have it as a control (maybe add the control next to the button on the menu)”
    - “A retry button and a main menu button for navigation between levels and restarting the levels would be great”
- Feedback from Live playtests:
    - We saw that people were having to either kill the player or refresh the whole page to even get to restart the level or go back to the menu

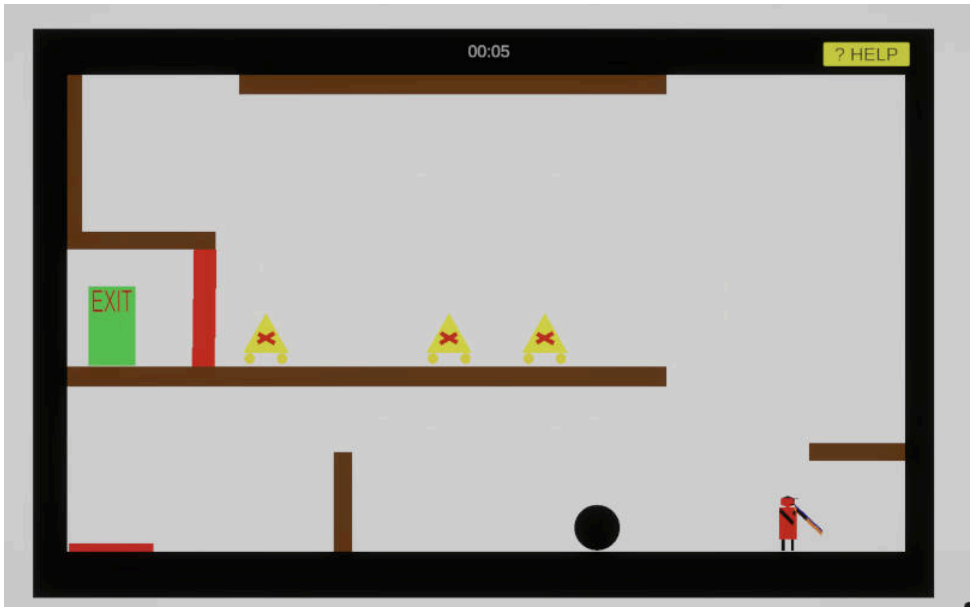
Potential Fixes

1. Add the buttons needed for Restart, Help and Menu in the game at the top right corner so people will be able to easily access them.
2. Introduce these buttons in the Tutorial levels so people have a better chance at remembering them.

Implemented Solution

We added the Restart, Menu and the Help button on the top right corner of each level. This makes it really easy to navigate through the game, hopefully alleviating the user’s pain from remembering too many buttons for basic game controls.

Before:



After:



Hypothesized Issue #3

Movable Obstacles once pushed against the Wall cannot be used and the Players find it hard to continue with the Level

Significance

The gates and the gate triggers are an important aspect of the puzzles in our game and players found it hard to use the movable obstacle to trigger the gate trigger. This results in player's not being able to complete the level or having to restart the level. We introduced the mechanic of pressing "O" to restart the obstacle during the tutorial but it was not intuitive enough with the rest of the controls to remember in the next few levels.

Feedback/Data points

- Feedback from Beta playtest:
    - "The ball gets stuck in a corner and I'm not able to place a portal there."
    - "The ball gets stuck too often"
    - "I get lock sometimes after moving the rocks to some ends..Need restart options"
    - "Once the ball/object gets pushed against a wall, I am not able to find a way to move it back and not able to place portals beneath or on side of it to move it back"
    - "If the ball gets stuck, then theres no way to restart the game."
    - "When balls are pushed at a corner, the only solution is restarting the level."
    - There was even more feedback from the beta playtest that mentioned that the ball/obstacle getting stuck at some wall is a terminal state for the level.
- Feedback from live play testing:
    - During the playtest, we saw a lot of players restart the level entirely instead of trying to reset the obstacle

Potential Fixes

1. Have a pop up show up on top of the obstacle for a few seconds to show that it can be reset by pressing "O"
2. Add a button up top near the 3 other buttons for restart, menu and help called "Reset Obstacle"

Implemented Solution

When the movable obstacle collides with a wall, we have a pop up which shows for 5

seconds the following text “Press O to reset Obstacle”. This helps the user to reset the obstacle and the timer for 5 seconds does not become an eye sore if the obstacle is supposed to be there due to a gate trigger.

Before:



After:



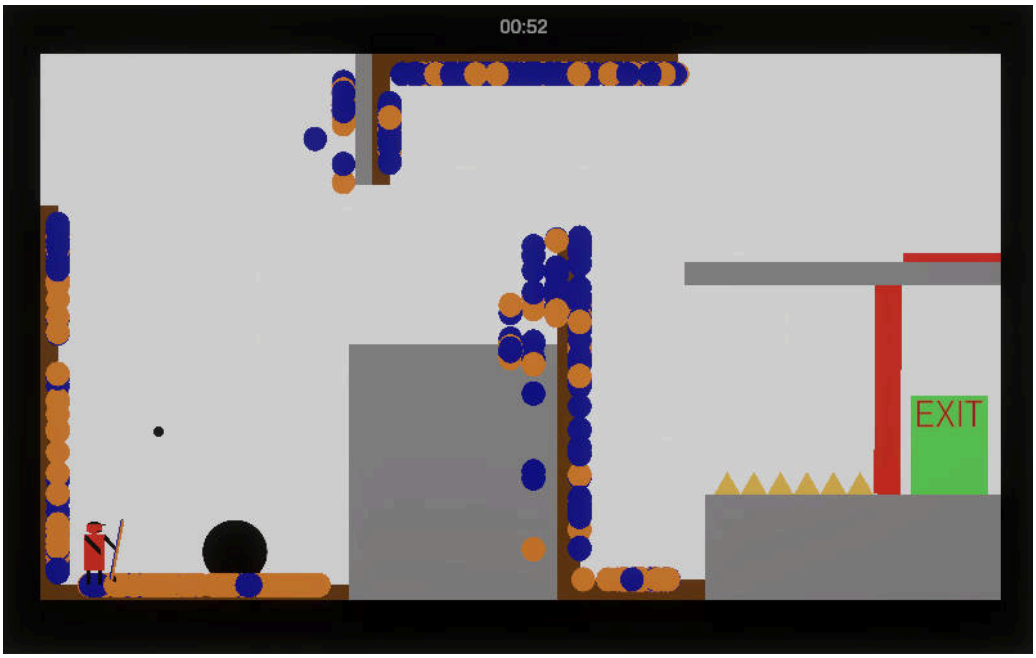
Hypothesized Issue #4

Level 4 difficulty jump is high without any written hints or help from directional art.

Significance

The difficulty jump from level 3 to level 4 is significant as level 4 relies on the momentum mechanic of the game which is not very obvious. We found players having difficulties with the level through playtests and analytics. This is a significant hindrance as the level does not provide enough context for the players on how to use momentum, which not only helps in clearing this level but also sets the player up with the required knowledge to get through the rest of the game.

Feedback/Data points

- Feedback from the Analytics:
    - The entire game focuses on being played in non-linear ways. However, we kept the game play for level 4 straight forward. There is only ONE way to trigger the gate and that is by using the momentum to launch the ball to the gate trigger ledge.
    - However, seeing the dot plots for the portals, it looks like players are trying their hand at everything because the dot plot should not be that spread out for this level. We see equal concentration of portals on all portal walls when we expect a higher concentration of portals on the bottom and the top walls of the level.
    - 

- Feedback from live playtest:
    - We saw people take a significantly larger amount of time in clearing level 4 compared to level 1,2,3 and even compared to some of the higher levels. The players were stuck mostly on what mechanic would come in handy in the level and when prompted about what needs to be done, we found that most of the players were able to clear the level easily.

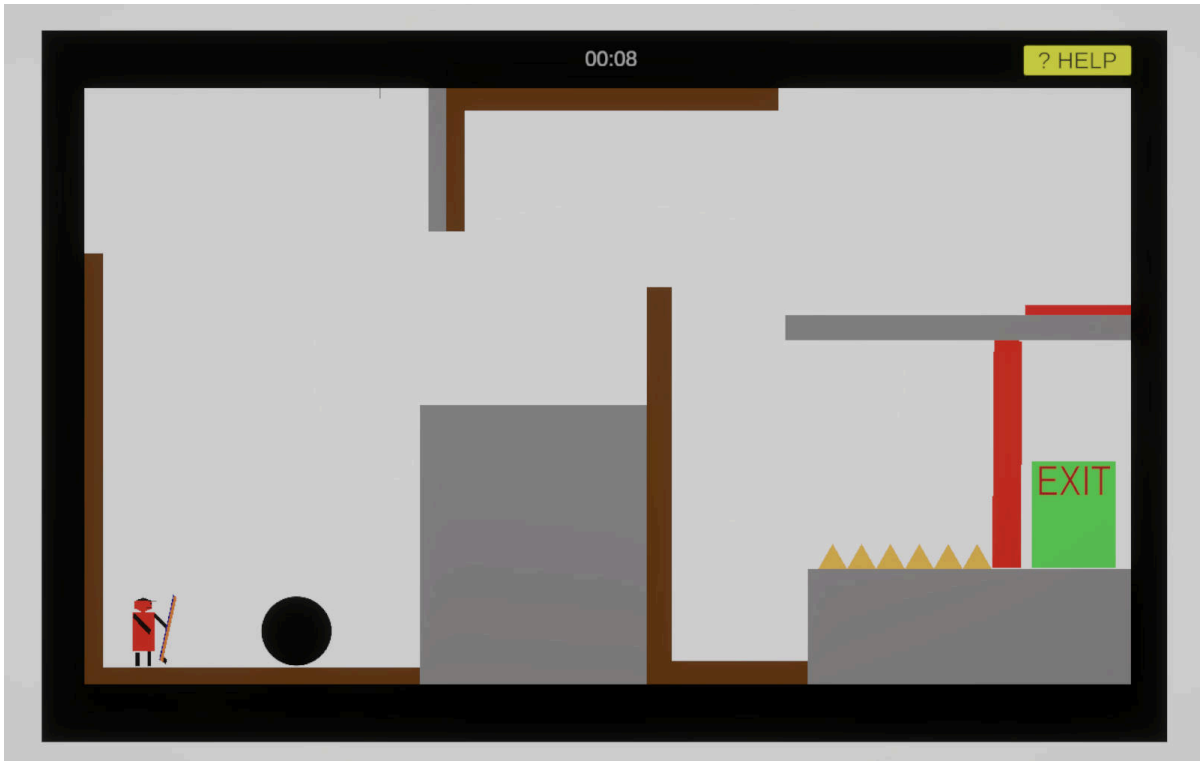
Potential Fixes

1. Make the level easier by having another way the players can trigger the gate.
2. Provide visual cues for the players on how momentum works and have the movable obstacle in a place that makes it clear but not too easy on how to clear this level

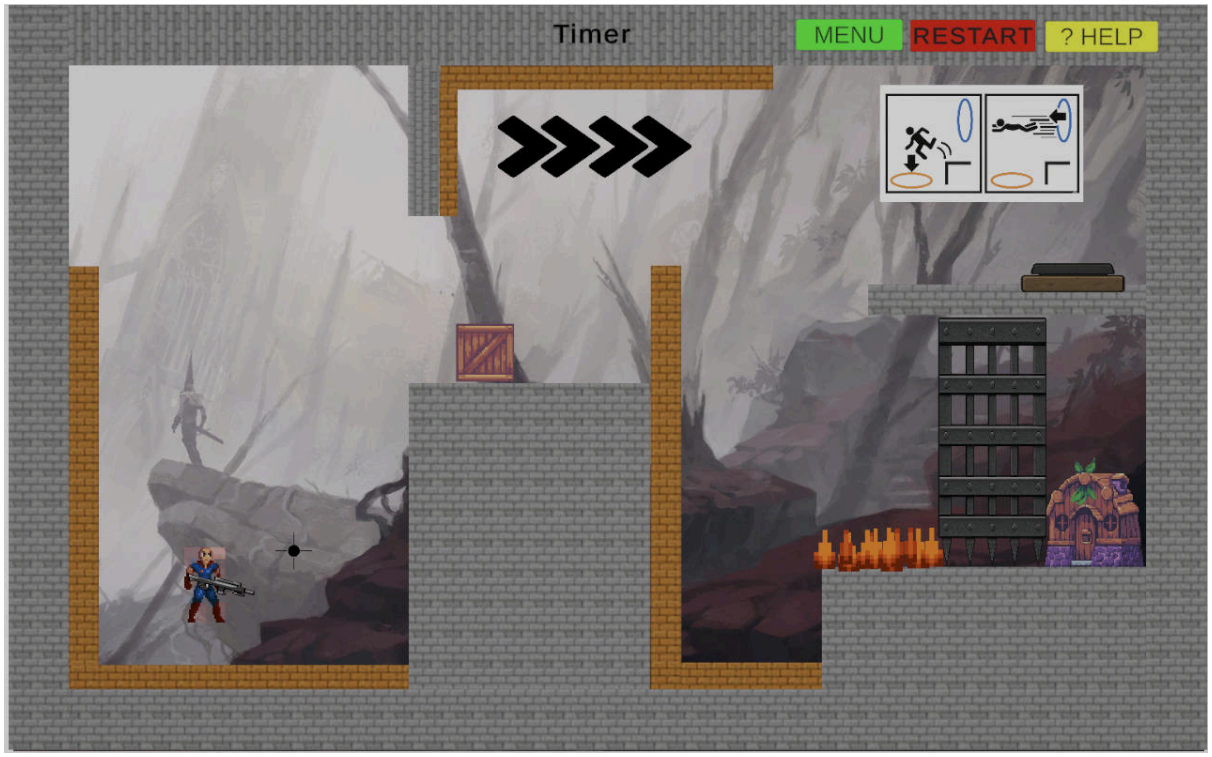
Implemented Solution

We added visual cues to the level which show how momentum will work and placed the movable obstacle on top of the ledge to indicate that this needs to be pushed from here. The visual cues make it more intuitive to understand what needs to be done but does not give away how to clear the level because there is another place for the usage of momentum which involves the user using the momentum mechanic to launch the player to the exit door. These visual cues only help in giving the player a hint but not the solution itself.

Before:



After:



External References

Link to any references (tutorials, assets, etc) from other sources used in your game here. Explicitly state how they are used.
*Note: You get credit solely for what you have developed yourselves. Assets will not count towards the grade and **must** be mentioned below.*
If external assets / code / tutorials are used without reference, you will incur negative consequences.

Name	**Link**	**Usage**
Wall texture	https://www.shutterstock.com/image-vector/2d-pixel-art-brick-wall-1353030242?dd_referrer=https%3A%2F%2Fwww.google.com%2F	*Put the wall texture to all our wall previously was just solid color*
Portal Texture blue and orange	https://f1xtach.itch.io/pixel-art-portal?download	*Put the Portal texture to both portals, previously was just solid color*
Gate and Gate Trigger	Created through AI Art: Google Gemini	*Put the new Gates and Gate triggers previously was just solid color*
Player Character	https://assetstore.unity.com/packages/2d/characters/axel-pixel-character-193747	Player character + Animations
Portal Gun	https://assetstore.unity.com/packages/2d/pixel-guns-310729	Portal Gun Asset
Laser Enemy and Common Enemy	https://assetstore.unity.com/packages/2d/characters/sunny-land-103349	Sprites and Animations for Enemies
Exit Door	https://assetstore.unity.com/packages/2d/characters/sunny-land-103349	Made the Exit Door into a Wooden House
Background Music	Created through AI Art: Google Gemini	Background tut1-lv4
Background Music	https://www.youtube.com/watch?v=c9B86DuVyIg&t=50s	Game music
Movement SFX	https://www.dropbox.com/scl/fo/mm3x18p8wqkewb7ezshye/AJutnYm09fil6iiqhSogp7k?rlkey=pgekxzquqgselmlc56mf8l8uv&e=1&dl=0	Sound effects when walking, jumping, landing, dying, reaching the end gate
Shooting SFX	https://assetstore.unity.com/package	Sound effect when shooting portals

	s/audio/sound-fx/shooting-sound-177096	
Teleporting SFX	https://www.myinstants.com/en/instant/ender-portal-2/	Sound effect when teleporting through portals
Pixel font	https://kenney.nl/assets/kenney-fonts	Pixel style font for the game
Background scene	gemini	Generate bg image for different scenes

Appendix

Misc

Other miscellaneous information if any.